

AN ARCHIVAL ALMA SEARCH FOR UNRESOLVED SPECTRAL-CARRIER TECHNOSIGNATURES TOWARD 60 STELLAR SYSTEMS WITHIN 40 PC AT 114–873 GHz

GLENN J. WHITE^{1,2} AND ROBIN DEY³

Version September 27, 2026

ABSTRACT

Experiment. We have searched the public ALMA archive for an unresolved, drifting spectral carrier from nearby stars: 402 fine-channel windows toward 60 systems over 47.7 GHz of unique sky frequency, plus 1249 coarse windows adding 70.4 GHz that cannot discriminate drift.

Sensitivity. Measured end to end, by injecting carriers into calibrated visibilities and recovering them through the unmodified pipeline. The median system’s best window reaches 1.5×10^{15} W EIRP; of 60 systems, 25 reach 10^{15} W, with a calibration uncertain by $\times 0.64$ – $\times 1.04$.

Result. **No independently confirmed carrier.** All 56 crossings are tabulated with dispositions: 16 are circumstellar or foreground CO, including the β Pictoris disc recovered blind; 40 are unattributed, of which 2 also outrank all 512 spatial controls, and neither recurs. *A crossing is not itself unusual, and we quantify why:* a flat 5σ is a threshold, not a false-alarm rate, and the independent searched cells per window span 4.8 decades, so a uniform per-window 1 per cent criterion needs 3.52–15.86 instead. Measured window by window, 27.3–37.6 of the 51 crossings are expected by chance at a flat 5σ (Class A 33–41 by three routes, against 50 observed), so the population needs no artefact hypothesis and the evidence is attribution, rank and recurrence.

Multi-epoch stack. Coherently combining epochs in the stellar frame over 78 stars deepens the limit $1.39\times$ in the median and reaches 5.15×10^{12} W for Proxima Centauri; the only survivors are β Pictoris CO in two bands. The gain is $\sqrt{N_{\text{eff}}}$, not $\sqrt{N}$, and reflex motion above 0.635 km s^{-1} de-registers a transmitter.

Limitations. Only 42 of 82 systems have epochs more than a day apart, so the result constrains persistent carriers and says little about intermittent ones; and the targets were observed for other purposes, so no inference to the 40 pc stellar population is attempted.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

A technological civilisation, signalling deliberately or going about its affairs, may imprint the electromagnetic spectrum detectably, a *technosignature* (Sheikh 2020; Vidal et al. 2026). Six decades of radio searches cover many thousands of stars yet sample only 6×10^{-18} of the haystack of transmitter frequency, bandwidth, power and duty cycle (Wright et al. 2018). Two gaps motivate this work. The millimetre and submillimetre regime is very sparsely searched, and everything above that axis’s ~ 115 GHz upper limit more sparsely still; and few searches reprocess *existing* interferometric archives, although those archives already contain a spectrum of every target they point at.

This paper is an archival ALMA search of nearby stellar systems over a largely unexplored frequency regime, with an explicit empirical characterisation of its sensitivity and selection function. It is not, and does not attempt to be, a survey of the local stellar population: the sample is whatever ALMA happened to observe, and §6.5 sets out how far that is from representative. The signal class searched is an *unresolved spectral-carrier technosig-*

nature: emission confined to a small number of spectral channels, spatially unresolved at the stellar position on the relevant baselines, and persistent enough to recur between epochs. Each qualifier is a real restriction, since a carrier broader than the searched channel width, one that fills the synthesised beam, or one that appears in a single epoch and never again would not be recovered here whatever its power. The fine channels used are 15.3–1953 kHz wide, orders of magnitude broader than a centimetre-wave narrowband backend, so “narrowband” throughout describes the assumed transmitter and never the resolving power of the data (§4); no hertz-scale spectral resolution is claimed. We mine the archive, correcting for proper motion and Doppler drift, for such carriers. Channelisation sets which of two search classes a window belongs to: 76 per cent of the searched windows carry no drift discrimination at all, and only 402 support the fine-channel search that “narrowband SETI” normally means.

Three contributions follow. The first is the systematic exploitation of archival ALMA stellar fields, 112 target/band datasets mined from public data with no new observing time. The second is spatial-control screen calibration for interferometric SETI, in which the same statistic is evaluated at the star and at 512 control positions so that the false-alarm rate is measured and not assumed. The third is a quantitative accounting of the frequency, power and drift domain actually searched, which gives the non-detection explicit boundaries. One worked

¹ School of Physical Sciences, The Open University, Walton Hall, Milton Keynes, MK7 6AA, UK; glenn.white@open.ac.uk

² RAL Space, STFC Rutherford Appleton Laboratory, Harwell Campus, Didcot, OX11 0QX, UK

³ VBRL Holdings, 50/5 Huay Kaew Road, Chang Phueak, Chiang Mai, Thailand

case runs through the paper, a crossing towards β Picoris, which is the largest star-versus-control contrast here and is attributed to the disc’s known circumstellar CO across two transitions and 9 execution blocks (§5.2). Figure 2 is the roadmap: it gives the selection from catalogue to searched sample on the left, and the decision chain from window to disposition on the right.

Two properties of the design bear on how every number below should be read. The statistic and the molecular-line mask were defined on these data, so every rate quoted here is post hoc; §4.1, §C.1.1 and Table 21 set out what was decided when, and against what had already been seen. And the statistic compares a single position with single positions, which rules out the obvious region-maximum alternative on grounds of exchangeability; both forms are scored for every window and released, and Appendix K.2 audits the comparison.

One convention holds throughout: **wherever this paper says “sensitivity” without naming another quantity, it means P_{90}^{sel}** . Four power scales appear below and only the last is a sensitivity: the trigger level P_{trig} , the power P_{eff} a transmitter must radiate to reach it, the trigger’s own 90 per cent recovery P_{90} , and P_{90}^{sel} , the power the *complete* automated selection recovers nine times in ten. P_{90}^{sel} is the only one of the four quoted in the abstract, the discussion and the conclusions; Appendix E defines them and every other term used here in a particular sense.

The targets were not chosen for a technosignature search but are the stars that archival disc and planet-formation programmes happened to observe, so the sample is enriched in F and G stars and depleted in M dwarfs relative to the 40 pc census, with only 21 of 5 908 catalogued M dwarfs within 40 pc in it at all. Every statistical statement below therefore concerns *the ALMA archive-selected stellar sample within 40 pc* and not the solar neighbourhood; in particular the survey cannot constrain the local habitable-zone planet population, which M dwarfs dominate (§6.5).

2. BACKGROUND

Radio searches, from Cocconi & Morrison (1959) and Drake (1961) through the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002), divide into wide-field commensal and targeted surveys (Zhang et al. 2020; Czech et al. 2021; Morrison et al. 2023; Li et al. 2026; Johnson et al. 2023), with wide-field low-frequency coverage from the MWA (Tremblay et al. 2024) and a single-dish C- and K-band programme at 6 and 18 GHz with the Sardinia Radio Telescope (Manunza et al. 2025), with machine-learning interference rejection now central (Ma et al. 2023; Choza et al. 2024; Jacobson-Bell et al. 2025; Poznanski et al. 2025; Garrett et al. 2026), a capability our classical threshold search lacks. Essentially all of that work lies below ~ 30 GHz, above which only Mason et al. (2025) has searched, in ALMA Band 3. There are physical reasons to go higher: interstellar scattering and dispersion fall steeply with frequency, a given aperture delivers higher gain and hence higher EIRP per watt of transmitter power, and directional communication, radar and beamed power are already engineered as coherent mm-wave emission. Atmospheric transmission, receiver noise and phase stability raise the cost without forbid-

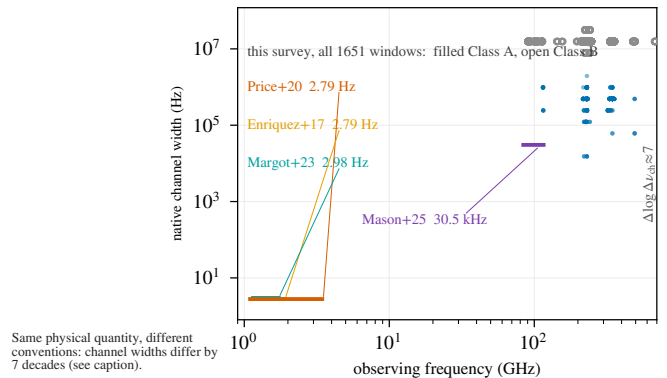


FIG. 1. — Where this survey sits in spectral resolution. Native channel width against observing frequency for every searched window (filled: Class A, fine-channel; open: Class B, coarse-channel), with the comparison programmes as horizontal segments: Enriquez et al. (2017), Margot et al. (2023), Price et al. (2020) and Mason et al. (2025). The distinctive point of this survey is its frequency regime, not its depth: the channel widths plotted span about seven orders of magnitude — which is what the $\Delta \log \Delta \nu_{\text{ch}} \approx 7$ annotation marks — so the channelisation, not the plotted power, is what differs most between programmes. The corresponding EIRP-versus-distance comparison, which requires that warning to be read correctly, is Fig. 8 in Appendix E.

ding it.

Mason et al. (2025) searched archival Band 3 windows toward 28 Gaia DR3 stars lying in the fields of 4 ALMA calibrators, observed for other science and usable for free, placing $\text{EIRP}_{\text{min}} > 6.9 \times 10^{17} \text{ W}$ for the nearest, at 1.01 kpc, and identifying the challenges of high-frequency work: drift rates scale with frequency, smearing erodes sensitivity without a drift search, and the dense forest of molecular lines raises the confusion background. Every window reported here was re-reduced and re-searched from the raw archive data with the pipeline of §4, so the numbers to cite for any star in common are this paper’s.

Figure 1 places this survey among published searches. The vertical axis is physically commensurable: for a carrier narrower than every programme’s channel the minimum detectable EIRP is a total radiated power, not a spectral power density, so the same quantity is plotted for each. What differs is not the unit but the convention — each programme adopts its own detection threshold and its own completeness level — and the difference runs one way here. The value plotted for this survey is P_{90}^{sel} , a 90 per cent recovery through the complete automated selection, whereas most published limits are nominal thresholds at which recovery is not stated. The figure also shows that ALMA occupies a different combination of frequency and spectral resolution from cm-wave SETI. Two numerical comparisons differ. Against centimetre-wave work, essentially all of which lies below ~ 30 GHz, a search spanning 89.6–873.1 GHz raises the frequency by about two orders of magnitude. Against the only prior millimetre search, Mason et al. (2025) in Band 3, the increase is about a factor of nine. Any EIRP advantage toward the nearest systems is a d^2 effect and not an instrumental one, decomposed in §6.2. This survey samples far fewer stars, far fewer epochs and a channelisation seven orders of magnitude coarser than a dedicated narrowband search; it is complementary to those surveys rather than a deeper version of them.

A non-detection must become a quantitative exclusion statement: Wright et al. (2018) formalised the “Cosmic

Haystack” search volume, Margot et al. (2023) its conversion into an upper bound on the transmitter-hosting fraction, and Sheikh et al. (2025) asked where Earth’s technosignatures would be detectable. That bound is valid only if the sample represents the underlying population, which interest-selected samples do not: most surveys select by proximity, spectral type, planets or Earth Transit Zone membership, and the ALMA-specific form, disc selection, skews the sample young and dynamically active. We instead take every star within 40 pc lying in the usable field of at least one public ALMA observation (§3). That sample is *work-list complete but epoch-incomplete*, and this paper uses no other completeness formulation: every star on the frozen work list that proves genuinely covered is searched, but not every execution block those stars hold is in this release, and no completeness is claimed for the local population itself. The 168 sample entries are ~ 1 per cent of the catalogued population within 40 pc (the 89 actually searched ~ 0.5 per cent), and the residual coverage selection is characterised as a completeness function (Table 1).

3. SAMPLE SELECTION

The sample is *work-list complete but epoch-incomplete* within 40 pc (Table 1), a much weaker condition than volume completeness. The execution-block accounting closes as follows, and every number in this paper refers to one of these populations.

1. 656 public execution blocks lie in the parent member observing-unit sets of the sample stars. This is the population in scope, and it splits in two.
2. 479 of them have been downloaded, calibrated and searched; 177 have not. Every one of the 177 is repeat coverage: each sits in an observing-unit set from which another block was already taken, and the de-duplication rule below keeps one block per star and tuning. 20 of them the archive offers as raw visibilities with no pipeline calibration at all. That is the first identity, $479 + 177 = 656$, which splits finer as $401 + 78 + 177 = 656$ into science-sample blocks, the reserved hold-out of §C.1.1, and blocks never searched.
3. 484 blocks were processed in total, 4.1 TB of raw data through one frozen pipeline. This is a *different* population from the 479: it is those plus 5 whose member-OUS progenitor link the archive does not expose (3 of them in the science sample), so they cannot be placed in the ledger above. That is the second identity, $479 + 5 = 484$. The two counts must not be added to each other.
4. Of the 484, 404 form the *frozen science sample* reported here, 77 are the reserved calibration hold-out of §C.1.1, and 3 left no surviving window after the quality cut.

No claim is made to have searched the whole ALMA archive, only every star on the work list; Table 28 gives the provenance of every dataset used anywhere in this paper.

The procedure admits every star within 40 pc of the Sun (Gaia DR3 parallaxes, distances direct inversions $d = 1/\pi$; every entry $\pi \geq 25$ mas with a parallax measured to better than 20 per cent, which is the condition the archive query applied) whose proper-motion-corrected position at the observation epoch falls in the

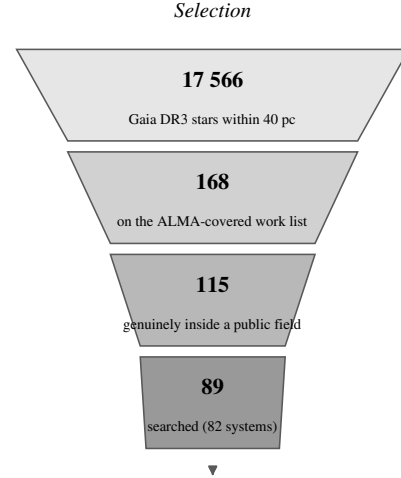


FIG. 2.— The selection funnel. The four denominators the text keeps apart are the catalogued population, the 168-entry ALMA-covered work list, the ~ 115 stars a public field genuinely contains, and the 89 searched. Every count is read from the generated macros, so the figure, the list of §4.3 and the released catalogue cannot disagree. The chain applied to what comes out of it is Fig. 5.

quoted field of view of at least one public ALMA execution block. The star need not sit at the pointing centre, and §A.3 shows the quoted field of view admitted 53 stars no observation in fact contains, so “covered” is a property of the selection and not a guarantee. Disc, planet and spectral type enter as descriptive metadata only, and units are defined in §A.2. This gives 168 unique target stars, processed nearest-first; because trigger thresholds scale with d^2 at fixed noise that order is sensitivity-ranked, so the prevalence statements of §6 are conditional on it.

Composition follows from the archival selection and is tabulated against the reference census in Table 14. The sample is F- and G-enriched and deficient in M dwarfs, and 285 of the 1651 windows come from 3 systems; §6.5 gives the figures and their cost. The spectral-class proxy is `teff_gspphot`, available for 53 of 89 sample stars and missing preferentially for the coolest objects, so the M fraction is a lower limit and the earlier-type ratios upper limits. Appendix K lists the cross-match and exclusion edge cases, including the “ γ Lupi” relabelling.

“Usable public ALMA coverage” is the five-criterion decision tree of Appendix K. No star or window was excluded by an unlisted criterion, and none for short integration.

4. DATA AND METHODOLOGY

Three abbreviations recur below: QA2 is ALMA’s second-stage quality assurance, the stage that delivers calibrated data; a member observing unit set (MOUS) is the scheduling container holding one or more execution blocks; and TDM and FDM are the correlator’s time- and frequency-division modes, giving coarse and fine channelisation. Symbols, including T_* , T_{trig} and the power scales, are defined in Appendix E.

One shared archive-ingestion stage feeds two independent search lanes, and Table 8 gives the data path of the primary lane in order. For each target the calibrated archival execution blocks covering the star’s Gaia-propagated position are mined within a bounded budget.

TABLE 1

THE SURVEY AS SEARCHED, AND THE PAPER’S HEADLINE NUMBERS IN ONE PLACE. EVERY COUNT QUOTED IN THIS PAPER RECONCILES WITH IT. THE TWO COMPLETENESS EXPERIMENTS MEASURE DIFFERENT QUANTITIES AND ARE KEPT IN SEPARATE TABLES. EVERY FREQUENCY UNION QUOTED HERE IS AN AGGREGATE OVER A CLUSTERED, REPEATEDLY REVISITED COVERAGE, WHOSE ACTUAL DISTRIBUTION IS FIG. 7; THE UNION SHOULD NOT BE READ AS CONTINUOUS COVERAGE OF ITS ENDPOINTS.

Quantity	Value
Stars within 40 pc with qualifying public ALMA coverage	168
genuinely pointed at; never observed (§A.3)	115; 53
Stars searched; independent systems	89; 82
Spectral windows searched	1651
12 m array; ACA 7 m	597; 1054
Class A fine-channel / Class B coarse-channel	402 / 1249 (76% Class B)
Execution blocks; distance range	404; 1.30–39.63 pc
$\Delta\nu_A$, <i>unique frequency union</i> , Class A (the drifting-carrier experiment)	47.7 GHz
$\Delta\nu_B$ -only, coverage Class B adds	70.4 GHz
$\Delta\nu_{A \cup B}$, total archival coverage, of which only $\Delta\nu_A$ is drift-resolving	118.1 GHz (46 intervals, 89.6–873.1 GHz; Fig. 7)
usable after the molecular-line mask	113.3 GHz
Native channel widths	15.3 kHz–31.25 MHz; median 15.625 MHz
90 per cent selection P_{90}^{sel} , Class A, per window	6.4×10^{13}–2.0×10^{17} W (median 2.7×10^{15}) ^a
per system, best window	6.4×10^{13} – 3.5×10^{16} W (median 1.5×10^{15})
Nominal trigger P_{trig} , Class A; Class B	2.2×10^{13} – 6.9×10^{16} W; 1.6×10^{13} – 2.8×10^{17} W
Threshold crossings ($T_* \geq T_{\text{trig}}$ at the stellar position)	56 windows (52 Class A, 4 Class B)
Star-exceeds-ring windows ($T_* \geq T_{\text{trig}}$)	12 windows (12 Class A, so all of them)
circumstellar or foreground CO	10
unattributed, absent in the repeat block	2
Persistent, independently confirmed unresolved spectral-carrier technosignatures, within the domain searched	0

^a Power and frequency quantities are given to two significant figures, except where a channel or sky frequency must be matched to the catalogue and is given to the precision the grid supports. The internal consistency checks quoted elsewhere to a per cent (§K.1.2) bound *relative* quantities within one dataset; the absolute flux scale rests on ALMA’s delivered calibration.

Their visibilities feed (i) the *primary* lane, a Doppler-drift-corrected spectral-carrier search following Mason et al. (2025) and Enriquez et al. (2017), which converts each non-detection into an equivalent isotropic radiated power limit $\text{EIRP}_{\text{trig}} = 4\pi d^2 S_{\text{min}} \Delta\nu$. The product $S_{\text{min}} \Delta\nu$ is an integrated power only if the channel’s effective noise bandwidth equals its separation, which for a smoothed correlator it does not; the response factor C_{response} corrects that. It is computed from the correlator’s own lag window and runs $\times 2.00$ – $\times 2.36$ with sub-channel position, median $\times 2.08$, and is validated against ALMA’s stated channel resolutions — including the effective resolution the archive reports for every window in this survey (Appendix G). So $P_{\text{eff}} = S_{\text{min}} \Delta\nu C_{\text{response}} C_{\text{smear}}$ recovers the injected power, with S_{min} the primary-beam-corrected trigger-level flux density defined below, $\Delta\nu$ the native channel width and d the distance (worked numerically in Appendix F). EIRPs are quoted to two significant figures, the precision the flux scale supports. Five terms enter the systematic budget, each multiplying S_{min} and

hence $\text{EIRP}_{\text{trig}}$ directly.

- (i) The ALMA absolute flux scale: 7 per cent band-weighted, up to 20 per cent at Bands 9 and 10, and epoch-dependent.
- (ii) Residual atmospheric phase after water-vapour correction and phase referencing: a coherent point-source loss $\exp(-\sigma_\phi^2/2)$, 5–20 per cent at 230–345 GHz. It is common to star and annulus within the isoplanatic patch and so no threat to exchangeability, but it is *not* common to the injected tones that calibrate P_{90} , which is why §5.4 carries it as a one-sided bias.
- (iii) The primary-beam model at the stellar offset: negligible on axis, 1.9 per cent at the worst pointing error.
- (iv) Visibility calibration: bounded empirically at 4.5 per cent. An inconsistent delivered amplitude scale would make the ratio of each window’s measured per-integration rms to its radiometric prediction scatter from block to block beyond what thermal noise explains. Over 2271 windows in 559 blocks the per-block median of that ratio is 0.956 with a robust scatter of 4.5 per cent, against 0.8 per cent within a block. It enters the combined budget of Table 12 in quadrature with the other two-sided terms.
- (v) The instrumental spectral response: median $\times 2.08$, carried explicitly as P_{eff} by the reading rule below. It is a definition rather than an uncertainty, and most of it is a change of units, as the next paragraph separates.

How much of the response factor is a real loss. The factor $\times 2.08$ is large, but it is not mostly a penalty. It factorises exactly into two parts. P_{trig} is quoted against the noise of the *delivered*, smoothed channel, whose rms is 0.612 of the raw-channel rms for the 0.25/0.5/0.25 kernel; converting to that convention accounts for $\times 1.63$ of the factor and is the same number for every window in the survey. What remains is the genuine inefficiency of reading a carrier in one smoothed channel: the peak channel holds 0.500 to 0.424 of the carrier while the noise has fallen by 0.612, so relative to an ideal single-channel detector on unsmoothed data the loss is $\times 1.22$ – $\times 1.44$, median $\times 1.28$. That is the number to have in mind when comparing this search with one on unsmoothed data; $\times 2.08$ is the number to multiply P_{trig} by, because P_{trig} is defined in smoothed-channel units.

Gaia distance errors are negligible, 0.2 per cent typically. Two survey classes are reported, distinguished by the dimensionless drift parameter

$$\eta_{\text{drift}} = |\dot{\nu}_{\text{max}}| \tau_{\text{track}} / \Delta\nu_{\text{ch}}, \quad (1)$$

the channel-width fraction the *maximum* searched drift $|\dot{\nu}_{\text{max}}|$ traverses over the full on-source track τ_{track} at native channel width $\Delta\nu_{\text{ch}}$. The track is the right time-scale because the de-drift step sums along it; τ_{track} is written throughout, leaving T for the search statistic alone, and η_{smear} (Appendix F) is the same ratio over one integration. A window is *drift-resolving* when $\eta_{\text{drift}} \gtrsim 1$, carrying a carrier across enough channels for its motion to be measured. The drift ceiling inside η_{drift} is a search-grid boundary, the widest trial drift the stack evaluates, and not a prior on transmitter drift rates: carriers drifting

faster are missed, and §6.1 gives the resulting selection function (Fig. 3).

Class A is the primary experiment, the fine-channel drifting-spectral-carrier search: the 402 windows of native width 15.259–1953.125 kHz (median 488.3 kHz; 60 per cent at the 488.3-kHz mode), spanning $\eta_{\text{drift}} = 0.17$ –422 (median 14.9), and the only class with measured end-to-end recovery of a drifting carrier. *Class B* is the coarse-channel spectral-excess search: the 1249 windows, 76 per cent of the sample, of native width 7.812–31.25 MHz (1209 at the 15.625-MHz mode), spanning $\eta_{\text{drift}} = 0.004$ –1.35 (median 0.51), sensitive to persistent near-stationary excess power but carrying no drift discrimination. It is the *secondary experiment* throughout, an unresolved spectral-excess search, and is never described as a carrier search here.

η_{drift} is the physical criterion and the classes are labelled by the instrument. Channelisation is bimodal, with no window between 1953.125 kHz and 7.812 MHz, an empty gap of factor 4.0, so any boundary inside the gap assigns identical membership. The instrumental split is the one used throughout; η_{drift} is released per window so any reader may reclassify on the physical criterion, which selects 427 windows of 1651 against 402 in Class A. *The two criteria do not partition the survey the same way, in either direction, and we do not claim that they do*: 13 of the 402 Class A windows (3.2 per cent) have $\eta_{\text{drift}} < 1$ and are therefore not drift-resolving, while 38 of the 1249 Class B windows reach $\eta_{\text{drift}} \geq 1$ (at the very bottom of the Class B range, whose maximum is 1.35). The 13 are not a class-boundary effect but a track-length one: every one is a short 60–181 s on-source track, against a Class A median of 2389 s, at the coarsest of the fine channel widths (488–977 kHz), and $\eta_{\text{drift}} = |\dot{\nu}_{\text{max}}| \tau_{\text{track}} / \Delta \nu_{\text{ch}}$ cannot reach unity for a one-minute track at 977 kHz even at the search grid’s drift ceiling. They lie on 8 execution blocks toward 6 stars and none of them contains a threshold crossing. Each window’s η_{drift} , η_{smear} , trial drift count and maximum searched acceleration $a_{\text{max}} = 3.60$ –4.00 m s^{-2} are released catalogue columns, so any reader may reclassify on the physical criterion alone. Results are given per class first and combined only where statistically appropriate (§5.2).

Catalogued molecular transitions are excluded: the search applies a $\pm 50 \text{ km s}^{-1}$ velocity exclusion mask (§5.2.6) to any threshold crossing, evaluated topocentrically against each crossing’s observed sky frequency. That is the frame the search ran, and it is proved from the released offsets rather than asserted (Appendix I); earlier versions of this paper described the mask as a stellar-frame one, which was a documentation error and changes no attribution. **The half-width was set after the β Pictoris crossings had been inspected, and is the one element of the analysis not fixed in advance.**

Table 2 therefore reclassifies all 56 crossings at ± 13 , ± 20 , ± 30 , ± 50 and $\pm 100 \text{ km s}^{-1}$, in both frames. The decisive entry is not a count but a set identity: at ± 50 and at $\pm 100 \text{ km s}^{-1}$ the rank-screened unattributed crossings are the identical 2 events, 61 Vir and CP–72 2713, in *both* frames. A half-width chosen to bury a candidate would show one reappearing when the tube was widened; here it can be doubled, in either

TABLE 2

CROSSINGS ATTRIBUTED, UNATTRIBUTED, AND UNATTRIBUTED WHILE ALSO OUTRANKING ALL 512 CONTROLS, AGAINST THE MOLECULAR-LINE MASK HALF-WIDTH, IN THE SKY (TOPOCENTRIC) FRAME THE SEARCH USED AND IN THE STELLAR FRAME. OVER ALL 56 THRESHOLD CROSSINGS; THE SKY COLUMN IS EVALUABLE FOR 53 AS RELEASED AND FOR ALL 56 ONCE THE ISOTOPOLOGUE KEY IS REPAIRED, WHICH CHANGES NO CELL, AND THE STELLAR COLUMN FOR 44, THE REMAINING 12 LACKING A RECOVERABLE PER-BLOCK BARYCENTRIC TERM OR A SYSTEMIC VELOCITY. THE ADOPTED HALF-WIDTH IS IN BOLD. WIDENING FROM ± 13 TO $\pm 50 \text{ km s}^{-1}$ COSTS 9 OF 11 IN THE SKY FRAME BUT ONLY 1 OF 3 IN THE STELLAR FRAME: THE WIDER TOLERANCE IS *not* FREE, AND WHAT IT REMOVES IS THE SURVEY’S OWN IDENTIFIED CO.

half-width	sky (topocentric)			stellar		
	attr.	unattr.	rank-fl.	attr.	unattr.	rank-fl.
± 13	1	52	11	13	31	3
± 20	3	50	10	14	30	2
± 30	12	41	3	14	30	2
± 50	16	37	2	14	30	2
± 100	16	37	2	14	30	2

frame, and nothing new appears. The generator asserts the set equality, not the counts.

The physical scale that motivates masking at all is the Keplerian bound, $\lesssim 13 \text{ km s}^{-1}$ where these molecules survive beyond 10 au, and it is a stellar-frame quantity.

The mask is a classification mask, not a detection mask. It is applied after the search, to decide how an event that has already been found should be interpreted, and it removes nothing from the detection stage: every masked window is searched exactly as an unmasked one is, and no threshold moves. What it defines is the frequency space *excluded from the exclusion statement*. The survey covers 118.1 GHz of unique sky frequency in 46 clustered islands (Fig. 7); the effective default quoted throughout is the $\approx 113.3 \text{ GHz}$ surviving the mask, and no constraint is placed on carriers inside the excluded velocity regions, not because they could not have been detected but because a detection there could not have been distinguished from the molecular emission the mask marks.

All flux densities and EIRP thresholds are total intensity, $I = (XX + YY)/2$, the QA2-weighted mean of the two parallel-hand cross-correlations of ALMA’s linear feeds. No cross-hand extraction was performed; the consequence for what the survey can and cannot discriminate is stated once, in the main limitations (§5.4), and is not repeated here. Incomplete parallactic-angle sampling, unequal flagging of the two hands and instrumental leakage all break the equivalence between a strongly polarised and an unpolarised signal of the same total power. The archive records both parallel hands for 198 of the 200 blocks whose polarisation setup it resolves, but not per-correlation flagging inside QA2; a window that lost a hand would be $\sqrt{2}$ less sensitive than its threshold implies, and that factor is carried as a one-sided term in Table 12.

Two power scales carry the main text, and Table 11 defines the rest. The trigger power P_{trig} is five times the window’s per-channel rms at its native channel width, for a transmitter at the star’s position matching the searched morphology: the level at which the search fires. It is an internal pipeline quantity, not a sensitivity and not a significance. P_{90}^{sel} , the power the complete automated selection recovers nine times in ten, is $\times 5.7$ the nominal trigger for Class A, and is the survey sensitiv-

TABLE 3

THE FOUR TERMS THIS PAPER COUNTS IN, AND NOTHING ELSE. EVERY COUNT IN THE MANUSCRIPT IS A COUNT OF ONE OF THESE FOUR OBJECTS, AND EACH IS STATED WITH ITS UNIT. EARLIER VERSIONS ALSO USED *stage-1 event*, *flagged window* AND *unattributed event* AS PARTIAL SYNONYMS; THOSE ARE RETAINED ONLY WHERE THE UNIT DIFFERS, AND ALWAYS WITH THE UNIT NAMED. “CANDIDATE” MEANS LEVEL 4 AND NOTHING WEAKER.

Term	Definition	This survey
Threshold crossing	A channel×drift cell at the propagated stellar position of one window of one execution block reaches $T_{\star} \geq T_{\text{trig}}$	56
Screened event	A threshold crossing whose window also ranks the star above all 512 control positions	12
Localised event	A screened or unscreened crossing for which the drift-following visibility fit shows emission at the stellar coordinates	40
Confirmed candidate	A localised event that recurs at the same tuning in an independent epoch, or is independently confirmed	0

ity. It comes from 8860 carriers injected into calibrated visibilities across 52 block-window configurations spanning both arrays, four ALMA bands and channelisations from 15.3 kHz to 31.25 MHz (§5.4, Appendix G); P_{95} remains *not determined*, because recovery saturates below it. Between them sits the response-corrected threshold $P_{\text{eff}} = 2.08 P_{\text{trig}}$, which converts the trigger from smoothed-channel units into transmitter power, and P_{90} , the trigger-stage completeness, ×3.0 the nominal trigger for Class A; both are released per window and are defined in Appendix E. Every headline EIRP is quoted beside its native channel width. Three signal morphologies are detectable, and §6.1 quantifies each against real windows: a continuous frequency-stationary carrier, a continuously drifting one, and an intermittent one.

4.1. Final candidate definition and statistical interpretation

The four terms are defined in Table 3 and are used with those meanings everywhere. Two cautions attach to the ladder. It is *not* a chain of increasing evidence in this survey: the order in which the tests are applied is set out below, the visibility fit is a check on position and epoch rather than a filter (§5.2.1), and the screen’s own status is fixed once, here: The star-versus-control rank is an empirically useful candidate-prioritisation statistic, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection. Its departure from exchangeability is empirical and its source is identified — the control annulus samples a different radial and noise geometry from the star (Appendix K.4.2) — and the parameter-free correction prospectively tested against that geometry does *not* restore a validated null and is not adopted.

The spatial screen ranks the stellar position against 512 single-position controls under one global noise normalisation. The controls lie in an annulus whose geometry is measured from each block’s own data: the dish diameter is read from the antenna table and the longest baseline

TABLE 4

WHAT EACH TEST CAN ESTABLISH. THE FIVE QUESTIONS ASKED OF A WINDOW, AND WHAT AN AFFIRMATIVE ANSWER TO EACH IS WORTH. THIS TABLE IS ABOUT THE *statistics*; TABLE 3 IS ABOUT THE *objects counted*. EVERY STATISTIC DEFINED IN THIS PAPER SERVES ONE OF THESE ROLES, AND THE ORDER HERE IS THE ORDER OF §4.1, NOT A RANKING BY STRENGTH.

Level	Question	Role
Detection	Does a cell exceed the trigger T_{trig} ?	generates crossings; expected 8.8% of the time at one control position
Attribution	Is there a catalogued transition within $\pm 50 \text{ km s}^{-1}$ in the stellar frame?	disposes of most crossings
Screen	Is the star above all 512 controls?	prioritisation only
Recurrence	Does it repeat in an independent epoch?	confirmation
Localisation	Is the emission on the star in the visibilities?	consistency check, reported alongside

from the visibilities, so the primary beam θ_{PB} and the synthesised beam θ_{syn} are both properties of the observation, and the annulus begins at $\max(0.14\theta_{\text{PB}}, 2.0\theta_{\text{syn}})$. The second term is what keeps a control off the star: at 230 GHz an ACA synthesised main lobe is 7.3 arcsec across, and a control inside it is not independent of the star, sitting on its beam and collecting its flux. $1/(512 + 1)$ is the resolution of the ensemble and not a p -value; what the screen is worth, and what it costs, is measured in Appendix J.

Throughout, p denotes a genuine significance level, such as the Kolmogorov–Smirnov tests on rank distributions; no probability computed against the screen’s own non-exchangeable null is called one.

Table 4 gives what each test can establish. What disposes of an event is the physical evidence, and the decisive test is astrophysical rather than statistical: coincidence with a catalogued transition in the star’s own frame, and recurrence in an independent epoch. The visibility domain answers a third question — whether the emission is at the stellar coordinates — and it is reported for every crossing, but it is a consistency check on position and epoch rather than a filter, for reasons measured in §5.2.1.

The selection logic is four steps, in order. The first generates crossings; the rest dispose of them.

- (i) Threshold crossing. A channel×drift cell at the stellar position reaches $T_{\star} \geq T_{\text{trig}}$. A fixed trigger power, not a calibrated p -value, and not a rare event: a single control position in a Class A window reaches the survey’s smallest crossing statistic 8.8 per cent of the time (§5.2.1).
- (ii) Astrophysical identification. The crossing is checked against catalogued molecular transitions in the stellar frame and against instrumental signatures.
- (iii) Spatial prioritisation. The stellar statistic exceeds all 512 control positions in the same window. A candidate-generation mechanism: it carries no significance of its own, for the reasons above.
- (iv) Independent recurrence. It repeats at the same tuning in an *independent epoch*, or acquires indepen-

dent confirmation; and its block is compared with the other blocks of the same star.

The drift-following visibility fit runs on every crossing and is reported beside this chain, not inside it. It asks a genuinely physical question — is the emission at the stellar coordinates? — on recalibrated raw data with no control ensemble of 512 positions; but it is evaluated at the position and in the cell where the search has already found an excess in the same visibilities, and its 12 controls resolve a rank of only 1/13. Its place in the argument, and the reference-epoch defect that once made it look decisive, are in §5.2.1. Every crossing is carried through all four steps in Table 5, and nothing short of (iv) is called a candidate.

All delivered channels of each spectral window were searched, with no edge-channel or bandpass-roll-off trimming before extraction, so residual bandpass structure near a window edge is carried in the calibration-residual term of the systematic budget rather than removed by flagging. The distance of every threshold crossing from its window edges is a released catalogue quantity, reported for the flagged windows in §5.2.

The primary statistic is formed on extracted spectra, never on images or visibilities, from the QA2-calibrated measurement set as delivered, per integration and per native channel, with the archive's flagging carried through untouched and no regridding (Table 8, step 3). What the extraction returns at a propagated sky position $\mathbf{x}$ is the real part of the weighted vector average of the phase-shifted visibilities, the natural-weighted dirty-map value there; there is no deconvolution, so a quoted S_{peak} in a field with extended emission is a dirty-map amplitude. The primary-beam correction enters the quoted thresholds and fluxes (§F.1) and leaves untouched the extracted amplitudes the statistic ranks. Each per-integration spectrum $I(\mathbf{x}, t, \nu)$ is then baseline-subtracted by a block median along the *frequency* axis, with no temporal median at any stage (§4.2). The residual dynamic spectrum is stacked along each trial linear drift rate $\dot{\nu}$ with inverse-variance weights, and the statistic is the maximum of that stack over the channel $\times$ drift grid,

$$T(\mathbf{x}) = \max_{\dot{\nu}, \nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot{\nu}t) / \sigma^2(t, \nu)}{[\sum_t 1 / \sigma^2(t, \nu)]^{1/2}}, \quad (2)$$

a matched filter whose denominator is the noise of the stacked spectrum. $T(\mathbf{x})$ is neither assumed nor shown to be Gaussian or χ distributed under the null, which is why T_{trig} denotes a trigger level and not a tail probability; Appendix J quantifies the departure. It reduces to a single-channel amplitude divided by one noise scale only when σ is the same for every integration and channel; otherwise the weights vary along the track. The scale $\sigma(t, \nu) = 1.4826 \text{ MAD}_{i=1\dots 512}[I_i(t, \nu)]$ is $1.4826 \times$ the median absolute deviation taken *across the control positions* at each integration t and channel ν , making it global in position, local in frequency and time, and shared by the star and every control. The across-position median enters only in forming that deviation and is never subtracted from the numerator, so T is an absolute stacked signal-to-noise carrying a cross-position noise scale, and measures no local contrast.

T is evaluated at one position, identically at the star

and at each control, which is what makes the rank interpretable: a statistic taking a maximum over several stellar-region positions while the controls remain single positions would not be exchangeable between the two sides. A line filling the primary beam drives star and annulus up together instead of cancelling.

The searched drift-rate range is the larger of (a) a generic frequency-scaled default ($12\text{--}13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$), the Mason et al. (2025) fiducial with a threefold safety margin, and (b) for confirmed exoplanet hosts, the maximum line-of-sight orbital acceleration of the innermost known planet, so a transmitter on a short-period planet cannot drift out of the window. Drift rate maps onto line-of-sight acceleration through the first-order Doppler relation

$$a_{\text{los}} = c \frac{\dot{\nu}}{\nu}, \quad (3)$$

so the ceiling is quoted in cross-band-comparable form $\dot{\nu}/\nu$: $12\text{--}13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ is $(1.2\text{--}1.3) \times 10^{-8} \text{ s}^{-1}$, that is $|a_{\text{los}}| = 3.6\text{--}3.9 \text{ m s}^{-2}$ independent of observing frequency, the acceleration at $\sim 0.04 \text{ au}$ around a solar-type star. Earth-motion contributions lie two orders of magnitude below and are subsumed by the grid, which is uniform in $\dot{\nu}$ and fine enough that a carrier midway between two trials drifts at most 0.26 of a native channel over the longest Class A track. Two limits of the model matter more. The rates searched are *apparent* topocentric ones, so the survey does not separate a deliberately chirped carrier from an accelerating platform and constrains only linear apparent drift inside the ceiling; and an emitter need not be tied to a known planet at all, a free-flying platform or rotating structure accelerating as it chooses. For circular orbits the ceiling covers a transmitter co-orbiting an Earth analogue by three orders of magnitude and Proxima b comfortably, while TRAPPIST-1 b ($a_{\text{los}} = 4.00 \text{ m s}^{-2}$) exceeds the default ceiling, 3.60 m s^{-2} , by 11 per cent, and c (2.13 m s^{-2}) reaches it at modest eccentricity, which raises the periastron value by up to $(1+e)/(1-e)^2$ (Fig. 3). TRAPPIST-1's own windows were searched on a widened grid, to 4.00 m s^{-2} , which falls 0.01 per cent short of planet b at its maximum line-of-sight acceleration. TRAPPIST-1 transits, so its inclination is measured and effectively edge-on, and the edge-on figure is the relevant one: against the default ceiling TRAPPIST-1 b lies above it for 29 per cent of its orbital phase, and against the widened one for 1 per cent; it is searchable through the rest.

Across the 1651 windows maximum searched drift rates span $1.1\text{--}10.5 \text{ kHz s}^{-1}$. The number of trial drift rates per window separates sharply by class and must be quoted that way: Class A windows search a median of 81 trials (interquartile 44–138, maximum 1945, the narrow and finely channelised β Pictoris Band 6 window, §D.2.2), while Class B windows search a median of 4 because a coarse channel needs almost no de-drifting. Residual smear separates in the same direction: all 320 Class B windows that carry a stored value have η_{smear} exactly 1, so every residual smear loss the survey records is in Class A, where 9 of 131 windows fall below 0.99 and the worst retains 0.574 of the signal. The threefold margin on the ceiling matters: at $1 \times$ the search would have been blind to two of its threshold-crossing windows, in-

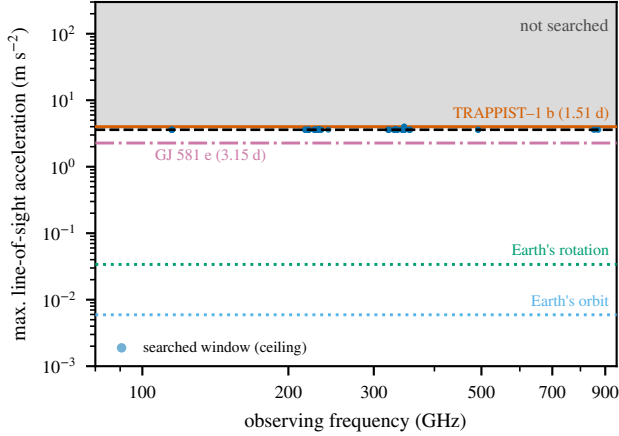


FIG. 3.— What the drift search reaches, as a physical selection function. The drift ceiling of each of the 402 Class A windows (points), converted to line-of-sight acceleration, $|a| = c\dot{\nu}/\nu$ (Eq. 3), against observing frequency; shaded above is unsearched. The boundary is flat because the grid was built at constant acceleration; in the units the search works in it is not, running 1.4 kHz s^{-1} at 115 GHz to 10.5 kHz s^{-1} at 872 GHz . Horizontal lines are reference accelerations: Earth’s rotation (0.034 m s^{-2} at the equator) and orbit (0.0059 m s^{-2}), and the two most demanding known planets of any target, TRAPPIST-1 b (4.00 m s^{-2}) and GJ 581 e (2.27 m s^{-2}). The boundary drawn is the default ceiling, 3.60 m s^{-2} , which is also the median; it runs to 4.00 m s^{-2} on the 5 widened TRAPPIST-1 windows, and it is the window’s own ceiling that decides whether a given orbit is contained.

cluding the AU Mic crossing, while at $5\times$ no disposition changes. Transmitters drifting faster than the $12.0\text{--}13.3 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ ceiling remain unconstrained. Channel widths are effective resolutions from the measurement sets’ spectral window tables (Appendix F).

4.2. Ancillary and validation analyses

Nothing here generates candidates; it is validation, and the detail is in the appendices. Two demonstrations carry the weight. Synthetic tones of known amplitude injected into real visibilities and recovered through the unmodified pipeline — 1200 trials across six window configurations, three bands and both classes (Appendix G) — establish that the machinery recovers a drifting unresolved spectral carrier, the searched morphology; and β Pictoris CO is recovered and localised in two transitions and 9 blocks (§D.2.2), establishing that it finds a real feature at a known stellar position. The baseline step is a per-integration median along *frequency* (Table 8, step 4) and so cannot remove a one- or two-channel feature. Appendix K reports the three quantified checks that bound undetected pipeline faults.

4.3. The experiment in summary

The experiment runs in the following steps, with the number surviving each: 656 public execution blocks whose fields contain a catalogued star within 40 pc; 484 of them delivered with a usable calibration and put through one frozen pipeline with no per-target tuning; 404 in the science sample once 77 are withheld by a rule fixed before any was looked at, a hash of the block identifier, and 3 yield no usable window; 1651 spectral windows, de-duplicated by identifier; 402 of them Class A, the primary experiment, the remaining 1249 supporting only a search for unresolved spectral excess; 56 threshold cross-

ings, where $T_\star \geq 5$ somewhere on the drift grid; 40 of those with no catalogued transition within $\pm 50 \text{ km s}^{-1}$ in the star’s own frame, the other 16 attributed, most of them the β Pictoris debris disc; 2 that also outrank all 512 spatial controls; and 0 that recur in an independent epoch. That chain is Fig. 5, and the survivor count at every step is a column filter on the released catalogue, reproduced by a script in the data release. The drift-following visibility fit is applied to every crossing and reported *beside* the chain rather than inside it, for reasons measured in §5.2.1.

5. RESULTS

The 1651 windows are the size of the processing exercise, not of the technosignature search. **The primary experiment is the Class A drifting spectral-carrier search: 402 fine-channel windows toward 60 systems. The secondary experiment is the Class B unresolved spectral-excess search: 1249 coarse windows with no drift discrimination.** That convention holds in the abstract, in the Results, in the Discussion and in the Conclusions: the combined 1651 appears only where it describes the archive processed, never as the extent of the carrier experiment, and never without “402 Class A” beside it.

The choice of detection statistic makes the non-detection more conservative, not less: the symmetric form of Eq. 2 flags a strict subset of what the region-maximum form flags, 12 windows against 68, with none flagged by the symmetric form alone (§K.2, Table 24). Per-target values, including each band’s searched frequency range, are in the released catalogue (Data Availability).

5.1. Barnard’s Star and Wolf 359

Barnard’s Star is the second-nearest stellar system and Wolf 359 the third if brown dwarfs are excluded, fifth if Luhman 16 and WISE 0855–0714 are counted; to our knowledge these are the first mm or submm technosignature limits for either, the searches checked being Enriquez et al. (2017), Price et al. (2020) and Margot et al. (2023). Both are non-detections, and only Band 6 data enter the limits. The nominal trigger thresholds are $6.2 \times 10^{13}\text{--}9.4 \times 10^{13} \text{ W}$ for Barnard’s Star over 12 windows and $7.4 \times 10^{13}\text{--}1.1 \times 10^{14} \text{ W}$ for Wolf 359 over 12.

5.2. Technosignature search

The screen used here ranks the stellar position against 512 controls under a single global noise normalisation. The star-versus-control rank is an empirically useful candidate-prioritisation statistic, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection. A radius-corrected variant, described in Appendix C, changes no conclusion: both give zero confirmed candidates and differ only in the count of screened events without an attribution, 2 against 1 windows.

All 112 star/band datasets yielded at least one carrier-lane result, 1651 spectral-window measurements after the catalogue audit removed 45 duplicate rows, Bands 6

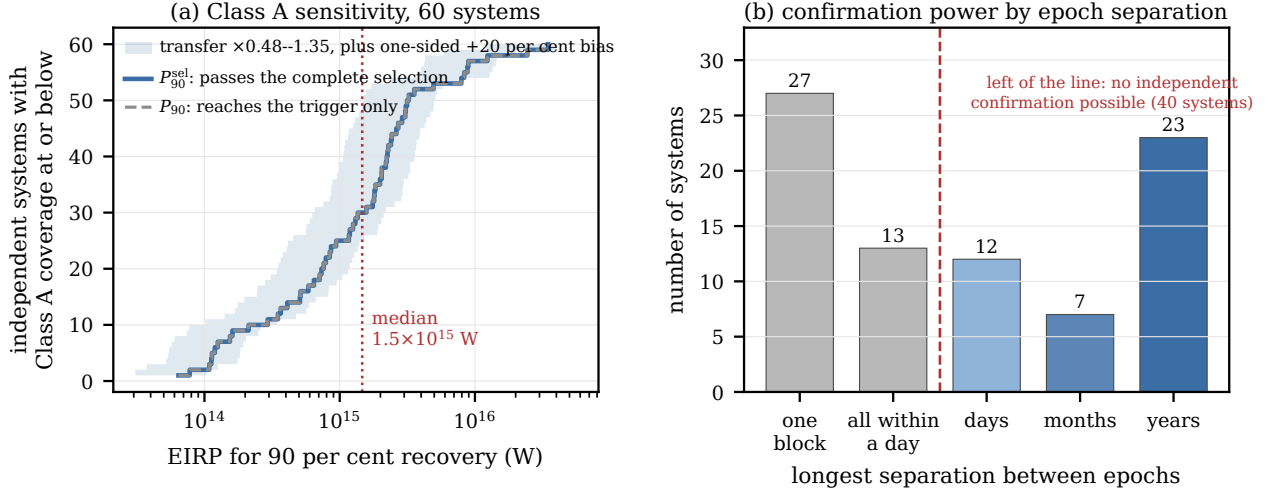


FIG. 4.— (a) Class A sensitivity. Cumulative number of independent systems whose best window reaches a given EIRP, for P_{90}^{sel} , the power passing the complete automated selection (solid), and for P_{90} , which reaches the trigger only (dashed). The shaded band is the systematic budget: the calibration bracket $\times 0.64$ – $\times 1.04$, plus the one-sided +20 per cent atmospheric-decorrelation bias, which is why it is asymmetric. (b) The temporal selection function: systems grouped by the longest separation between their searched epochs, an independent epoch being defined as a separation of more than 1 day. **For the 27 systems with a single block and the 13 whose blocks all fall within one day — left of the dashed line — independent confirmation was not possible at all, whatever the sensitivity in panel (a).** Recurrence over years can be tested for 23 systems.

and 7 dominating (Table 16). Nominal trigger thresholds span 1.6×10^{13} – 2.8×10^{17} W at a median of 7.1×10^{14} W (Figs. 1 and 7), the deepest reached toward the nearest systems — 1.6×10^{13} W for the UV Ceti pair at 2.7 pc — though depth tracks integration time and channelisation as well as distance.

The statistics come strictly before the astrophysics, in the levels of Table 3, which do not promote one another. A *threshold crossing* is one channel $\times$ drift cell at the stellar position reaching $T \geq 5$: 56 of the 1651 windows contain one — 56 crossings in all, counting a window of one block once — and 93 per cent of those windows are Class A, a trials effect rather than a physical one, since a median Class A window holds 2.6×10^5 channel $\times$ drift cells against 468 for a median Class B window. A *screened event* is a crossing whose window also ranks the star above all 512 controls: 12 windows do, in 4 systems and 5 star-band pairs, and those windows are referred to below as the stage-1 flagged windows where the distinction between a window and a crossing matters. A local scramble test, which asks only whether a feature exceeds the noise at its own position, clears the control-ring maximum too in 3 of the 4 windows, so screening turns on the star-versus-control comparison. A *confirmed candidate* is a localised event that recurs or is independently confirmed. There are 0.

The screened events fall into two groups. 10 of the 12 stage-1 windows sit within $\pm 50 \text{ km s}^{-1}$ of a molecular transition in the star’s own frame and carry independent CO evidence: 9 are β Pictoris, in 5 Band 3 and 4 Band 6 windows drawn from 9 execution blocks; one is the published foreground emission at HD 48370. That is 4 astrophysical sources, not 12 independent events, and the recurrence is the point: a frozen pipeline recovers the same disc in two transitions and 9 independent blocks without being told it is there.

The remaining 2 stage-1 windows have no astrophysical attribution, and what disposes of them is the absence of a second epoch rather than anything the visibilities

say. *This is a change from earlier versions of this paper*, which reported both as showing no emission at the stellar position; that finding was produced by a visibility estimator referred to the wrong epoch and is withdrawn (§5.2.1). Under the corrected epoch one of the two, towards 61 Vir, returns $\text{Re}/\sigma = 5.26 - 0.97$ of what a compact source at the star producing its trigger would give — and the other has not yet been re-fitted. Neither recurs, and neither is called a candidate on that ground alone.

The chance expectation of an unattributed stage-1 event, its conditioning on the trigger as well as the rank, and the size of the observed count against it are calibration and are given in Appendix J.

STELLAR FLARES, EXCLUDED BY MEASUREMENT RATHER THAN BY ARGUMENT. The hosts are not a quiet sample. 2 of the 2 carry activity indicators in SIMBAD (Table 23, queried 2026 September 26) and 0 is catalogued as multiple. Previous versions said 1. That count tested SIMBAD’s *aggregated* object-type bag, which accumulates every type any catalogue row for an object has ever carried, and 61 Vir’s bag holds the double-star flag only because the star appears as component A of four historical visual-pair catalogues. Its principal type is high-proper-motion; it has no parent system and no sibling in SIMBAD’s hierarchy, and its only hierarchical children are its three planets. CP–72 2713 in particular is a K7Ve star flagged as an emission-line, rotationally variable X-ray source, the profile of a star that flares. Magnetically active stars do produce coherent bursts at centimetre wavelengths. Stellar activity is therefore the leading astrophysical alternative to an instrumental origin for the crossings — and, unlike in earlier versions of this paper, it is *not* disfavoured by the localisation test, because 40 of the 50 fitted crossings do localise at the stellar position and stellar emission is expected to.

It is excluded instead by a broadband prediction with no free parameter. A millimetre flare raises every chan-

nel of a window by the same flux density, so a flare able to produce a 5σ single-channel crossing must produce a $5\sqrt{N}\sigma$ continuum event in the same integrations — 320σ in a 4096-channel window, 57σ in a 128-channel one. Over the 48 crossings with a recoverable continuum measurement the prediction runs 49–4,717 (median ≈ 170), while the observed continuum statistic in the crossing’s own time window has median -0.29 and $|Z| < 3$ in 45 of 48; the largest continuum excursion anywhere in those blocks, at any duration, corresponds to a single-channel contribution of median 0.05σ and at most 1.97σ . The crossings are also not concentrated in time: the de-drifted track’s boxcar maximum minus the same bank’s maximum at the controls has median -0.53 , and 1 of 48 exceeds $+2$.

One block settles it directly. AU Mic A002_Xa42f75_X319 contains both a threshold crossing ($T_\star = 5.27$) and the published 2015 June 24 millimetre flare of MacGregor et al. (2020). In the crossing’s own time window the continuum is raised at $Z = 10.3$ — the flare — while the crossing’s de-drifted track reaches only $Z_{\max} = 2.1$, below its own controls. *A real, bright, independently published millimetre flare in the very same data contributes 0.17σ to the single-channel statistic.* The statistic has the power to see temporal concentration when it is there: β Pictoris CO, the survey’s strongest real feature at $T_\star = 14.7$ – 30.8 , has a boxcar maximum of 3.2 – 4.4 against 3.8 – 4.2 at its own controls, because a steady line is steady.

Both have a persistence test and neither passes it. Each has 1 to 4 further execution blocks of the same star whose tuning covers the event’s frequency, 5 in total, and the feature recurs in none: the largest stellar statistic anywhere in them is $T_\star = 5.04$, 0 exceed their own window’s control maximum and 0 are stage-1 flagged windows. The baselines are days rather than years: 2 of the repeats are separated from their event by a resolvable interval, 2–769 d, and the remaining 3 fall within the same scheduling block. *These tests exclude a transmitter persistent over hours to days and bound no duty cycle measured in months or years.* Repeat coverage is matched on the window rather than on the crossing frequency, which the release records only where the pipeline stored a peak. CP–72 2713 is examined in detail (§A.3.1).

5.2.1. The crossing ledger, and what the visibility fit is for

Every threshold crossing has been carried through the same chain, and the drift-following visibility fit is reported beside that chain rather than inside it. Table 5 is the resulting ledger: all 56 crossings, each with its search statistic, its stellar-frame line attribution, the 512-position rank flag, the visibility fit under *both* reference epochs, and its disposition. Table 6 sets the counts against what chance would give. The fit is no longer conditional on the image-plane rank: a crossing earns the physical test by crossing the threshold, and the rank is a column beside the result rather than a gate in front of it, because a screen that decides which events are examined cannot also be the evidence that they are real.

The reference epoch, and a defect in the estimator. The fit follows the carrier along its recovered drift rate, which means it must decide the time at which the recovered frequency is the crossing frequency. The implementation used first took the median of the timestamps it loaded;

the search de-drifts to its own first retained integration. The two differ by $\dot{\nu}(t_{\text{med}} - t_0)/\Delta\nu_{\text{ch}}$ channels, which is nothing for a short window and several channels for a long one, and the question is not one of convention: the injection campaign of §5.4 fitted every triggered tone *both* ways. Of 1833 injected carriers in 19 block-window configurations, the 167 displaced by more than 2.8 channels are recovered at a median 0.293 of their injected flux when referred to the search’s own epoch and -0.012 when referred to the median, and are localised in 76.0 per cent of cases against 0 of 167. Below a quarter of a channel the two are the same experiment, which is why a pilot on a short window could not have found this. The paper adopts the search’s epoch; the ledger reports both, because the difference between them is one of the results.

Two consequences follow. The rejections produced by the earlier implementation are *withdrawn as evidence* — an estimator evaluating the wrong channel rejects almost everything, and 33 of the 50 fitted crossings change verdict between the two epochs, every one at a displacement above a channel and none below 0.25 . And the molecular lines offered as the test’s positive controls are withdrawn as validation of the estimator: they are *resolved* emission an order of magnitude wider than a channel, and a track scanning across a line peaks near its centroid whichever epoch is used, so they never exercised the failure mode. A validation has to be sensitive to the error it exists to catch.

The criterion was fixed before any fit was run and recorded at commit `edfcc0d3d7db`: a crossing is localised at the stellar position when $\text{Re}/\sigma \geq 4$ there, $|\text{Im}/\sigma| < 3$, and Re/σ exceeds the maximum returned by the same fit at eight annulus control positions and four off-event control frequencies. 50 of the 56 crossings (89 per cent) have been fitted; the remaining 6 in 4 execution blocks have not, and are recorded as untested rather than as non-detections. Of the fitted crossings 11 are localised under the earlier epoch and 40 under the adopted one, 38 of them with the residual to the extraction epoch verified from the retained products and 2 with it bounded rather than measured.

This is a consistency check on position and epoch, not a candidate filter, and the reason is measurable rather than rhetorical. Of the 44 crossings reaching $\text{Re}/\sigma \geq 4$ under the adopted epoch, the imaginary-part and control clauses reject 4 — all of them BD+05 1668, one grossly displaced source with $|\text{Im}/\sigma|$ of tens — and on the epoch-verified subset 1 of 39. The control clause never rejects on its own. That is what should be expected of it: the fit is evaluated at the position and in the cell where a threshold crossing was, by definition, already found, in the same visibilities; and its 12 controls resolve a rank of only $1/13$, with an expected maximum of 1.63 for pure noise against 3.04 for the 512-position ensemble, so the spatial clause is passed by anything above about 2σ . It is the weakest screen in the chain, not the strongest. With the epoch wrong it behaved like a powerful independent test by accident, because it was looking where nothing was.

What the fit does establish is worth keeping. Phase-rotating every visibility to the star turns emission centred there into a constant real offset, while emission elsewhere is suppressed in the real part and in general leaves a non-zero imaginary part. The two parts do different

work: the real part carries the detection, and the imaginary part discriminates against *displaced* emission only. A small imaginary part is not itself evidence that emission is present, because for pure noise it is centred on zero whether or not a source is there. The average is taken over the measurement set's own rows, which carry one point of each conjugate pair rather than the Hermitian completion of the uv -plane; with the conjugates included the imaginary part would vanish identically. BD+05 1668 is the test working in that direction, and so is HD 48370, whose large real part comes with an imaginary part far from zero — the signature its published identification as foreground cloud emission predicts.

BD+05 1668 (GJ 273, Luyten's Star): A DEFECTIVE EXECUTION BLOCK, DIAGNOSED. 4 threshold crossings belong to this star, with the highest statistics in the survey outside β Pictoris and HD 48370 ($T_\star$ up to 17.52), and they deserve a disposition rather than a sentence. *All 4 are in one ACA execution block, A002_Xc7fa6f_X28a8*, of the 25 this star contributes; the 24 sibling executions of the same field, in the same project with the same array and the same spectral setup, reach a maximum of $T_\star = 4.26$ and produce no crossing at all. In the defective block the *whole field* is above threshold: 512 of 512 control positions exceed $T = 5$, with a median of 10.9 and a maximum of 19.5, against a survey median control of 2.75 over 1476 windows. The single-integration noise is exactly as predicted (0.1609 Jy measured against 0.1611 Jy), but the residual does not integrate down: a jackknife on the channel mean gives 0.030 Jy against 0.0048 Jy thermal, a factor 6.2, a binned-mean test is flat, and the statistic scatters by 4.15 where it should scatter by unity. A slow, smooth temporal drift common to every channel and every position inflates the statistic roughly uniformly, which turns an empty field into 512 detections. The alternatives fail separately: line emission, because off-frequency controls at the star's own position reach $|\text{Re}/\sigma|$ up to 97.9; a common gain or atmospheric mode, because the median channel-to-channel correlation of the star's time series is 0.007; a field source, because the statistic is flat in position angle and radius across the annulus and because a source on the sky cannot appear in one execution and be absent from 24 others; and a signal at the star, because the star is outranked by a third to nine tenths of its own controls in three of the 4 windows and the visibility fit returns an imaginary part tens of σ from zero. Two of the 4 sit 8 channels from a band edge at the identical fractional position 0.928 in basebands exactly 2.000 GHz apart, which is a fixed intermediate frequency and not a sky frequency.

The quality criterion this survey lacked is the median of the control ensemble, not its maximum. The survey median is 2.75 with a 99th percentile of 4.76, and this block's windows are the extreme of the distribution at 11.20 against 2.60–2.75 for its siblings. We report it as a *diagnostic* and deliberately do not adopt it as an exclusion, because a cut strict enough to remove this block would also remove the HD 48370 CO positive control.

The fit does have the power to see a compact source, and that is why it does not settle these crossings. An event of statistic $T_\star$ in a window of trigger flux $S_{\min} = 5\sigma$ is a flux $T_\star S_{\min}/5$, and the fit reports its own flux uncertainty in the same units, so a compact source at

the stellar position producing the trigger would return $\text{Re}/\sigma = 5.4$ at the rank-flagged unattributed crossing that has been fitted; no free parameter enters that statement. The value it returns is 5.3, 0.97 of the expectation. The visibility fit therefore neither excludes nor establishes anything there: under the corrected epoch the crossing looks exactly like what a threshold crossing at the stellar position is defined to be. *Earlier versions of this paper reported the opposite* — an exclusion at several σ — and that exclusion was a product of the reference-epoch defect and is withdrawn.

The chain that decides is therefore the one drawn in Fig. 5: crossing, then stellar-frame line attribution, then the 512-position rank, then recurrence and the block-versus-position test. The frozen $\pm 50 \text{ km s}^{-1}$ mask attributes 16 of the 56 crossings against 2.3 expected by chance from its 4.1 per cent occupancy of the searched band, so the mask carries information; 40 are unattributed, 2 of those also outrank all 512 controls, and of those 0 recur. *Nothing survives among the crossings tested*, and §5.2.4 says what that qualification covers.

What a threshold crossing is worth on its own. The trigger is not a significance level for a single position in a fine-channel window, and the survey's own control positions say so. Over the 332 Class A windows whose 512-element control vectors are retained, a single control position reaches the smallest crossing statistic in the survey, $T = 5.0059$, 8.8 per cent of the time (median 6.5 per cent), against 0.11 per cent in the coarse windows. Summing each window's own rate predicts 29 Class A windows whose star reaches that value by chance, and 49 are observed; extrapolated to all 402 Class A windows the chance expectation is of order 35. *A crossing at the trigger is an expected event, not a rare one*, which is why the chain does not stop at one and why the rank screen exists.

5.2.2. A flat 5σ is not a uniform criterion, and that explains the crossing population

The preceding paragraph is a statement about the control ensemble. There is a second, independent statement to make about the threshold itself, and it is the more general one: *5σ is a threshold on a statistic, not a false-alarm rate, and in this survey the two are very far from interchangeable.* The number of searched (channel, drift) cells per window runs from 112 to 6.87×10^6 — 4.8 decades, median 354 — because channel widths span 15.3–31.25 and the drift grid is sized per window. A common threshold applied across that range buys wildly unequal protection.

We therefore measured, for each window separately, the scale of its own null and the number of *independent* cells it searched, and we publish both (Data Availability: columns `scale_route_a`, `scale_route_b`, `n_search_cells`, `n_ind_cells`). Two independent estimators of the scale are used: route A from a zero-drift inverse-variance stack over the retained control positions, route B from the released 512-position search maxima, each calibrated against matched-shape Gaussian Monte Carlo of the window's own published noise map, so that unity is demonstrated rather than assumed. Coverage is 1637 of the 1651 released windows (398 Class A, 1239 Class B), carrying 51 of the 56 crossings; the remainder are windows whose retained products can no longer be

re-reduced.

The null itself is sound. Route A has median 1.026 and route B 0.986, the two agreeing window by window with median ratio 1.038 and the combined s_w lying within 8 per cent of unity from the 1st to the 99th percentile. Requiring *both* routes to exceed a threshold fixed in advance flags exactly 1 window of 1637, and the same one at every cut from 1.10 / 1.15 / 1.20 / 1.50 (1 / 1 / 1 / 1 by both routes, 102 / 18 / 3 / 2 by either): BD05 1668 A002_Xc7fa6f_X28a8, at $s_A = 4.13$, $s_B = 4.10$, the defective ACA block already diagnosed in §5.2.1. It carries 1 measured crossing, or 4 of the 56 by block attribution. *No rescaling for noise is required anywhere else:* the amplitude of the published noise map is right to a few per cent in every window.

What is not sound is the flat threshold. Writing the requirement for a fixed false-alarm probability α as

$$T_w = s_w \Phi^{-1} \left[(1 - \alpha)^{1/N_{\text{ind}}} \right], \quad (4)$$

a common per-window $\alpha = 1$ per cent needs a statistic of 4.00 in the median window and anything from 3.52 to 15.86 across the catalogue (Class A median 5.38, Class B 3.94); a 1 per cent false alarm over the whole catalogue, i.e. $\alpha = 6.09 \times 10^{-6}$ per window, needs 5.53 in the median and 4.97–22.18 across it (Class A 6.61). The flat 5σ is wrong in *both* directions: it over-protects 1256 windows, by a median 1.05σ , while under-protecting 381; against the survey-wide target it under-protects 1636 of 1637. Its realised per-window false-alarm probability has median 1.01×10^{-4} but 84th percentile 0.047 and maximum 1.000.

The consequence is the crossing population itself. Summing each window's own realised false-alarm probability at $T = 5$ gives 27.3 expected crossings with an ideal unit null and 37.6 with the measured scale, against 51 observed. For Class A alone, three independent routes — the empirical exceedance rate of the released 512-position ensembles (33.5), the Monte-Carlo null maximum over each window's own cell count and realised scale (36.5), and the same re-run on a deliberately looser catalogue join (41.1), spread 1.23 — bracket the expectation at 33–41 against 50 observed. *Between roughly half and four fifths of the crossings this survey reports are expected by chance at a flat 5σ , and 30 of the 51 measured crossings lie below their own window's per-window requirement, 40 below the survey-wide one.* That is not a defect discovered late: it is why the paper's chain never treated a crossing as a detection, and it removes the need for any of the artefact hypotheses a reader might otherwise reach for to explain 56 crossings.

It also settles the smallest of them. η Crv A002_X122b6ff_X1041e has $T_\star = 5.0059$ in a window of 6.71×10^4 cells with $s_w = 1.008$, and Eq. 4 requires 5.157 of it at a per-window 1 per cent and 6.420 survey-wide. *It is below both* — so the event discussed in §5.2.3 is, on its own window's terms, not a significant excursion at all.

Two things are deliberately *not* changed. The trigger is left at 5σ , because moving it would redefine the crossing list, the frozen catalogue and every downstream test in the middle of a search that has already been run; Eq. 4 is published so that a reader can apply whichever α they prefer to the shipped columns. And the rank screen

and the recurrence test are untouched, because both are relative to the same window and therefore carry no dependence on N_{ind} or s_w at all — which is precisely why they, and not the trigger, are where this paper's negative result rests.

5.2.3. The crossings that are unattributed and localised

Under the adopted epoch 30 crossings are both localised by the fit and unattributed by the frozen mask, and under the earlier epoch 1 is: *this paper's earlier statement that there were none was wrong.* The clearest case is also the clearest demonstration of the preceding paragraphs.

η Crv, A002_X122b6ff_X1041e, Band 7, 357.281464 GHz localises under *both* reference epochs, $\text{Re}/\sigma = 4.46$ then 4.64 with $\text{Im}/\sigma = -1.07$ against control maxima 1.43 and 2.01, the displacement between the conventions being only -0.38 channels so that the epoch defect could not reach it; and it is unattributed, the nearest frozen-mask transition CO(3→2) lying 9929 km s^{-1} away in the stellar frame. *It also fails the rank screen* — 11 of 512 control positions equal or exceed the star, an add-one rank of 0.023 where the screen requires zero — *with $T_\star = 5.0059$ against a trigger of 5, the smallest search statistic of all 56 crossings.* Those two facts belong with the localisation every time it is quoted. Its one covering repeat block is 2.2 times deeper (1.85 against 4.10 mJy), 86.7 d earlier, and returns $T_\star = 1.59$ maximised over its own drift grid at $-0.25 \pm 1.82 \text{ mJy}$ against the discovery's 20.0 mJy , excluding a persistent source at the discovery flux at 11.2σ ; it is 1473 channels from the nearer window edge, and 0 of the 9 other blocks covering the same sky frequency carry a crossing.

A caution about line lists, which generalises beyond this survey. Against the wide Splatalogue harvest rather than the frozen mask this crossing does have a near coincidence, SO₂ at $+5.3 \text{ km s}^{-1}$, and it means nothing: at 357 GHz that harvest holds 30 transitions per GHz, so a $\pm 50 \text{ km s}^{-1}$ window contains 3.6 on average and a *uniformly random* frequency in the spectral window falls within the mask half-width of some harvested transition 77.7 per cent of the time. A hit is the default outcome, so the wide list can neither attribute a crossing nor be cited to say that nothing lies near one. The frozen mask is the only line test in this paper that carries information, and every count above is computed against it. Any survey quoting a line coincidence at these frequencies should quote the occupancy of the list it used.

5.2.4. The end of the chain, and what it still rests on

2 of the 56 crossings are both unattributed and rank-flagged, and the terminal count of 0 is what is known about those two. 61 Vir A002_Xc079b5_X82f ($T_\star = 5.96$) is the crossing whose visibility verdict the epoch correction moves furthest, from $\text{Re}/\sigma = -0.97$ to 5.26 at a displacement of 10.4 channels — that is, correctly referred it *does* look like emission at the star — and it is disposed of instead by 7 covering repeat blocks in which the largest statistic at its own channel and drift is $T_\star = 2.94$. CP-72 2713 A002_Xff0235_X4a6d ($T_\star = 5.81$) is one of the 6 crossings whose visibility fit is outstanding; its 5 covering repeats return at most 3.60 (§A.3.1). Recurrence has now been evaluated for 10 crossings over 51

repeat blocks and the largest statistic returned anywhere in them is $T_\star = 3.94$, against the trigger of 5.

The bound on that statement is stated rather than absorbed. 6 crossings in 4 blocks have no visibility fit, so the honest form is *zero among the crossings tested*. Under the argument of §5.2.1 that bound is weak — the fit is a consistency check and not what disposes of anything — and it cannot manufacture a detection: fitting them can only produce a result that has to be reported alongside the chain, as 61 Vir’s was. The counts of untested crossings and of crossings without a recurrence test are both assertions in the generator that builds Table 5, so neither can change without stopping the build and forcing this paragraph to be rewritten with it.

5.2.5. Dispositions of the remaining outlier windows

The third stage-1 flagged window, towards HD 48370 in Band 6, has $T_\star = 26.83$ against a control-ring maximum of 18.36 — star and ring rising together, as emission filling the primary beam does. The velocities agree: Cataldi et al. (2023) report prominent CO(2→1) towards HD 48370 at $v_{\text{bary}} \approx 41 \text{ km s}^{-1}$ and flag it as possible foreground cloud contamination, and our own frame chain returns $v_{\text{bary}} = +41.2 \text{ km s}^{-1}$ for this crossing. Three checks close the circumstellar alternative: the same authors report the disc itself undetected in CO and in carbon; the sight line runs through the outer Galaxy, where a flat rotation curve reaches the measured velocity only at a kinematic distance of 2.07 kpc; and the block’s $^{13}\text{CO}(2\rightarrow1)$ window has a control maximum of 10.8 against 4.4 at the star, so the field carries bright ^{13}CO that the star does not, resolved 2.9 synthesised beams away.

5.2.6. Line exclusion: definition and cost

The exclusion regions are defined before any crossing is classified, and those frequencies are *excluded from the experiment* rather than vetoed within it. Each catalogued transition of 11 species is masked to $\pm 50 \text{ km s}^{-1}$ in the stellar frame; Appendix I gives the kinematic justification for that half-width and the species criterion, and Appendix K.4.3 the cost species by species. The statistic is computed inside the masked channels and the mask applied afterwards, at the reporting stage: the threshold and rank tests run over every delivered channel, and the mask then removes the masked intervals from the *reported* search domain, because a crossing coincident with a catalogued transition cannot be dispositioned as artificial on these data alone.

That ordering makes a robustness check possible, and removing the mask entirely adds no candidate. Of the 56 threshold crossings, 16 fall inside a tube, of which 10 are stage-1 flagged windows already dispositioned as CO in Table 5 and the remaining 6 do not exceed their own control ensembles. The mask costs 4.1 per cent of the unique-frequency union, 4.7 per cent of gross bandwidth, and nothing else. One caveat of principle survives: those rest frequencies are the ‘magic’ channels Cocconi & Morrison (1959) argued a communicator might choose, so a transmitter deliberately parked on one lies in unsearched reported space and this survey would not claim it either way. Those channels are retained in the release for anyone bringing independent spatial or time-domain discrimination.

5.3. A stellar-frame multi-epoch stacked search

A persistent carrier is coherent across epochs; noise is not. 78 of the surveyed stars have observations in more than one execution block, so a search that combines epochs coherently reaches deeper than any single one — provided the epochs are first brought into a common frame. No published millimetre technosignature search has done this, and we report it here as a limit rather than as a detection.

Method. Each epoch’s frequency axis is transformed to the *stellar* rest frame before stacking, per execution block rather than per star, because the barycentric term itself moves: the measured correction swings by a median 6.9 km s^{-1} between the epochs of one star and by as much as 53.9 km s^{-1} , while within a block it moves only 0.072 km s^{-1} , so a single mid-time per block suffices. Epochs are combined by inverse variance over channels of common stellar-frame coverage, with a minimum of 2 blocks and 64 common channels; 27 of the 2838 retained products (1.0 per cent) carry anomalous weights up to 10^{11} times the median and are excluded, which is a data-release defect rather than an analysis choice. The detection criterion, the control ensemble and the null were fixed before the star positions were looked at, in the form of §5.2: stacked $Z \geq 6.0$ at a common channel, above all stacked control maxima, with no single epoch supplying more than 0.60 of the weighted sum and at least 0.50 of the signal surviving removal of the strongest epoch. The controls are stacked identically, and every clause is evaluated at the control positions as well as at the star — which matters: evaluated at the star alone the null appeared to breach its own 1 per cent ceiling at 0.79 per cent, and applied consistently it is 0.19 per cent. No threshold was moved.

Registration works, and it was demonstrated on real lines first. The per-block transform collapses HD 285968’s molecular line from a 19.26-channel topocentric spread across 4 epochs spanning 44 d in 488 kHz channels down to 0.798 channels, an rms of $0.351 \text{ channels} = 0.223 \text{ km s}^{-1}$ — a factor 24 — and places β Pictoris’ CO(2–1) at $+3.52 \text{ km s}^{-1}$ in its 244 kHz windows against $+3.55 \text{ km s}^{-1}$ in its 15.3 kHz ones, agreeing to 0.087 km s^{-1} across a factor 16 in channel width.

The gain is $\sqrt{N_{\text{eff}}}$, not $\sqrt{N}$, with $N_{\text{eff}} = \sigma_{\text{min}}^2 \sum_e \sigma_e^{-2}$ the effective number of equal-depth epochs; the median N_{eff}/N is 0.67 and the 10th percentile 0.33. Measured against $\sqrt{N_{\text{eff}}}$ the realised improvement is 0.991 — the combination is ideal to one per cent — and against $\sqrt{N}$ it is 0.79, the shortfall being unequal epoch depth and not registration error. N_{eff} must travel with any quoted N : HD161868 stacks 12 epochs for a gain of 0.94, because its N_{eff} is 1.1.

Limits. Over 394 groups, 2277 epoch-windows and 1454 h, the flux limit deepens by $1.39\times$ in the median over the best single epoch ($2.32\times$ at the 90th percentile, $4.45\times$ at most), or $1.21\times$ carrier-equivalent after the measured sub-channel registration loss of $1 - 0.875$. The median stacked EIRP limit is $5.61 \times 10^{14} \text{ W}$ against $7.38 \times 10^{14} \text{ W}$ single-epoch. The deepest is *Proxima Centauri*: 63 blocks ($N_{\text{eff}} = 13.9$), 84 h spanning 6.3 yr, gain $3.97\times$, S_{min} falling from 6.45 to 1.63 mJy and hence $\text{EIRP} = 5.15 \times 10^{12} \text{ W}$ at 1.30 pc — some two orders of magnitude below this survey’s median and below the

TABLE 5

THE CROSSING LEDGER. EVERY ONE OF THE 56 THRESHOLD CROSSINGS: ITS SEARCH STATISTIC T_* , THE 512-POSITION RANK FLAG (*scr*), THE STELLAR-FRAME LINE ATTRIBUTION UNDER THE FROZEN $\pm 50 \text{ km s}^{-1}$ MASK, AND THE DRIFT-FOLLOWING, CONTINUUM-SUBTRACTED VISIBILITY FIT EVALUATED AT ITS OWN RECORDED CROSSING CELL UNDER *both* REFERENCE EPOCHS — THE COMMITTED $t_0 = \text{median}(T)$ AND THE ADOPTED $t_0 = T_0$, THE SEARCH'S OWN — WITH D THE CHANNEL DISPLACEMENT BETWEEN THEM. *ctrl* IS THE LARGEST REAL PART THE SAME FIT RETURNS AT THE TWELVE CONTROL POSITIONS AND CONTROL FREQUENCIES. BOLD MARKS THE CROSSINGS MEETING THE PRE-COMMITTED CRITERION ($\text{Re}/\sigma \geq 4$, $|\text{Im}/\sigma| < 3$, AND ABOVE EVERY CONTROL), FIXED AT COMMIT **EDFCC0d3d7db** BEFORE ANY FIT WAS RUN. THE CRITERION IS A CONSISTENCY CHECK ON POSITION AND EPOCH AND NOT A FILTER (§5.2.1); THE CHAIN THAT DISPOSES OF A CROSSING IS THE *chain* COLUMN, AND FIG. 15 PLOTS THE TWO EPOCHS AGAINST ONE ANOTHER. **A** ATTRIBUTED **U1** UNATTRIBUTED, BELOW RANK SCREEN **U2** UNATTRIBUTED, RANK-FLAGGED, TESTED AND DOES NOT RECUR. FREQUENCIES MARKED * ARE WINDOW CENTRES, THE CROSSING FREQUENCY NOT BEING RECORDED. BLOCK IDENTIFIERS OMIT THE CONSTANT **A002_** PREFIX. ROWS MARKED *n.f.* ARE UNTESTED: THE VISIBILITY FIT IS OUTSTANDING, NOT NEGATIVE. $\dagger$ MARKS AN ADOPTED-EPOCH VERDICT WHOSE RESIDUAL TO THE EXTRACTION EPOCH IS BOUNDED RATHER THAN VERIFIED.

Star	block	B	ν (GHz)	T_*	scr	line	$\Delta\nu$	Re/ σ	Im/ σ	ctrl	committed $t_0 = \text{median}(T)$	Re/ σ	Im/ σ	ctrl	D	off ch
β Pic	Xf5d76d_Xeb8	3	115.2604*	30.765	Y	CO(1-0)	-27.8	8.12	0.75	1.14	6.73	1.28	1.27	-6.01	0.072	A
β Pic	Xf5d76d_Xe19	3	115.2605*	27.684	Y	CO(1-0)	-27.7	9.45	-0.12	2.09	6.67	0.82	0.74	-6.08	0.072	A
β Pic	Xf5d76d_Xcc5	3	115.2605*	17.288	Y	CO(1-0)	-27.6	6.16	0.47	1.77	1.53	0.41	1.86	-6.09	0.072	A
β Pic	Xf4d6ff_Xc7	3	115.2622*	17.907	Y	CO(1-0)	-23.3	2.29	0.86	1.35	-1.67	-0.33	0.92	-9.12	0.078	A
β Pic	Xa7a216_X23e4	6	230.5430*	7.988	Y	CO(2-1)	-14.0	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	-	-	A
β Pic	Xf5d76d_X32f1	3	115.2606	14.654	Y	CO(1-0)	-26.9	7.94	-1.21	1.64	3.18	-0.09	1.20	-3.58	0.071	A
β Pic	Xd9668b_Xa9df	6	230.5160	8.948	Y	CO(2-1)	-28.6	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	-	-	A
β Pic	Xd9668b_X3a90	6	230.5161	11.681	Y	CO(2-1)	-28.4	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	-	-	A
β Pic	X7116f1_X1785	6	230.5272	9.677	-	CO(2-1)	-14.1	5.98	1.63	2.25	10.31	-0.92	2.72	-2.39	0.020	A
β Pic	X6f1341_X1484	6	230.5284	9.832	Y	CO(2-1)	-12.5	9.11	1.23	4.40	9.42	0.78	3.73	0.72	0.020	A
β Pic	Xd9668b_X3a90	6	241.7512	5.420	-	-	-3898.0	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	-	-	U1
β Pic	Xd9668b_Xa9df	6	241.8367	5.079	-	-	-3793.5	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	-	-	U1
η Crv	X75bfbf_X1430	7	345.5404	5.013	-	-	-221.6	-0.92	0.24	2.68	4.66	-0.88	2.09	-6.29	0.624	U1
η Crv	X122b6ff_X1041e	7	357.2815	5.006	-	-	9957.5	4.46	-1.22	1.43	4.64	-1.07	2.01	-0.38	0.037	U1
τ Cet	Xba460f_X7e6	6	231.0334	5.198	-	-	644.2	-0.13	-0.35	0.86	4.74 $\dagger$	-0.60	1.39	-2.13	0.164	U1
61 Vir	Xc067f7_X822	7	345.5599	5.038	-	-	-204.7	-0.80	-0.04	1.41	4.19	-0.40	1.45	3.43	0.062	U1
61 Vir	Xc079b5_X82f	7	346.3138	5.958	Y	-	448.9	-0.97	0.38	2.88	5.26	-0.35	0.96	10.37	0.066	U2
ALMA J153702653-33192492	Xf6e2c1_X398	6	216.1844	5.228	-	-	-1271.2	1.01	1.07	1.41	-0.56	-0.15	1.59	-8.50	5.746	U1
ALMA J153702653-33192492	Xf8f6a9_X10359	6	217.7292	5.091	-	-	-676.9	1.79	0.70	1.28	1.79	-0.81	2.15	7.26	5.750	U1
ALMA J153702653-33192492	Xf8f6a9_X10359	6	218.6711	5.048	-	-	617.0	-1.10	-1.17	1.96	4.49	0.52	1.49	7.26	5.750	U1
ALMA J153702653-33192492	Xf6e2c1_X398	6	218.8729	5.090	-	-	894.2	-0.47	0.22	0.63	4.69	0.09	1.05	-8.50	5.746	U1
ALMA J153702653-33192492	Xf0235_X8392	6	230.5222	5.405	-	CO(2-1)	-20.6	4.44	0.96	3.27	4.44	0.96	3.27	-0.14	5.746	A
ALMA J153702653-33192492	Xd9d7c7_Xbb64	7	347.2179	5.009	-	-	1232.8	-1.05	-0.21	1.59	4.64	-0.28	1.49	-7.27	5.721	U1
AU Mic	Xa42775_X319	6	230.6864	5.274	-	-	192.9	0.79	-0.58	1.39	5.50	1.09	1.32	7.18	0.036	U1
AU Mic	X899169_X1838	6	230.8252	5.221	-	-	373.5	-0.28	1.07	0.62	5.39	0.21	0.94	2.62	0.035	U1
BD+05 1668	Xc7fa6f_X28a8	6	224.7422	11.868	-	-	5907.7	60.37	48.61	55.17	38.15	21.99	36.66	-0.34	0.195	U1
BD+05 1668	Xc7fa6f_X28a8	6	226.7422	10.136	-	-	-4936.1	52.35	-34.11	17.81	43.00 $\dagger$	-22.50	19.63	-0.34	0.195	U1
BD+05 1668	Xc7fa6f_X28a8	6	240.1016	17.519	-	-	-5917.2	94.66	-49.21	34.99	89.37 $\dagger$	-41.21	33.07	0.34	0.195	U1
BD+05 1668	Xc7fa6f_X28a8	6	242.1953	8.784	-	-	-3354.5	39.51	33.85	25.16	47.30 $\dagger$	21.25	25.23	0.34	0.195	U1
CD-57 1054	Xcf525f_X5aa4	7	346.5099	5.108	-	-	618.9	2.33	0.78	1.71	4.04	0.26	2.20	1.58	0.067	U1
CP-72 2713	Xff0235_X4a6d	7	344.2697	5.809	Y	-	-1323.2	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	<i>n.f.</i>	-	-	U2
GJ 849	Xd9668b_X841a	6	230.8769	5.090	-	-	440.7	1.77	-1.05	2.73	5.35	1.76	1.14	5.33	0.037	U1
HD 10647	Xff294f_Xa7f0	7	331.6570	5.120	-	-	-12258.0	-0.09	1.56	1.33	5.16	-0.35	1.69	12.14	0.049	U1
HD 110411	Xd45e29_Xb7d4	6	219.4538	5.393	-	-	-145.0	-0.17	-1.45	1.04	5.20	0.54	1.39	3.81	1.287	U1
HD 14055	X1001fdc_X29ae	7	331.1559	5.082	-	-	-12692.5	-0.17	0.52	1.93	4.41	0.12	0.77	16.15	0.014	U1
HD 14055	Xff99e1_X24be0	7	331.7078	5.277	-	-	-12214.0	0.76	0.05	1.01	5.13	1.08	0.58	-3.94	0.011	U1
HD 14055	Xff348_Xd446	7	331.7699	5.575	-	-	-12160.1	0.56	-0.90	1.10	4.98	-1.14	1.37	14.93	0.012	U1
HD 14055	Xff99e1_X2b2ee	7	344.7007	5.175	-	-	-949.6	-1.39	-0.93	0.90	4.77	0.49	1.73	-11.05	0.011	U1
HD 14055	Xfffd1_X3835	7	345.0099	5.177	-	-	-681.5	-0.49	0.57	1.20	4.62	-0.28	1.28	6.75	0.012	U1
HD 14055	Xff99e1_X2c3bb	7	345.5256	5.186	-	-	-234.4	1.21	0.17	2.27	5.05	-0.95	1.59	16.49	0.011	U1
HD 161868	Xff45a6_X10ca8	7	345.6338	5.481	-	-	-140.6	0.10	0.35	0.79	4.93	0.69	1.89	15.98	0.059	U1
HD 170773	Xff0235_X8c42	7	331.0101	5.029	-	-	-12818.8	-1.46	-0.74	1.31	4.39	0.86	1.05	3.08	0.027	U1
HD 207129	Xc19d6f_X5706	6	230.4226	5.433	-	-	-150.0	-1.08	0.95	0.85	4.88 $\dagger$	0.03	1.51	9.86	0.025	U1
HD 23484	X10b22d7_X2445	6	230.7904	5.520	-	-	328.2	-0.59	-0.32	3.41	4.96	0.17	1.57	-2.31	0.061	U1
HD 285968	Xd98580_X3bf6	6	230.5021	5.956	-	CO(2-1)	-46.7	5.64	0.74	3.23	5.64	0.74	3.23	-0.17	0.104	A
HD 285968	Xdb4b9a_X108f	6	230.5100	5.835	-	CO(2-1)	-36.4	5.66	2.31	2.44	5.66	2.31	2.44	-0.16	0.068	A
HD 285968	Xdb7ab7_X5b4e	6	230.5116	5.868	-	CO(2-1)	-34.4	5.94	2.76	2.82	5.94	2.76	2.82	-0.16	0.061	A
HD 31392	X129754b_X6946	6	231.1998	5.289	-	-	860.7	0.13	-1.91	1.57	5.17	-0.11	1.62	4.28	0.033	U1
HD 48370	Xc26103_X155a	6	230.5117	26.834	Y	CO(2-1)	-34.2	11.92	4.27	7.82	5.81	1.30	3.58	-7.08	0.028	A
HD 69830	X12277ed_X178c6	7	321.2005	5.046	-	-	-21323.4	0.54	0.59	1.50	4.78	0.07	3.16	-11.22	0.059	U1
HD 69830	X12209f7_X16edb	7	322.4325	5.259	-	-	-20255.3	0.60	-0.19	1.97	5.22	0.77	1.48	6.09	0.064	U1
HIP 10679	Xa95c04_X14cd	6	220.4185	5.456	-	13CO(2-1)	26.5	-0.25	0.55	1.87	4.94	0.89	1.39	30.00	0.231	A
HR 1010	Xc66ce1_Xaa9	6	230.8348	5.044	-	-	385.9	0.68	0.05	1.74	0.68	0.05	1.74	0.10	0.086	U1
HR 1010	Xc6a3db_X7b5b	7	345.4782	5.054	-	-	-275.5	-1.81	-0.53	0.82	4.65	-0.07	1.21	13.13	1.208	U1
HR 1010	Xc66ce1_X1345	7	345.6206	5.582	-	-	-152.1	-2.46	0.86	1.75	4.86	-0.85	2.75	3.13	1.199	U1
LHS 1140	Xd9d7c7_X6fd2	6	230.6355	5.136	-	-	126.7	2.42	0.50	1.63	4.69	-0.09	1.98	1.49	0.004	U1

EIRP of Arecibo's planetary radar, which is, as far as we are aware, the deepest millimetre technosignature limit placed on any star.

Outcome. 2 groups pass the full criterion, and both are β Pictoris CO: the $J = 1-0$ line at $+3.41 \text{ km s}^{-1}$ over 7 epochs ($Z = 56.3$) and $J = 2-1$ at $+3.55 \text{ km s}^{-1}$ ($Z = 12.1$), agreeing to 0.087 km s^{-1} in two different bands. That is the positive control passing. Attributing every survivor against the survey's own $\pm 50 \text{ km s}^{-1}$ mask leaves 0 unattributed, and the candidate count is unchanged at zero. The test also supplies an *independent* check on the events of §5.2.3: the 9 stars whose crossings the chain examined stack to $Z = 3.26-4.43$ in 8 cases, with 0 of the 9 exceeding their own control ensembles and 0 passing the criterion. The exception is HD 285968, which reaches $Z = 12.89$ on the same frame-locked foreground line used above as the registration standard and is rejected because its own controls reach 14.19: the emis-

sion is extended, which is what a foreground cloud does and a transmitter does not.

The restriction that travels with every number above. The stack assumes a transmitter whose velocity is steady in the stellar frame, and a channel is finite: at CO(2-1) a 488 kHz channel is 0.635 km s^{-1} , so a reflex motion of $\pm 30 \text{ km s}^{-1}$ — a transmitter on a short-period giant planet — is 23.1 MHz, about 47 channels, and *de-registers itself completely between epochs*. The same applies to any intrinsic drift larger than roughly $1.5 \times 10^{-1} \text{ Hz s}^{-1}$ over the median inter-epoch span of 38 d, or $2.0 \times 10^{-3} \text{ Hz s}^{-1}$ over the longest (2857 d). *This search therefore constrains persistent, reflex-stable emitters and says nothing about the rest*; the single-epoch limits of §5.4 remain the ones that apply generally.

5.4. *What this experiment excludes, and what it does not*

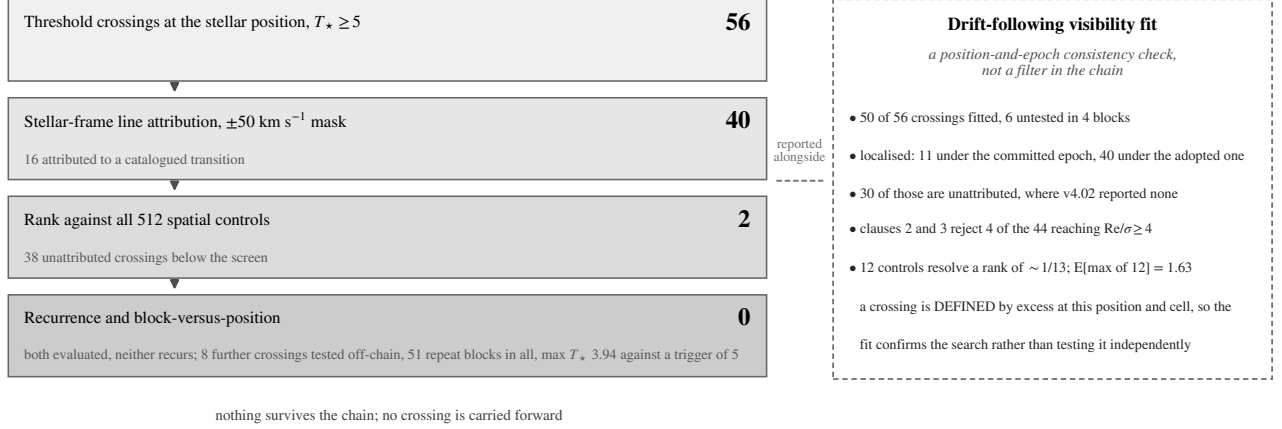
The chain that decides

FIG. 5.— The chain that disposes of a threshold crossing, with the number surviving each step, and the drift-following visibility fit reported beside it rather than within it. Every count is read from the ledger, so the figure and Table 5 cannot disagree. The fit sits outside the column because it is evaluated at the position and in the cell where the search already found an excess, and because its twelve controls resolve a rank of 1/13 (§5.2.1); it confirms the search rather than testing it independently.

TABLE 6

THE LEDGER IN COUNTS, EACH AGAINST CHANCE. THE LINE-MASK EXPECTATION IS THE MASK’S 4.1 PER CENT OCCUPANCY OF THE SEARCHED BAND TIMES THE 56 CROSSINGS; THE LOCALISATION EXPECTATIONS ARE THE MEASURED FALSE-ALARM RATE OF THE FULL CRITERION ON 577 EVENT-FREE CONTROL SAMPLES — 0.69 AND 0.17 PER CENT UNDER THE TWO EPOCHS, BOTH INSIDE THE 1 PER CENT CEILING FIXED BEFORE THE CAMPAIGN — TIMES THE NUMBER FITTED. THE $\text{Re}/\sigma \geq 4$ CLAUSE ALONE FIRES AT 3.29 AND 2.77 PER CENT, SO THE IMAGINARY-PART CLAUSE IS CARRYING THE CRITERION.

Quantity	Observed	Chance
Threshold crossings	56	—
Fitted in the visibilities	50	—
Untested (fit outstanding)	6	—
Line-attributed ($ \Delta v \leq 50 \text{ km s}^{-1}$)	16	2.3
Unattributed	40	53.7
Localised, committed epoch	11	0.35
Localised, adopted epoch	40	0.09
epoch residual verified	38	—
provisional († , residual bounded)	2	—
of those, unattributed	30	—
Rank-screen flagged	12	—
Unattributed and rank-flagged	2	—
Unattributed, rank-flagged, recurring	0	—
Crossings with a recurrence test	10	—

Excluded. For a continuously transmitting, unresolved carrier lying outside the molecular-line mask, within the searched frequency range and inside the linear-drift grid, and present during one of the archival epochs, the complete automated selection recovers the signal at least nine times in ten above $P_{90}^{\text{sel}} = 6.4 \times 10^{13}$ – $2.0 \times 10^{17} \text{ W}$ of EIRP per window, and per system, on each system’s best window, 6.4×10^{13} – $3.5 \times 10^{16} \text{ W}$ with a median of $1.5 \times 10^{15} \text{ W}$. That per-system median is the figure quoted in the abstract, in Table 1 and in the conclusions. No such carrier was found toward any of the 82 systems searched.

That number is measured, not assumed. Carriers of known amplitude, convolved with the instrumental response and given a random sub-channel phase and a random drift rate within each window’s own grid, are added to the *calibrated visibilities* and recovered through the unmodified pipeline — extraction, baseline removal, de-drift stack, trigger and the 512-control rank — and scored on the criterion the analysis actually applies: peak

de-drifted statistic at the injected cell above the trigger and localised at the stellar position by the drift-following visibility fit.

That criterion changed in this version, and the reason is a measurement. Earlier versions scored the campaign on the trigger *and* the 512-control rank, giving $P_{90}^{\text{sel}} = 5.70 P_{\text{trig}}$ in Class A. But §5.2.1 establishes that the rank screen rejects nothing here, so it is a consistency check and not a filter, and charging the published limit for a gate the analysis does not apply overstates it by $\times 1.98$ in flux. Measured through the chain the paper does use, the Class A multiplier is $2.88 P_{\text{trig}}$ (central interval 2.84–2.91) against 2.87 for the trigger alone — *localisation costs essentially nothing*, which is the same finding stated as a number — and 4.62 in Class B. The superseded rank-gated value 5.70 is retained here and nowhere else. The campaign behind it is 6 107 tones over 40 scored block-window configurations of 42 usable, recovering 0.995 of tones at twenty times the trigger and localising 0.089 per cent of uninjected cells against a 1 per cent ceiling fixed before the campaign ran. The predecessor campaign, on the rank criterion, was 8 860 tones over 52 block-window configurations; it recovered 1.000 of tones injected at twenty times the trigger and *none* of the 9 543 cells scored on the uninjected passes, so neither criterion fails on strong signals or fires on noise. The tones were deposited through a channel model that over-attenuates a carrier at a channel boundary by 13 per cent (Appendix G), so these powers are pessimistic by at most that factor and are quoted unaltered rather than corrected by a model. The control rank is what makes the selection expensive: Class A reaches ninety per cent *trigger* recovery at $\times 3.0$ the nominal trigger power against $\times 5.7$ for the selection, a further factor of 1.9, while Class B, whose coarse channels leave the control ensemble far less disturbed, pays only 1.1.

Two properties of that measurement bear on how far it should be pushed. Recovery does not reach unity: at the top rung the trigger fires for 1.000 of Class A tones and 0.945 are recovered, because a bright source at the stellar position leaks into the control ring through the dirty beam and raises the control maximum with it, from 4.8σ

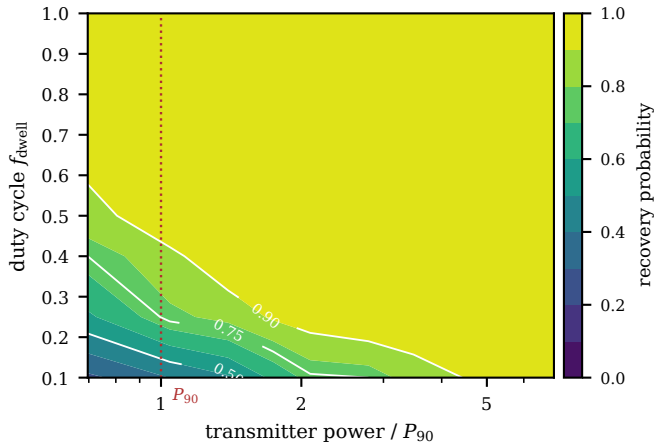


FIG. 6.— The duty-cycle selection function, measured. Recovery probability against transmitter power, in units of P_{90} , and duty cycle f_{dwell} , from 3888 injection trials. The abscissa is in units of P_{90} , the measured trigger completeness of §5.4, which for Class A is $\times 3.0$ the nominal trigger power. Recovery at $f_{\text{dwell}} = 0.10$ is 71 per cent for the fine class and 83 for the coarse, rising to 91 and 100 at $f_{\text{dwell}} = 0.25$ (Appendix G). The dwell injections are zero-drift, so this selection function is measured for a stationary carrier and transferred to the drifting primary experiment; the joint drifting-and-intermittent cell of Table 13 remains unmeasured.

on the uninjected pass to 8.8σ at the top rung. There is a ceiling on selection that transmitter power does not remove, and it is a property of the control geometry rather than of the sky. And the Class A ninety-per-cent point lies close to the end of the measured ladder: in 26 per cent of the unit-level bootstrap resamples the Class A curve does not reach ninety per cent within the range injected, so $\times 5.7$ should be read with its interval, $\times 3.7\text{--}\times 5.9$, and as the point at which the measurement runs out rather than a sharp threshold. Figure 6 shows the same completeness against duty cycle and Fig. 4 the per-system spread and the epoch coverage that limits confirmation; Appendix E defines the chain from P_{trig} to P_{90}^{sel} .

Read every such value as an upper bound on reach rather than a central estimate. The campaign measures 52 of the 402 Class A windows directly, for a computational reason — every injected window is re-extracted and re-searched end to end over its full drift grid, once per amplitude — so the ratio measured on those is applied to the nominal trigger of every window, and the window-to-window spread of that ratio is carried as an uncertainty rather than removed. The interval quoted above is a bootstrap over configurations, not over tones, because tones injected into one window share its noise realisation, its control ensemble and its beam. Widening the campaign is the recommendation of §L that would most directly improve the headline number.

Table 12 is the budget, one line per term: P_{90} is linear in every factor that builds it and in d^2 , so each entry is a percentage of the answer, the independent terms combine to ± 9 per cent dominated by the injection statistics, and three terms of a different kind are carried separately — the carrier’s unknown position within a channel, the one-sided atmospheric decorrelation the injected tones do not suffer, and the window-to-window transfer spread, which is the largest. The second significant figure of P_{90} is therefore meaningful for a single injected window and not for the survey.

Not excluded. A transmitter absent during the archival

observation, or on for less than about a tenth of the time. Emission inside the $\pm 50 \text{ km s}^{-1}$ molecular-line mask, at any strength. Drift outside the searched grid, or drift curved enough to leave a channel within the track. Signals broad enough to be removed with the continuum baseline. Anything below the window’s own completeness threshold. And any emission from the 17 566 stars within 40 pc that ALMA has not observed, which is almost all of them.

THE SURVEY’S OWN ON-SOURCE TIME, IN BOTH DIRECTIONS. Two defects of opposite sign were found in the bookkeeping of the exposure, and neither may be quoted without the other. First, the extractor kept a single FIELD row per execution block. In 42 of the 404 released blocks — 171 windows (10.4 per cent) toward 18 stars — the discarded rows were not calibrators but further in-beam pointings on the same star, so 19.1 h of real on-source time out of about 30 h available, 6.8 per cent of the survey, was thrown away before a visibility was read. Recovering it would deepen those windows by a median factor 1.73 in flux density (3.0 in EIRP), and up to 2.83. Nothing published is wrong as a result — every rms and every S_{min} was measured from the data actually used — but the *achievable* sensitivity in those windows is understated, and a re-extraction of the affected blocks is in progress. Second, and in the opposite direction, the released `on_source_s` is $n_{\text{int}} \times \text{median}(\Delta t)$ rather than a sum of integration times, so sub-scan boundary dumps shorter than the median are over-counted: 486 of 1938 released windows fall by more than one per cent under the exact computation and 0 rise, taking the survey total from 279.0 to 260.5 h on the aggregation the published figure uses (280.3 h) and from 272.2 to 259.3 h on the alternative one, that is -13.0 to -18.5 h (4.8–6.6 per cent). *The first defect costs the survey about 19.1 h it had; the second credits it with 13.0–18.5 h it never had.* The published exposure is therefore approximately right by accident, to within $+0.6\text{--}+6.2$ h, and both statements belong in the same paragraph because either alone misrepresents the experiment in the opposite direction. The reason the loss survived seven rounds of review is that nothing in the products recorded it: a selection that discards data must say how much, and this one did not.

THE OMITTED ANNUAL PARALLAX. The extraction phase-rotates the visibilities to each star’s *barycentric* direction, while the visibilities see the apparent one, so the annual parallax is omitted from the assumed phase centre. The resulting amplitude loss is the normalised dirty beam evaluated at (parallax displacement)/(synthesised beam) — parameter-free, multiplicative and shape-preserving — and it was identified by the external flux-scale check of Appendix D.2. Measured over the 1654 retained windows that carry the astrometric keys, the loss is 0.13 per cent in the median window, 2.98 per cent at the 90th percentile, 32.4 per cent at the 99th and 51.5 per cent at worst; 76 windows lose more than 10 per cent. It is negligible throughout the ACA data that dominate this survey and matters only for nearby stars observed in long-baseline 12 m configurations. *We state it and do not apply it.* Correcting it would move the median per-system P_{90}^{sel} from 2.91 to $2.93 \times 10^{15} \text{ W}$ (0.69 per cent) and the median Class A

window from 5.33 to 5.46×10^{15} W, and would leave the number of systems reaching 10^{15} W unchanged at 14, because a long-baseline block is never a system’s most sensitive window. Two things about the direction must be said explicitly. The sign is *optimistic*: every limit quoted here is too deep by $1/(1 - \text{loss})$ in its own window. And the completeness campaign cannot see the term at all, because its tones are deposited at the same assumed position the estimator evaluates at — the third instance in this work of a validation being insensitive to the error it exists to catch. The defect is repaired in the extraction code and the repair is carried by the re-extraction described above; the released catalogue is not retrospectively corrected by a model.

Two blind spots remain. The search is Stokes I only: a fully polarised carrier is not missed, but the discrimination polarisation would offer against unpolarised astrophysical emission is unavailable. ALMA’s feeds are linear, so a per-hand search would help only a carrier aligned with one feed, and a circularly polarised one — the electron-cyclotron maser case of §D.2.2 — divides equally between XX and YY and gains nothing; its circularity is in the cross-hands, which this analysis does not extract. And the frequency coverage is the archive’s, not a choice: none of the “magic frequency” arguments that have guided targeted searches — the 1420 MHz line, its multiples, the water hole, or π and e times them — falls in an ALMA band, so a transmitter deliberately placed on one is outside this experiment by construction and its absence here is not evidence against it.

6. DISCUSSION

Everything below is conditional on what the archive chose to observe. 90 per cent of the searched stars were observed for debris-disc or planet-formation science, which skews the sample young, dusty and dynamically active and puts circumstellar CO — the source of 10 of the 12 stage-1 events — in the field of a large fraction of the windows (§6.5). The quantities that follow describe *these* stars and not the 40 pc population.

6.1. What signals could ALMA archival data actually detect?

Three morphologies set the reach of the archive, each illustrated with a real window and with the completeness entries of Table 13 and Fig. 9 behind it. Denominators count the 82 systems whose deepest threshold in the class lies at or below the case power.

(i) *A continuous, frequency-stationary carrier* in a coarse channel is the best case. The UV Ceti pair at 2.7 pc has a deepest window whose trigger is 1.6×10^{13} W and whose effective threshold is 3.4×10^{13} W, and a channel-confined carrier there is recovered at near-ideal $\sqrt{N}$ gain (§G.1). Read on the effective threshold, the comparison a transmitter power calls for, 0 systems of 82 reach the numerical EIRP of the Arecibo planetary radar and 1 twice it; on the trigger the figures are 1 and 4. The survey’s best case is a few times that numerical EIRP.

(ii) *A continuously drifting carrier* in a fine 488-kHz channel is the only morphology with a measured end-to-end recovery curve (Appendix G): above $\sim 8 \times 10^{13}$ W nominal it is recovered in 83 per cent of trials with its

TABLE 7
TEMPORAL COVERAGE OF THE 60 CLASS A SYSTEMS. RECURRENCE IS WHAT A CARRIER CLAIM WOULD REQUIRE, AND IT CAN ONLY BE TESTED WHERE THE ARCHIVE SUPPLIES EPOCHS FAR ENOUGH APART TO BE INDEPENDENT. 18 SYSTEMS WERE OBSERVED ONCE AND 10 HAVE ALL THEIR BLOCKS INSIDE A SINGLE DAY, SO FOR 18+10 OF 60 THE EXPERIMENT CANNOT ADDRESS RECURRENCE AT ALL. THE POOLED FIGURE OVER ALL 82 SEARCHED SYSTEMS IS 51 PER CENT (§6.5).

Longest separation between searched epochs	Systems	Per cent
Single epoch ($N_{\text{epoch}} = 1$)	18	30
Repeats within < 1 d	10	17
1–30 d	8	13
30–365 d	5	8
> 365 d	19	32
<i>Able to test recurrence at all</i>	32	53

drift inside the searched ceiling, a depth 10 systems reach in Class A.

(iii) *An intermittent carrier, on for a quarter of the track*, above $\sim 2 \times 10^{14}$ W is detected toward 28 mid-distance systems, and recovery falls rapidly at smaller dwell fractions (Table 13): below $f_{\text{dwell}} \sim 0.1$ only the nearest windows carry information.

6.2. What does the null result mean?

No persistent, independently confirmed unresolved spectral-carrier technosignature has been found within the domain searched (§5.2). *The null result primarily constrains persistent spectral carriers.* For 40 of the 82 systems the archive provides no independent epoch capable of testing recurrence at all — 27 were observed in a single block and 13 have all their blocks within one day — so for close to half the sample the experiment places little constraint on a low-duty-cycle emitter, whatever its power. Only 23 systems can be tested for recurrence over years. The $\text{EIRP}_{\text{trig}}$ values are nominal trigger powers (§4), and completeness-calibrated exclusion holds only in Class A, the 402 fine-channel windows where the measured recovery curve applies; the 1249 coarse Class B windows provide spectral-excess coverage only and cannot discriminate drift.

The drift ceiling cuts the other way. It is derived from line-of-sight Doppler acceleration — of the 50 planets known around the 23 hosts only TRAPPIST-1 b reaches it in its edge-on worst case — and does not model transmitter-side oscillator drift, so a transmitter whose own reference drifts faster than the kinematic budget falls outside the searched grid (Fig. 3), as does one parked inside the line mask (§5.2.6).

Set against Wright et al. (2018)’s haystack, only Band 3 falls inside that axis’s span, which their §3.2.2 sets at 10 MHz–115 GHz — the ionospheric cutoff to the top of the Green Bank Telescope’s instrumentation — (20.6 of 115 GHz); the remaining 97.5 GHz in Bands 4–10 is new territory, though the median threshold detects a 10^{13} W transmitter only within ~ 0.9 pc. The sole prior ALMA-archival search (Mason et al. 2025, 28 stars in 4 calibrator fields at ≥ 1.01 kpc) is four orders of magnitude more distant, so an EIRP ratio between the two would be almost entirely the distance term. The quantity that does not depend on how far the target happens to lie is the minimum detectable *received* flux, $F_{\text{min}} = S_{\text{min}} \delta\nu$, and it must be quoted at a matched channel width, be-

cause $S_{\min}$ falls as $\delta\nu^{-1/2}$ while $F_{\min}$ rises as $\delta\nu^{1/2}$. Mason et al. reach 3.8×10^{-23} – $5.8 \times 10^{-23} \text{ W m}^{-2}$ across their four fields, on a 30.52 kHz channel at a bare 5σ threshold. Rescaled to that channel width and on that same threshold convention — this paper’s P_{trig} , not P_{90}^{sel} — the Class A windows here reach $5.2 \times 10^{-24} \text{ W m}^{-2}$ at best, a factor 9 deeper, and $7.1 \times 10^{-23} \text{ W m}^{-2}$ at the median, a factor 1.5 shallower. Carried through to this survey’s own completeness convention, the median becomes $2.1 \times 10^{-22} \text{ W m}^{-2}$, a factor 4 shallower than their threshold; Mason et al. quote no recovery-based completeness, so that last comparison is not like for like and is given only to show the size of the convention. Comparison is in general better made axis by axis than on a single figure of merit: this survey differs from centimetre-wave searches in frequency, spectral resolution, number of systems, EIRP completeness, drift range and temporal coverage at once, and those axes are coupled, so a product of marginals would misstate the overlap by orders of magnitude. Figure 1 makes the central point, that the novelty is the frequency regime and not depth; on the conventional Continuous Waveform Transmitter Figure of Merit (Appendix B) this survey is about four orders of magnitude less competitive than major centimetre-wave programmes.

How far the null reaches is set by recurrence rather than by the rank statistic. No single event in this design can clear the survey-wide threshold (Appendix J), so a genuine but weak transmitter seen once is not separable from the 0.89 chance events the survey expects over its Class A windows — exactly the position CP–72 2713 leaves us in — and what separates them is a second epoch or an independent astrophysical identification. A repeat block is not necessarily a second epoch: **only 42 of the 82 systems, 51 per cent, have searched epochs separated by more than 1 day** (Table 7), and only 23 systems, 28 per cent, have baselines extending over years — so recurrence on a year time-scale can be tested for fewer than a third of the sample. Three figures are easily confused here and each is quoted with its criterion: 55 systems (67 per cent) have more than one *searched* execution block; 65 (79 per cent) have more than one *archived* block, searched or not, the extra 10 holding a second block that was never processed; and 42 (51 per cent) have searched epochs more than a day apart. Extending the search to every archived block therefore cannot deliver 79 per cent of the sample as a recurrence test: only 76 per cent of multi-block systems clear the one-day bar on the blocks already searched, so the realistic projection is about 50 systems, 61 per cent, with no new observing time.

6.3. The two array configurations

The 1054 ACA windows are 64 per cent of the survey and are screened on the same footing as the 12m windows, each with a control annulus scaled to its own primary and synthesised beams (§4.1). The array configuration does not enter the sensitivity at all: the control geometry enters the *screen*, while P_{trig} , P_{eff} , P_{90} and P_{90}^{sel} are set by the noise in each window and by the injection campaign. The deepest trigger and the deepest P_{eff} in the survey are both 12m windows. What the rank is worth in each array, and why no rank against 512 controls can carry survey-level significance, is in Appendix J.

6.4. The null result has not been checked against an independent pipeline

Every validation reported here tests this pipeline against itself, and the only prior ALMA-archival search (Mason et al. 2025) targeted stars beyond 1 kpc, disjoint from this sample, so no external comparison is available. **The null result therefore rests on internal validation alone.** The release includes 4 known-answer test vectors generated from first principles — Gaussian noise plus a closed-form drifting sinusoid, with no import from the search or injection modules — and a reimplementation of Eq. 2 run on them meets all 4 stated expectations: the null reaches only $T = 4.0$ anywhere on the grid, the drifting 5σ carrier returns $T = 5.8$ at its own drift trial against 1.0 at zero drift, and the off-grid drift returns 4.4, 24 per cent lower, which is the grid-loss term the completeness curve has to carry. The stationary case returns 3.3 rather than 5: the injected cell carries one unit-variance noise draw and this realisation’s is low, which is the scatter a single vector can show and not a defect.

6.5. What this sample can and cannot speak for

Selection is on archival coverage, and ALMA’s archive was not built to sample stars. 80 of the 89 searched stars were observed under proposals categorised *Disks and planet formation* (Fig. 12), and 74 carry the *Debris disks* science keyword. Among the 53 searched stars with a Gaia temperature, 21 are M dwarfs, 40 per cent, against 69 per cent of the reference census (Table 14), so the sample is F- and G-enriched and M-dwarf deficient; and window counts are concentrated, three systems — BD+05 1668, Proxima Cen and HD 33793 — carrying 17.3 per cent of the searched windows.

Two consequences follow, and they run in opposite directions. A debris-disc-dominated sample is the worst case for astrophysical false positives at millimetre wavelengths, which is where the stage-1 flagged windows of §5.2 concentrate, so the vetting burden here is heavier than a blind sample would carry. And M dwarfs host most of the habitable-zone planets in the solar neighbourhood while only 21 of the 5908 within 40 pc carry a searched window, so this result carries very little demographic weight for the local habitable-zone planet population. Both counts are restricted to the temperature-classified sample of Table 14, and illustrate the selection rather than measure it.

The point is sharper in targets than in percentages. Six of the 21 searched M dwarfs host a planet that some published analysis places in the habitable zone — Proxima Centauri, GJ 273, HN Lib, GJ 581, TRAPPIST-1 and LHS 1140, the last named in this paper’s catalogue as BD+05 1668 and carrying GJ 273 b (Astudillo-Defru et al. 2017), which earlier versions of this list omitted — and since habitable-zone membership is model-dependent and two of these are contested, 6 is an order of magnitude rather than a census. The list, the count and the requirement that every star in it carry a searched window are generated from the released catalogue rather than typed; GJ 273 (4), LHS 1140 (1) do carry threshold crossings, in 2 execution blocks, and those are the events of §5.2.1 and §5.2.3 rather than new ones. That is this survey’s coverage of the class of

target most current technosignature programmes prioritise, against 5 908 M dwarfs within 40 pc. Sun-like stars are sampled relatively heavily by comparison, but still through a highly non-random, disc-selected archive: the survey constrains transmitters around the stars ALMA happened to point at, not the local stellar population.

Useful subsamples still fall out of archival selection. 23 of the 89 processed stars (50 known planets) are confirmed exoplanet hosts, among them Proxima Centauri — the subject of the most closely scrutinised radio technosignature candidate to date, the Breakthrough Listen signal of interest blc1, which was ultimately attributed to local interference (Sheikh et al. 2021) — τ Ceti, ϵ Eridani, β Pictoris and the seven-planet TRAPPIST-1 system, while HN Lib and LHS 1140, each with a habitable-zone planet, now carry an ALMA technosignature non-detection. Four are white dwarfs (Gentile Fusillo et al. 2021), among the few mm/submm SETI constraints for that class (Huang et al. 2026). Earlier versions of this paper gave 19 hosts and 42 planets. Those were a lower bound rather than a count: the census flag they were drawn from is set only for entries admitted through the exoplanet selection route, so 96 of the 168 census rows carried it blank and every blank was silently counted as a non-host. The figures above come instead from a positional join of the in-sample census to the confirmed-planet table; all but one match lies within 6 arcsec, and the exception is BD+05 1668 = GJ 273, at 60 arcsec because it moves 3.7 arcsec per year — so a join that filtered on distance agreement would throw away precisely the nearest and fastest-moving of the additions. On the stricter criterion that a companion must lie below the deuterium-burning limit, 2 of the 23 fall out and the figures are 21 and 48; one of the 2 was already inside the published 19, so that criterion subtracts as well as adds.

7. CONCLUSIONS

1. **Search performed.** 402 fine-channel spectral windows toward 60 independent stellar systems, covering $\Delta\nu_A = 47.7$ GHz of unique sky frequency, searched for an unresolved spectral carrier whose frequency drifts. A secondary coarse-channel spectral-excess search covered a further $\Delta\nu_{B\text{-only}} = 70.4$ GHz in 1249 windows but cannot discriminate drift. Both come from a uniform reduction of 484 public execution blocks, 4.1 TB of raw data.
2. **Sensitivity.** Measured by injection and recovery through the complete automated selection, not inferred from a nominal threshold: per system, on each system's best window, $P_{90}^{\text{sel}} = 6.4 \times 10^{13}$ – 3.5×10^{16} W of equivalent isotropic radiated power with a median of 1.5×10^{15} W. This is an upper bound on reach rather than a central estimate, the calibration being uncertain by $\times 0.64$ – $\times 1.04$, and a one-sided +5 to +20 per cent decorrelation bias can only make the figure optimistic.
3. **Result.** All 56 threshold crossings are reported with their dispositions (Table 5). 16 are attributable to circumstellar or foreground carbon monoxide, among them the β Pictoris debris disc recovered blind in two transitions; 40 are unattributed and 2 of those also outrank all 512 spatial controls. **Neither of those two is independently confirmed: neither**

recurs in the available repeat observations, and 0 crossing survives the chain among those tested. The bound on that statement is stated rather than absorbed: 6 crossings in 4 blocks are not yet fitted in the visibilities (§5.2.4), a test which in any case disposes of nothing. Non-recurrence is the load-bearing evidence, and it is weak in time: the repeats are separated from the event by no more than 769 d, so they constrain intermittent emission poorly. The screen assigns no calibrated false-alarm probability: no rank against 512 controls can carry survey-level significance, whatever its null (§J.3).

And the crossing population itself is expected. A flat 5σ is a threshold on a statistic, not a false-alarm rate, and the number of independent searched cells per window spans 4.8 decades, so a uniform per-window 1 per cent criterion requires 3.52–15.86 (median 4.00) and a 1 per cent criterion over the whole catalogue requires 4.97–22.18 (median 5.53). Summed window by window with each window's own measured null scale, 27.3 to 37.6 of the 51 measured crossings are expected by chance at a flat 5σ , with three independent routes bracketing the Class A expectation at 33–41 against 50 observed; 30 of the 51 lie below their own window's per-window requirement and 40 below the survey-wide one, η Crv among them. *This is the simplest available explanation of the crossing population and it requires no artefact hypothesis at all* (§5.2.2). It leaves the negative result intact and better supported, because the rank screen and the recurrence test are both relative to the same window and carry no dependence on the cell count or the noise scale; the trigger is deliberately left where it was, and Eq. 4 with s_w and N_{ind} per window is published so a reader can impose any α they choose. *A stellar-frame multi-epoch stack extends the limits and finds the same answer.* Coherently combining 2277 epoch-windows over 78 stars and 1454 h deepens the flux limit by $1.39\times$ in the median — the realised gain being $\sqrt{N_{\text{eff}}}$ to 0.991, not $\sqrt{N}$ — and reaches 5.15×10^{12} W for Proxima Centauri over 6.3 yr. 2 groups survive the pre-registered criterion, both β Pictoris CO in two bands agreeing to 0.087 km s^{-1} , and 0 are unattributed (§5.3). It applies only to emitters that are stable in the stellar frame to 0.635 km s^{-1} . *The visibility-domain test does not carry this conclusion, and in earlier versions of this paper it was reported as doing so* (§5.2.1).

4. **Interpretation.** No persistent, independently confirmed unresolved spectral carrier was found within the frequency, drift-rate, duty-cycle and sensitivity domain searched. **This cannot be converted into an occurrence constraint on technological civilisations within 40 pc.** The sample is archive-selected and strongly incomplete: 90 per cent of the searched stars come from disc and planet-formation programmes, only 21 of 5 908 catalogued M dwarfs within 40 pc are covered, and the median system is searched over 1.84 GHz rather than the survey's 47.7 GHz union. Occurrence figures, hosting fractions and figures of merit require a denominator this sample does not supply, and are confined to Appendix B, conditional on these stars, frequencies and epochs

and on nothing wider.

These EIRP limits apply specifically to carriers unresolved by ALMA’s 15.3–1953 kHz Class A channels, and should not be read as equivalent in signal class or spectral resolution to Hz-resolution narrowband SETI limits.

We find no independently confirmed unresolved spectral carrier among the 60 systems searched with drift-resolving Class A data. The result constrains transmitters only within the observed frequency intervals, the searched acceleration and drift range, the measured completeness limits of §5.4 and the archival temporal sampling; and because the targets are strongly archive-selected — chosen by other astronomers for disc and planet-formation science, and deficient in M dwarfs — it does *not* constitute an occurrence-rate constraint on technological transmitters in the 40 pc stellar population.

AUTHOR CONTRIBUTIONS

G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript. R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection–recovery campaigns, and produced the released catalogue and its regeneration scripts. Both authors designed the candidate vetting procedure, carried out the statistical validation reported in §4.1 and Appendix J, and agreed the dispositions of Table 5.

FUNDING

This work received no dedicated funding. It used public archival data and no observing time was awarded for it.

COMPETING INTERESTS

The authors declare no competing interests. VBRL Holdings is a privately held company through which R.D. conducts independent research, named here solely as his affiliation; it had no role in the design of the survey, the selection of targets, the analysis, the decision to publish or the content of this paper, and provided no funding for the work.

ACKNOWLEDGEMENTS

This paper uses ALMA data from the public ALMA Science Archive. The 404 execution blocks searched span 2013 October 6 to 2025 June 11 across 63 proposal codes, which in the standard ADS/JAO.ALMA#*code* form are: 2012.1.00142.S, 2012.1.00198.S, 2012.1.00268.S, 2012.1.00385.S, 2012.1.00437.S, 2012.1.00993.S, 2013.1.00359.S, 2013.1.00588.S, 2013.1.01147.S, 2013.1.01214.S, 2015.1.00307.S, 2015.1.00633.S, 2015.1.00783.S, 2015.1.01015.S, 2015.1.01260.S, 2016.1.00104.S, 2016.1.00358.S, 2016.1.00515.S, 2016.1.00637.S, 2016.1.00803.S, 2016.1.00817.S, 2016.1.00880.S, 2016.2.00015.S, 2016.2.00200.S, 2016.A.00013.S, 2017.1.00167.S, 2017.1.00200.S, 2017.1.00215.S, 2017.1.00561.S, 2017.1.00698.S, 2017.1.00704.S, 2017.1.00786.S, 2017.1.00986.S, 2017.1.01583.S, 2017.1.01644.S, 2018.1.00045.S, 2018.1.00072.S, 2018.1.00126.S, 2018.1.00461.S, 2018.1.00470.S, 2018.1.00947.S, 2018.1.01149.S, 2018.1.01545.S, 2018.1.01833.S, 2019.1.00189.S,

2019.1.01443.T, 2019.1.01517.S, 2019.1.01603.S, 2019.1.01681.S, 2019.2.00208.S, 2021.1.00485.S, 2021.1.01209.S, 2021.2.00097.S, 2022.1.00134.S, 2022.1.00338.L, 2022.1.00793.S, 2022.1.01163.S, 2023.1.00390.S, 2024.1.00165.S, 2024.1.00595.S, 2024.1.00722.S, 2024.1.01095.S, 2024.A.00040.S. Per-block epochs are in the machine-readable release. The archive metadata exposes no proposal code for 2 of the 404 blocks (A002_X11d9ce7_X5257, A002_Xbf792a_X26ec), which are therefore not represented in that list. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (CDS, Strasbourg, France).

DATA AVAILABILITY

The release carries every reduced data product, the full target-selection and analysis pipeline, the per-window catalogue, the injection records and the control statistics. What has been demonstrated, rather than claimed, is narrower and is stated in Appendix A: 123 generated products regenerate byte-identically from the frozen inputs, and 41 headline quantities are re-derived from the released catalogue alone, without the pipeline. *The reference of record is the Zenodo deposit*, which is frozen, carries a DOI minted on acceptance, and is the version any reproduction should cite; the repository at <https://github.com/Tilanthi/SETI> is a working mirror and may move on.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; CASA Team et al. 2022), Python 3 with astropy (Astropy Collaboration et al. 2022), numpy (Harris et al. 2020) and scipy (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through astroquery (Ginsburg et al. 2019); all are free and open-source, with exact versions recorded in the frozen release.

APPENDIX

WHAT THIS PAPER DID, AND WHERE THE DETAILS ARE

In plain terms, this paper asks whether any nearby star is transmitting a narrow radio signal that ALMA happened to record while observing something else. ALMA’s public archive holds thousands of hours of millimetre and submillimetre observations taken for quite different reasons — mostly studies of dust discs and planet formation — and every one of those observations also contains a spectrum of the star at the centre of the field. A transmitter would show up in such a spectrum as a spike confined to one or two channels, sitting at the star’s position and not anywhere else in the field, and reappearing if the star were observed again. We searched the archive for exactly that, with a single pipeline applied without human

intervention. Of the 82 systems covered, 60 have channels fine enough to resolve the drift a transmitter on a moving planet would show; those 402 windows are the primary experiment, and the remainder can register an unexplained spike but cannot tell a drifting signal from a stationary one.

Nothing survived. 56 threshold crossings were found; 16 of them fall within $\pm 50 \text{ km s}^{-1}$ of a catalogued transition, most of them carbon monoxide from the β Pictoris debris disc, which the pipeline recovered without being told it was there. 12 windows were also picked out by the automatic spatial screen; 10 of those carry a line attribution and the remaining 2 do not, and neither of those two reappears in a later observation, though the repeats available span days rather than years and so test intermittent emission only weakly. An earlier version of this paper added that neither is a compact source at the stellar position in the raw interferometric data; that finding came from a visibility estimator referred to the wrong epoch and is withdrawn (§5.2.1). The result is therefore a non-detection, and the useful part of the paper is the calibration: how deep the search actually reached, for what kind of signal, and over what fraction of the frequencies and epochs a real transmitter would have had to use.

Two cautions matter more than the limit itself. The targets were chosen by other astronomers for other purposes, so the sample is not representative of the solar neighbourhood and cannot speak for M dwarfs, which dominate that population but are barely represented here. And the star-versus-control ranking used to pick candidates is a prioritisation tool, not a significance test: it cannot assign a probability to any individual window, which is why every disposition here rests on physical evidence instead.

The appendices are reference material and none is needed to follow the result. They cover, in order: how frequency, linewidth and smearing are defined (F); the injection campaign that measures the sensitivity (G); the per-window catalogue (H); the line mask (I); the continuum lane false-alarm accounting (J); provenance, extraction geometry and robustness checks (A); the spatial normalisation and its validation (C); the β Pictoris positive control and the epoch extension (D); the conventional figure of merit (B); and the notation (E).

REPRODUCIBILITY AND PROVENANCE

Every number, table and figure in this paper comes from the submitted-analysis snapshot, the repository commit tagged at submission, which freezes the pipeline code and the input data as run. Deleting every generated table and figure and rebuilding from the scripts alone reproduces every product exactly.

The machine-readable catalogue, `per_target_results_v3.99.csv`, carries one row per searched window, 1651 rows in 60 self-described columns: the observation; the sensitivity reached, including P_{90} with the $\times 0.48$ – $\times 1.35$ bracket the injection campaign measures; the stellar and control peak statistics with the add-one rank under both the released and the radius-corrected normalisations, the crossing frequency, the nearest catalogued transition; three disposition columns, described below; and the control

geometry. Everything quoted in this paper follows from it: the sample sizes, the candidate list under either statistic, the frequency union and every P_{90} .

The survey's published dispositions were assigned by hand, not computed from the line mask. Through the previous version they came from a 5-entry literal keyed on (star, band) and were written into the catalogue only where the rank screen had already flagged the window, so the released column carried 12 strings and answered nothing about the 56–12 crossings it did not flag. This version therefore ships three columns rather than one. `disposition_computed` applies the frozen 15-transition mask topocentrically at $\pm 50 \text{ km s}^{-1}$ to all 56 crossings with no rank gate, and attributes 16 of them, of which 10 are also rank-screened — the two counts the results quote, each now with its population attached. `disposition_hand` carries the authors' evidence-based label, re-keyed on (star, execution block, window) because (star, band) is not a key: β Pictoris Band 6 holds 7 crossings in two basebands 12 GHz apart, and applied to all of them the old literal labels 2 of them, $3,793$ – $3,898 \text{ km s}^{-1}$ from CS(5→4), as circumstellar CO. The rank screen happened to hide that; one control fluctuation and the catalogue would have shipped it. `disposition` is unchanged, so that nothing published moves silently. Table 9 lists every crossing on which the computed and the hand-assigned columns differ. Two assertions in the build enforce the pair: every crossing the mask attributes must carry a disposition, and any disposition naming a molecule must agree with the mask. The first fails on the released column with 6 offenders and the second fails twice on the ungated hand literal; both pass on the computed column.

A script in the release opens the catalogue and nothing else, re-derives 41 of the headline numbers, compares each against the value typeset here and exits non-zero on any disagreement. The statistical unit is `system_id`, with 8 bound pairs sharing one identifier. The known-answer test vectors of §6.4 ship in `testvectors_v399/` with a manifest, checksums and the expected recovery for each case.

The prospective extension of §D.2.1 and the epoch-extension campaign of §C.1.1 lie outside the frozen release. The extension ships separately, with its own 463 window records in the same schema and the full 512-element control vector for each. Those blocks were searched after the statistic and the mask were fixed, so merging them into the catalogue would change the denominator on which every rate here is calibrated; pooling is left to the reader as an explicit choice.

Besides the catalogue, the Zenodo deposit holds the per-window search products for all 1651 windows with their 512 control statistics, so that every rank and p -value is recomputable; the line mask in frozen and repaired forms; the execution-block metadata and pointing ledger; the 1200 injection, 3888 dwell and 4032 stratified trials; the local-scramble and visibility products behind the CP-72 2713 disposition; and the change-log that timestamps the promote and retire criteria before the data were seen. Data products carry the Creative Commons Attribution 4.0 International licence (CC BY 4.0) and the pipeline code the MIT licence. Raw ALMA visibilities are public from the ALMA Science Archive at <https://almascience.org>.

TABLE 8

THE STATISTICAL DATA PATH, IN ORDER. STEPS 1–8 RUN IDENTICALLY AT THE STELLAR POSITION AND ALL 512 CONTROLS; ONLY STEP 9 ONWARD DISTINGUISHES THEM.

1	calibrated measurement set from the ALMA archive
2	stellar position propagated to the observation epoch (Gaia DR3)
3	spectral extraction at that position, native channelisation
4	per-integration baseline removal: block median along frequency ($W = 65$)
5	stack along each trial linear drift rate
6	robust noise map $\sigma(t, \nu)$: MAD across the 512 controls
7	$T(\mathbf{x})$ from Eq. 2, an inverse-variance-weighted stack
8	repeat 3–7 at 512 control positions on a ring
9	threshold: $T_* \geq 5$ (a crossing)
10	rank test: $T_* > \max_i T_i$ (stage-1 flagged window)
11	molecular-line mask, $\pm 50 \text{ km s}^{-1}$, topocentric (sky) frame
12	vetting: recurrence and astrophysical attribution

TABLE 9

EVERY CROSSING WHERE THE COMPUTED AND THE HAND-ASSIGNED DISPOSITION DO NOT SAY THE SAME THING ABOUT THE SAME WINDOW. THE FIRST 6 ARE CROSSINGS THE FROZEN MASK ATTRIBUTES THAT THE RELEASED CATALOGUE LEAVES BLANK: NOT ERRORS OF ATTRIBUTION — EACH IS CORRECTLY IDENTIFIED BY THE COMPUTED RULE — BUT A COVERAGE GAP, BECAUSE ONLY RANK-SCREENED WINDOWS WERE EVER DISPOSITIONED. THREE OF THEM ARE THE HD 285968 CO(2→1) EVENTS LOCALISED IN THREE INDEPENDENT BLOCKS (§5.2.1). THE LAST TWO ARE THE RANK-SCREENED CROSSINGS THE MASK DOES NOT ATTRIBUTE; THERE THE HAND STRING AND THE COMPUTED RULE AGREE IN CONTENT, AND THE HAND STRING CARRIES THE EXTRA SECOND-EPOCH EVIDENCE THE MASK CANNOT KNOW. Δv IS TOPOCENTRIC, AGAINST THE NEAREST TRANSITION OF THE FROZEN MASK.

Star	block	B	T_*	mask	Δv	shipped disposition
ALMA J153702653-33192492	Xff0235_X8392	6	5.41	CO(2-1)	-20.6	(blank)
HD 285968	Xd98580_X3bf6	6	5.96	CO(2-1)	-46.7	(blank)
HD 285968	Xdb4b9a_X108f	6	5.83	CO(2-1)	-36.4	(blank)
HD 285968	Xdb7ab7_X5b4e	6	5.87	CO(2-1)	-34.4	(blank)
HIP 10679	Xa95c04_X14cd	6	5.46	$^{13}\text{CO}(2-1)$	+26.5	(blank)
β Pic	X7116f1_X1785	6	9.68	CO(2-1)	-14.1	(blank)
61 Vir	Xc079b5_X82f	7	5.96	CO(3-2)	+448.9	unattributed
CP-72 2713	Xff0235_X4a6d	7	5.81	CO(3-2)	-1323.2	unattributed; absent in epoch 2

No inference to the overall 40 pc stellar population is attempted in this paper. The question this experiment answers is what the existing ALMA archive already constrains, for the stars it happens to contain; the question of how common powerful transmitters are within 40 pc cannot be answered robustly from a sample selected this way, and we do not attempt it.

Extraction, control geometry and position-correlated systematics

The baseline filter takes the median in contiguous blocks of 65 channels and interpolates linearly between block centres, using $\max(2, n/65)$ blocks over n usable channels. Its effective width is 65 channels in the 133 windows long enough to hold more than two blocks and $n/2$ in the short coarse windows, capping the linewidth it passes at ≈ 32 MHz for the fine class and ≈ 0.9 GHz for the coarse one (Appendix F). The crossing channel is not excluded from the scale. The median runs across positions at a fixed channel, so what matters is how many of the 512 controls carry the feature: a compact feature at the star appears in none and the 50-per-cent breakdown point holds, while emission filling the primary beam appears in all of them, as for HD 48370 (§5.2).

$T_* \equiv T(\mathbf{x}_*)$ is evaluated at the proper-motion-propagated stellar position, and the same operation, drift grid, channelisation and noise map give $\{T_i\}$ at all 512 control positions. `probe_positions()` in the released extraction code adds its offsets to the star’s own direction cosines, so the annulus is centred on the *star* wherever the star sits in the field. It holds 512 positions drawn uniform in area, $r = \sqrt{U(r_{\text{in}}^2, r_{\text{out}}^2)}$, and uniform in azimuth, from the fixed seed 20260825 re-applied per window, so all windows share one pattern in units of θ_{PB} .

Radii run $0.14\text{--}0.78 \theta_{\text{PB}}$ with θ_{PB} evaluated at each window’s own median frequency, and an inner check-ring of 135 positions serves the point-source test. The geometry ships as catalogue columns but is absent from the frozen products, so it is reconstructed from the code; the reconstruction recovers the check-ring’s 135 positions, and the control statistics fall with reconstructed radius where extended CO fills the beam (Spearman $\rho = 0.05$, HD 48370 Band 6).

Placing controls off-centre costs sensitivity. A star near the pointing centre sits at primary-beam response ≈ 1 while the annulus samples 0.95 at $0.14 \theta_{\text{PB}}$ falling to 0.19 at $0.78 \theta_{\text{PB}}$, area-weighted mean 0.47. Star and controls are therefore not exchangeable in flux density. Forming $T(\mathbf{x})$ on uncorrected amplitudes against one shared noise scale removes that particular asymmetry from the ranked quantity, but it does not establish exchangeability and earlier versions of this paper wrongly said it did: the out-of-sample test of Appendix C.1.1 shows the stellar ranks are displaced even on windows the pipeline had never seen. The star-versus-control rank is an empirically useful candidate-prioritisation statistic, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection. Extended sky brightness does not cancel: smooth emission lives on short baselines whose phases barely change under a rotation of a few arcsec, so star and annulus rise *together*, as for HD 48370, while emission offset from the star raises particular controls above it, as in the same star’s ^{13}CO window, whose ring maximum sits 15.9 arcsec from the star (§5.2). The two residual asymmetries run in opposite directions: a suppressed control makes a star-exceeds-ring outcome more

likely under the null, while a multiplicative calibration error scales with the continuum flux at the extracted position and so favours the star.

Neighbouring controls are correlated below the synthesised-beam scale, so a window’s annulus carries $N_{\text{eff}} \simeq 30\text{--}43$ independent spatial trials. That is an analytic beam-geometry estimate and a worst case, the annulus containing far more resolution elements than the 512 positions drawn. The rank test needs exchangeability rather than independence, and correlation among controls costs resolution without touching the $1/513$ floor. Radii are quoted in arcsec and in θ_{PB} because the frozen products carry neither the synthesised beam nor the array configuration; the 64 per cent of windows taken with the ACA 7 m array are treated in Appendix J.

Residual continuum, self-calibration residuals and position-correlated calibration error all concentrate at or near the phase centre, where the pointed science target sits, and σ is built from the annulus and carries none of it. These stars are not all blank there — an ancillary continuum lane detects 17 of 57 target/bands — and a residual bandpass ripple of fractional amplitude ϵ leaves a channel-confined artefact of order ϵS_{cont} at the extracted position and essentially nothing on the annulus. Three measurements bound the effect. The star’s own residual variance equals the controls’ to better than a per cent in the flagged windows (§K.1.2). Recomputing the noise scale locally, in a band around each feature and excluding the feature itself, gives local-to-global ratios of 0.97 (CP–72 2713), 1.29 (HD 48370), 0.93 (β Pic B3) and 1.02 (β Pic B6); the global scale is therefore good to a few per cent except towards HD 48370, where the field is filled with CO, and rescaling both the statistic and its ensemble there leaves the window flagged ($T_{\star} = 20.8$ against 14.2). A badly misstated σ would also displace the control distribution, which the uniform pseudo-star ranks exclude. A per-crossing local scale inside the search remains an open item (§L).

The statistic evaluates one position at the star and one at each control, because a rank means something only when the identical operation runs everywhere; §K.2 scores every window under the asymmetric region-maximum form as well and audits the difference.

Statistical units and independence

Six units appear in this paper and they are not interchangeable. A *catalogue entry* is one star as SIMBAD designates it, 89 searched. A *physical system* is the set of entries gravitationally bound together, 82 of them, the difference being seven designation-linked pairs. A *target/band* dataset is one entry in one ALMA band, 112 in all. An *execution block* is one archival observation, 404 searched, and is the unit that shares weather, calibration and correlator setup. A *spectral window* is one window of one execution block: 1651 catalogue rows, reducing to 1616 distinct (execution block, window) datasets once repeated archive products are collapsed. An *independent frequency trial* is none of these, the channel $\times$ drift grid being oversampled on both axes, so a window’s 177–6 869 740 cells carry about a factor 5.3 fewer independent trials than cells (Appendix J). An *independent epoch* is a separate execution block at the same tuning; blocks of one scheduling block executed on one night are not independent epochs in any useful sense.

The survey-level flag expectation uses 1651 windows, which is conservative because 1616 is the smaller number and because windows within a block are correlated. Statements about how common transmitters are use 82 systems, never 89 entries and never 1651 windows; statements about persistence use epochs, of which most stars here have one.

Data-quality exclusions and open items

112 target/bands are searched to completion and reported here. Four classes of entry are absent from the released catalogue and are itemised in Appendix K.

A further 68 candidate target/bands were selected but never delivered a searchable product, and the cause lies in the selection step. 60 of them were never observed at all, sitting a median 1.6 degrees from the nearest field centre: those observations were pointed at solar-system targets whose quoted field of view is enormous, and a crossmatch that trusts that quantity admits stars far outside any real primary beam. The remaining 8 entries were pointed at but yielded no search product. The per-star pointing ledger is released so that the selection can be audited. 53 of the 168 candidate stars fail on pointing in every band, so the sample is 89 of at most about 115 genuinely covered stars rather than 89 of 168, and the coverage completeness of Table 1 should be read against the smaller denominator.

One automated quality check precedes any sensitivity number. The radiometer relation makes $q \equiv \sigma \sqrt{t_{\text{on}}} \Delta \nu_{\text{ch}} \propto \text{SEFD} / \sqrt{N_{\text{bl}}}$ a quantity physics predicts, so a window whose q falls far below the sample median cannot be a noise fluctuation. 13 coarse time-division windows fail by about three decades, all of them channel-averaged auxiliary windows riding on a science baseband and carrying a constant WEIGHT column, so they never received that baseband’s calibration. The deflation is common mode in data and weights alike, so it misstates a threshold but cannot manufacture a flag: those windows are excluded from the sensitivity accounting only, and their statistics and ranks were carried through candidate generation unchanged (Appendix K).

45 of the 1721 extracted rows are duplicate entries for one physical window, the same (star, execution block, frequency range, channel width) key appearing twice in the archive query, so one row per key is the right statistical unit. The retention rule — the key’s first-listed row in the query’s own identifier order — consults no measured value and cannot prefer a crossing or a deeper threshold. Repeats need not agree: two reductions of the same τ Ceti Band 6 window differ by 44 per cent in threshold, which exceeds the flux-scale systematic of §F.1 and is the most direct empirical bound here on threshold reproducibility, and the two HR 1010 Band 6 rows straddle the trigger at $T_{\star} = 5.10$ and 4.95, so the crossing count is rule-dependent at the level of one window. The four flagged windows are not, no discarded row being both a crossing and a stage-1 flagged window. The line-excluded continuum lane is exploratory, bears on no technosignature limit, and is reported separately.

CP–72 2713: an unconfirmed event, absent in the repeat block

CP–72 2713 is a worked example of the full vetting chain rather than a result: at the survey’s one chance

expectation (Appendix J) 0.89 such events are expected over the Class A windows and 2 are observed. In Band 7 it reached $T_\star = 5.81$ against a ring maximum of 5.68, with no transition of the frozen mask within $\pm 50 \text{ km s}^{-1}$ in the stellar frame. A wider catalogue does put one nearby — SO $N_J = 8_8 - 7_7$ at -12.3 km s^{-1} in this block’s own stellar frame — and by the calibration of §5.2.3 that is the default outcome rather than evidence: over this window the harvest holds 39 transitions per GHz and a uniformly random frequency falls within $\pm 50 \text{ km s}^{-1}$ of one of them 91 per cent of the time, the nearest here being an unidentified line and not SO. Two measurements close it independently of any line list: the feature is 3 channels wide at the recovered drift against 2 for the instrumental response, where a warm circumstellar SO line would be resolved; and CO(3→2) lies in this same window and is absent, below a 5σ limit fainter than the crossing itself. Sulphur-bearing species are not in the frozen mask and were not in it before the search. It is one of the 6 crossings whose visibility fit is outstanding (§5.2.4); the value earlier versions of this paper quoted for it came from an estimator referred to the wrong epoch and is withdrawn. It acquired no astrophysical attribution and was recorded as an unattributed stage-1 flagged window under promote and retire criteria committed 9 hours before the second block was searched (2026 September 11, 119a68a6ab50).

That second block matches the first in tuning, channel width, channel count, integration count and on-source time, registers to 0.18 of a channel, and is the deeper of the two, so a persistent emitter at the first block’s flux would have returned $T_\star = 6.21$. Pinned to the first block’s own frequency and drift it returns 3.60, and read as two measurements of one constant amplitude the pair is inconsistent at 4.6σ . The blocks are 2.05 h apart, so this is a within-night test: it excludes a persistent emitter of that strength on that baseline and excludes nothing about an intermittent one. The disposition rests on that recurrence test and not on the rank; the frequency, drift, control and continuum diagnostics are in Appendix K.

Sample bookkeeping: entry counts, coverage criteria and pointed versus serendipitous

The 168 stars are complete against, and conditional only on, the 40-parsec reference catalogue intersected with the public-archive snapshot of 2026 September 8. Against that Gaia DR3 catalogue of 17 566 stars they are $\lesssim 1$ per cent, and the 82 systems searched are 0.47 per cent, fewer still once §A.3 removes those never observed (Fig. 2). The two parallax cuts are not interchangeable, the sample’s binding condition being $\sigma_\varpi/\varpi < 20$ per cent against the census’s 10 per cent; the realised sample reaches 2.48 per cent at worst.

Stellar counts throughout are catalogue-entry counts. The 89 searched entries comprise 82 systems, differing by seven designation-linked component pairs, so the 112 star/band datasets, 1651 windows and 404 execution blocks all derive from 82 systems. The 1651 rows comprise 1616 distinct (execution block, spectral window) datasets; the larger number is the trials denominator throughout, which is conservative and makes no practical difference, predicting 3.2 chance outliers against 3.1 for the smaller.

ONE POINTING, TWO CATALOGUED STARS, AND ONE OF THEM NOT IN THIS RELEASE. The field named after HD 139084B contains both components of V343 Nor, 10.3 arcsec apart, at 38.72 and 39.31 pc; each passes the distance cut on its own parallax, and each lies 5.09–5.21 arcsec from the phase centre of a 18.1–18.8 arcsec primary beam, where the response is 0.79–0.82. Two census stars inside one beam is legitimate and occurs in 35 windows of this survey. What was not legitimate is that the primary had inherited the ALMA *field* name, so a display-name collision was disambiguated by Gaia source id and the pair read as two solutions of one star — the 3 collisions in this census that really are one star span 0.44–1.55 arcsec, so a 3 arcsec tolerance tells the two cases apart; and that the re-extraction of §J.2 matches products on (execution block, window) without a star, which spread one component’s measurement over 8 rows in 4 windows. This release therefore carries those 4 windows once, under HD 139084 at 38.72 pc, which is the star they measure — the two extractions differ in every window, 2.34, 2.72, 2.47, 5.66 against 2.41, 2.49, 2.33, 4.15 in $T_\star$, and the threshold powers the 4 removed rows carried belong to HD 139084: relabelling them to HD 139084B would have multiplied every one by 1.0308. HD 139084B is covered, and was extracted, but its own extraction is *not* represented here: it contains a $T_\star = 5.66$ crossing at 345.8241 GHz, 24.3 km s^{-1} from CO(3→2), which its own control ring exceeds at 17.7 ($p_{\text{rank}} = 0.094$, so it is rank-screened and is not a stage-one event), and which was a counted crossing at $T_\star = 5.66$ in `per_target_results_v3.84.csv` before v3.99; its own threshold there is $1.75 \times 10^{15} \text{ W}$. Restoring it requires re-extracting the companion under the corrected control geometry and is deferred to the next catalogue release; 82 systems, the crossing list of Table 5 and every per-system limit are unchanged either way, because the two components were already collapsed into one system.

Admission to the sample required the following.

1. At least one public execution block whose visibilities are the archive’s delivered QA2 products, taken as delivered and not re-calibrated, covering the star’s epoch-propagated position. Calibration vintage therefore varies across the 2013–2025 span; the flux-scale term this contributes is small beside the $\times 2.08$ spectral-response correction. There is no minimum integration, sensitivity, antenna count, mosaic-tile count or array configuration beyond QA2.
2. That position inside the field’s primary beam. Extraction aborts beyond $2\theta_{\text{PB}}$ and the largest retained offset is $0.46\theta_{\text{PB}}$.
3. Each spectral window admitted individually, on QA2 vetting alone.
4. No programme-category restriction: science, calibration and observatory projects enter on identical terms.
5. A fixed archive snapshot date, 2026 September 8.

Completeness with respect to the work list holds in targets and tunings, not in epochs or integration time: the survey takes at most 25 blocks per target and deduplicates spectral windows by identifier, so 404 of the 656 public execution blocks those targets hold are in this release, 61.6 per cent, with 326 further blocks reserved as an external calibration sample. ALMA project metadata

quantify pointed against serendipitous coverage: almost every spectral-window-covered star is the pointed science target in at least one execution block, leaving Wolf 219 the sole serendipitous star, so the residual selection function of a work-list-complete sample is the union of ALMA target lists.

THE CONVENTIONAL FIGURE OF MERIT

These are the standard occurrence metrics, computed so the survey can be placed on the axes the literature uses. Following Enriquez et al. (2017), the Transmitter Rate is $(N_\star \nu_{\text{rel}})^{-1}$ and the Continuous Waveform Transmitter Figure of Merit is

$$\text{CWTFM} = \zeta_{\text{AO}} \frac{\text{EIRP}}{N_\star \nu_{\text{rel}}}, \quad \nu_{\text{rel}} = \frac{\Delta \nu_{\text{tot}}}{\nu_{\text{mid}}}, \quad (\text{B1})$$

with $\zeta_{\text{AO}} = 5.0 \times 10^{-11}$ fixed by their normalisation (CWTFM = 1 at EIRP = 10^{13} W, $\nu_{\text{rel}} = 1/2$, $N_\star = 1000$) and lower values better. On the Class A experiment, $N_\star = 60$ systems and $\Delta \nu_{\text{tot}} = 47.7$ GHz about $\nu_{\text{mid}} = 494$ GHz give $\nu_{\text{rel}} = 0.097$, a Transmitter Rate of 0.17 and, at the median per-system $P_{90}^{\text{sel}} = 1.5 \times 10^{15}$ W, CWTFM = 1.3×10^4 (3.0×10^5 at the worst-case system). That figure is reported for comparison only. It uses the survey-wide frequency union, which no single star was observed over; the quantity that answers whether a transmitter’s frequency was actually observed is the per-system version below. For scale on the conventional axis, Enriquez et al. (2017) reach 0.85 and Price et al. (2020) 0.11 at the GBT. This survey is therefore some four orders of magnitude worse than cm-wave SETI on the standard continuous-carrier metric, and that is the correct conclusion: 60 systems at a 10^{15} W floor cannot compete with hundreds of stars at 10^{13} W. What the metric does not score is the frequency axis, the one on which this survey is first, since Eq. B1 credits fractional bandwidth and not location, and 47.7 GHz at 494 GHz is a small fraction of a large number.

Two caveats are not presentational. First, the transmitter-hosting fraction. Zero confirmed detections in 60 Class A systems bounds the fraction hosting a continuously transmitting carrier above each system’s own P_{90}^{sel} at $f < 5.0$ per cent, the 95 per cent Poisson limit; the $1/N$ convention common in the literature would say 1.7 per cent, and is optimistic. That bound holds only for carriers inside the searched frequency islands, drift grid, duty-cycle domain and archival epochs, on a sample ALMA selected for disc science, and it is not a statement about the 40 pc population.

Second, and more seriously, ν_{rel} in Eq. B1 is a survey-level quantity, and *a civilisation around one star does not have access to the union of frequencies observed toward all 82 systems*. System by system the median Class A coverage is 1.84 GHz, from 0.11 to 19.08 GHz over the 60 systems that have any — 3.9 per cent of the survey’s own union. Substituting that median into Eq. B1 while holding $N_\star = 60$ gives $\nu_{\text{rel}} = 0.0037$ and a factor $\times 26$; we label that an illustrative rescaling and not a CWTFM, because it is not one. The product $N_\star \nu_{\text{rel}}$ in Eq. B1 is a *star-bandwidth* product — the number of independent (star, fractional-band) opportunities the survey bought — and a per-system ν_{rel} with $N_\star$ left at 60 counts neither the survey’s opportunities, which are heterogeneous, nor

one system’s, which would need $N_\star = 1$. It also divides one system’s union by the *survey’s* midpoint; against each system’s own midpoint the median per-system fractional bandwidth is 0.0076, about twice as large.

The figure that needs no substitution is the star-bandwidth product itself, which reduces to Eq. B1 when coverage is uniform:

$$\text{CWTFM} = \zeta_{\text{AO}} \frac{\text{EIRP}}{\sum_i \nu_{\text{rel},i}}. \quad (\text{B2})$$

Measured in Eq. B2, $\sum_i \nu_{\text{rel},i} = 0.628$ against $N_\star \nu_{\text{rel}} = 5.80$ under the uniform convention, so the convention over-credits this survey by $\times 9.2$ — not $\times 26$, which charges the median system’s narrow coverage to all 60 — and the survey’s star-bandwidth product is equivalent to 6.5 uniformly covered systems rather than 60. At the median per-system P_{90}^{sel} that gives CWTFM = 1.2×10^5 (2.8×10^6 at the worst-case system), and *that* is the number to place on the conventional axis. It does not change the conclusion: on this metric the survey remains some four orders of magnitude behind cm-wave SETI. We therefore take the per-system quantities as the scientifically meaningful ones and the survey-union figure as a literature convention: it measures aggregate bandwidth, not the probability that a given transmitter frequency was observed. Under any frequency prior broader than the searched islands the occurrence bound vanishes, which is a property of the archive and not of the method.

THE SPATIAL NORMALISATION: EVIDENCE AND VALIDATION

This appendix collects the three measurements behind §J.1 and the prospective test of the correction.

Prospective validation of the radius correction

The radius-corrected normalisation was derived from the calibration data, so on its own it establishes nothing about data it has not seen. We therefore froze it — the probe radii, the bin edges and the standardisation are those of Appendix K, with no free parameter to tune — and applied it unmodified to 778 windows that played no part in its construction: the hold-out and the extension of §D.2.1. The level for consistency with uniformity, $p \geq 0.05$, was written down before that test was run (Table 21).

Two rank distributions can be tested and they answer different questions. The *pseudo-star* distribution ranks an inner-annulus probe against the other probes; it passes both before and after correction (uncorrected $D = 0.005$ at $p = 0.16$, corrected $D = 0.006$ at $p = 0.09$, over 49792 probes). That is expected and is not evidence for the correction: every probe is drawn from the same radial distribution, so probe-against-probe ranks are near-uniform however they are normalised. The discriminating test is the *stellar* distribution, because the star lies at the phase centre, inside the probe annulus, and that asymmetry is what the correction exists to repair.

On that test the correction fails. Out of sample the uncorrected stellar rank has median 0.409 against the expected 0.5, with $D = 0.105$ at $p < 10^{-4}$; after correction the median is 0.407 with $D = 0.097$ at $p < 10^{-4}$. The departure from uniformity is essentially undiminished, and

the median moves further from the exchangeable value. The result holds separately in both samples (hold-out $0.405 \rightarrow 0.392$, extension $0.409 \rightarrow 0.417$).

The interpretation we draw is narrow and, we think, unavoidable. The radial gradient is real and is one mechanism behind the displacement, but removing it does not leave an exchangeable statistic, so the residual is not radial. Neither normalisation yields a calibrated tail probability, and the spatial screen is therefore a candidate-generation device under both. **We consequently retain the frozen screen as primary — it at least has the virtue of having been fixed in advance — and rest every disposition on the physical evidence rather than on either rank.**

The external calibration sample

Two distinct out-of-sample sets are used in this paper and they must not be conflated. The **hold-out** is 77 execution blocks (315 windows toward 36 stars) reserved by a hash of the block identifier, committed `c75069040eab` on 2026 September 14, and it sits inside the block ledger of §3 as the 78-block term of the second identity there. The **external calibration sample** described in this subsection is a different object: 326 execution blocks toward 56 stars, assembled in processing order from the host’s own campaign log rather than by the hash rule, and reserved by a rule fixed and committed before the archival search finished and before any reserved block was searched. *It is not a subset of the 656 in-scope blocks of §3* — it includes blocks toward stars outside the 40 pc work list, which is why its block count cannot be placed inside that ledger and must not be added to the 484 processed there. It enters no count of the science sample; it is used only to calibrate the rank statistic. Appendix K.4.1 gives both reservation rules, their commitment dates and the full result, and Table 28 gives the provenance of each. Three findings carry into the main line of the paper.

First, the pipeline behaves on unseen data: over 1322 windows it returns 7 stage-1 flagged windows, of which 4 are β Pictoris CO and 3 are unattributed, against 3.7 expected at the measured tail rate. Second, and adversely, the stellar rank is *not* uniform out of sample: its median is 0.405 where 0.5 is expected, with a block-clustered 95 per cent interval of 0.341–0.458 and $D = 0.109$ at $p = 0.0011$ over 315 windows. A displacement that survives a hold-out chosen before the data is a property of the statistic and not of this sample, which is why the spatial rank is used as a screen and never as a significance (§4.1). It reproduces a third time, at 0.409 over 463 prospectively searched windows (§D.2.1). Third, the false-alarm tail factor — the ratio of the measured rate at which a no-signal position ranks first to the rate exchangeability would give — is 1.2 on these blocks, with a bootstrap over the 77 reserved blocks giving 0.6–1.9. That factor and its width set the accuracy of every chance expectation in this paper.

Where the displacement comes from: radius, not stars

The displacement has a mechanism, and it is instrumental. Images are corrected for the primary-beam response, so the true noise rises with distance from the pointing centre, while the search divides every position by one scale estimated over the whole window. The control probes therefore sit at systematically different effec-

tive noise from the star, which is at the phase centre. Appendix K.4.2 sets out the evidence: the control level varies with radius across the annulus, pseudo-stars carrying no star at all are displaced as far as real ones, and rescoring the survey with the gradient removed changes which windows are flagged but not what the paper concludes.

Two numbers matter here. Standardising each control probe within its own radius bin — the probes come from one fixed seed, so their radii are known, and the correction has no free parameter — takes the stage-1 list from 12 windows to 11, losing 2 and gaining 1, all of them marginal and all already dispositioned on other grounds. And the median add-one rank over the 1614 windows that retain a control vector does *not* move toward 0.5 under the correction: it goes from 0.485 to 0.493, slightly further away. The radial term is one demonstrated mechanism for the displacement, not the whole of it, so we do not adopt the corrected statistic as primary and we do not present it as a validated null.

EXTERNAL CHECKS AND THE POSITIVE CONTROL

No independent pipeline check

The consequence of validating this pipeline against itself is stated in §6.4; the measurement behind it is the absence of shared data with the only prior ALMA-archival technosignature search. The 28 “stellar bycatch” targets of Mason et al. (2025) all lie at or beyond 1.01 kpc while every star here is within 40 pc, so the two distance ranges are disjoint by a factor of about 25 and no target, hence no execution block, is common to both. The 112 Band 3 windows searched here are in their band but in different fields. Neither of the two external checks that would carry weight — an independent re-implementation of the search statistic, or a blind injection by a third party before delivery — has been performed; the second would be the more informative, and is the check we would recommend to anyone extending this work.

An end-to-end check of the absolute flux scale

One external check is available, and it is the one that found the positional defect of §5.4. Two stars in this sample have millimetre photometry published by other groups from the *same* archival execution blocks, which tests our extraction end to end rather than against itself. For AU Mic, MacGregor et al. (2020) measured the two 2015 June 24 flares by fitting point-source models to the visibilities with CASA’s `uvmodelfit`, an estimator that solves for source position as well as flux and is therefore immune to the term below; for Ross 154, Burton (2024) measured two 2022 September flares from the same ACA visibilities used here. AU Mic’s impulsive millimetre flaring at 106 GHz is described in White et al. (2026).

Our extractor returns 0.98–0.99 of the published Ross 154 amplitudes and 0.81–0.82 of the published AU Mic ones. The two comparisons differ in exactly the way a *positional* error predicts and no other candidate does. The extraction phase-rotates to the star’s barycentric direction and so omits the annual parallax; the resulting amplitude loss is the normalised dirty beam at (parallax displacement)/(synthesised beam). For the Ross 154 blocks that ratio is 0.04–0.05 of a 7m-array

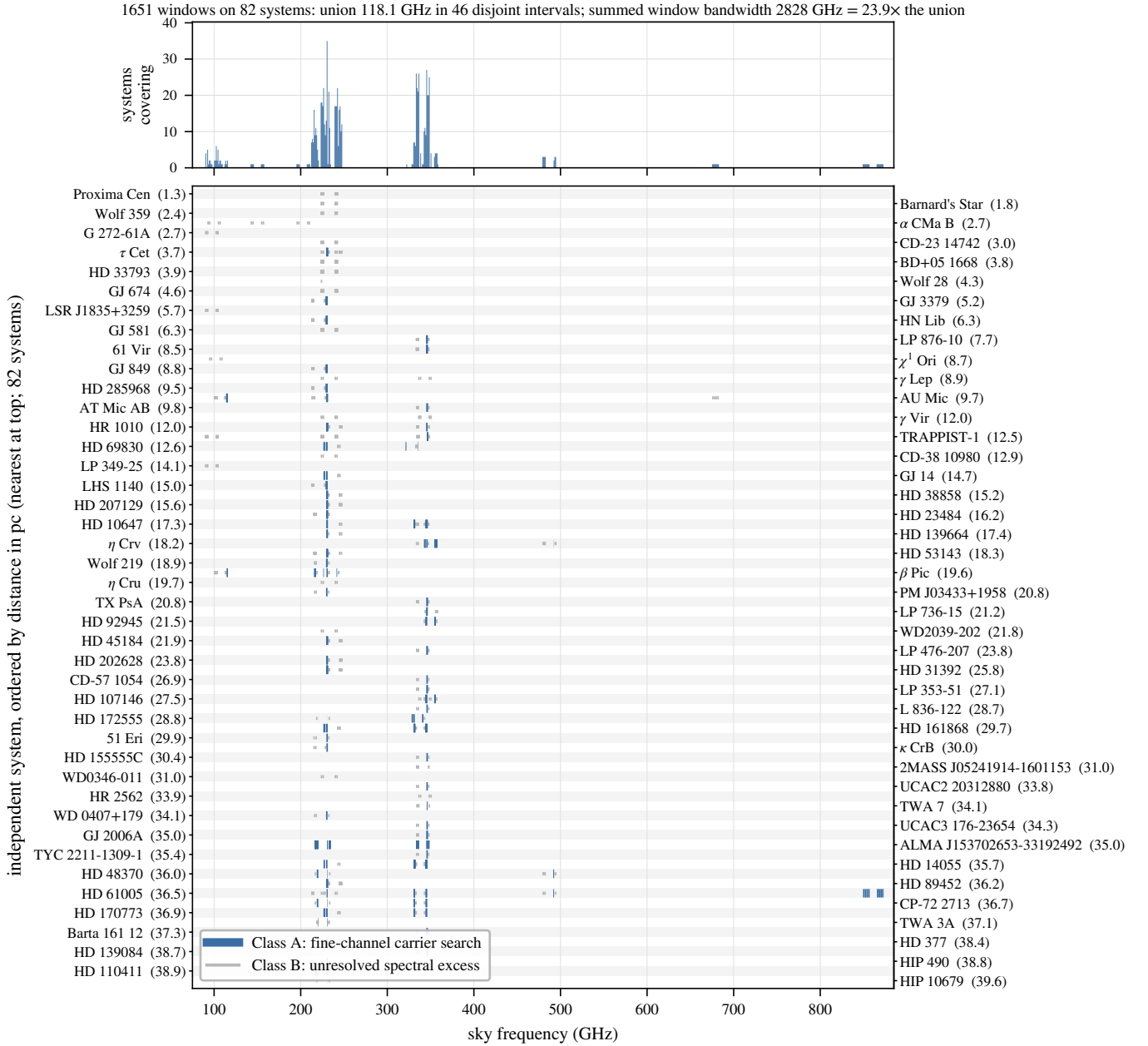


FIG. 7.— Where this survey searched. *Lower panel*: each searched window is a horizontal segment on its host system's row, rows ordered by distance and labelled alternately on the left and right axes; heavy blue marks the fine-channel carrier search (Class A) and light grey the coarse-channel spectral-excess search (Class B). *Upper panel*: the number of systems covering each gigahertz of sky frequency; the threshold distribution is Fig. 1. The coverage is narrow, clustered and repeatedly revisited: 46 islands concentrated near 230 and 345 GHz, with summed window bandwidth 23.9× the union. The union measures frequency coverage, the summed bandwidth integrated search effort.

beam and the predicted loss is 1.0 per cent; for the AU Mic block it is 0.245 of a $0.255''$ beam and the predicted retained amplitude is 0.89 (0.86–0.93 over the 17 real 12 m configurations whose dirty beam we measured directly). A calibration scale error, a polarisation convention, a channel-weighting error or a time-averaging loss would all apply to the Ross 154 comparison equally, and none does.

The residual is stated, not explained away. At the measured-beam prediction the parallax term accounts for about 58 per cent of the AU Mic deficit, leaving 7–8 per cent unaccounted for. That is comparable to the combined quoted uncertainties (ours ≈ 5 per cent, theirs ≈ 2.5 per cent), and the block's own measurement set has since

been deleted, so its dirty beam cannot be computed. The honest statement is therefore that this check bounds any *further* systematic in our absolute flux scale at roughly the 10 per cent level — not that no second term exists. The identification of the mechanism rests on the *pattern* across the two comparisons, not on the size of either.

A prospective epoch extension

The work list was built against an earlier archive map, which had not exposed 177 public in-scope execution blocks. 136 of them, 77 per cent and 1.4 TB, have since been processed; 115 are searched with the frozen pipeline and reported here, *after* the detection statistic and the molecular-line mask were fixed (snapshot 2026-

09-22 08:18 UTC). The 41 that remain are accounted for individually rather than left as a remainder: 17 expose no calibration product in the archive, 11 are Cycle 0/1 blocks whose legacy calibration-script replay this work does not maintain, 4 (158 GB) have a calibration working set larger than the volume available and cannot be processed here at any scheduling, and 9 (449 GB) are reachable in principle and were not attempted. We therefore do not claim the archive has been exhausted, and the recurrence statements below rest on the epochs actually in hand. They are reported here and not merged into the science sample: every statistic above rests on a denominator and a candidate list fixed before the data were examined, and merging would destroy that property. The frozen sample is unchanged.

The extension is 463 windows toward 16 stars in 16 systems, 67 fine-channel and 396 coarse, 57.6 GHz of frequency of which 13.2 GHz is Class A. Every block is repeat coverage of a star and a tuning the survey had already searched, so it adds no star, no system and no frequency; what it adds is epochs. 32 further windows in 8 blocks were withheld under the survey’s own rule, being ϵ Eridani Band 6, where the star sits near 0.7 primary-beam FWHM and the Gaussian beam form used for the correction does not hold.

The extension ledger closes as follows, and the reconciliation is stated because it does not close on its face. 136 in-scope blocks had been processed by the time the block-fate ledger above was cut; 123 of them are in the extension export, which was frozen earlier, on 2026-09-22, while the mining campaign was still running. Of those, 8 are withheld by the rule just stated and 115 are analysed here. The remaining 13 were processed after the extension snapshot and are *not* in this analysis: $136 = 115 + 8 + 13$. They are repeat coverage like the rest and would be folded into a future extension, not into the frozen sample. Separately, two counts of uncalibrated blocks appear in this paper and are not in conflict: 20 is over all 177 in-scope blocks the original work list missed, and 17 is over the 41 of those that remain unsearched; the second is a subset of the first.

The epoch gain is small. The number of systems with more than one searched epoch rises from 55 to 56 of 82, the number with a single epoch falls from 27 to 26, and 1 system acquires a second epoch: the extension deepens the time coverage of systems that already had several rather than repairing the confirmation gap of §6.5.

Three results carry. The departure from exchangeability reproduces on data the statistic had never seen: the median add-one rank is 0.409 over 463 windows where 0.5 is expected, with $D = 0.116$ at $p < 0.01$. That is a third independent measurement of the effect §C.1.1 and §C.1.2 report, and it is why the spatial rank is used as a screen and never as a significance. The positive control recovers: 3 windows reach stage 1 and all 3 of them are the β Pictoris disc in CO(1–0), at $T_\star = 23.17$, 19.39 and intermediate against ring maxima below 6.21, within 8.8–9.0 MHz of the transition, in three *different* previously unsearched execution blocks. And no new unattributed event appears: 13 threshold crossings (2.81 per cent of windows), 4 rank-first against 1.3 expected, and 0 unattributed stage-1 events against 0.21 expected, that expectation computed on the 78 extension windows whose control maximum reaches the trigger (§5.2). The

TABLE 10
THE PROSPECTIVE EPOCH EXTENSION AGAINST THE FROZEN SCIENCE SAMPLE. EVERY FROZEN-COLUMN NUMBER IS COMPUTED FROM THE RELEASED CATALOGUE BY THE SAME DEFINITION APPLIED TO THE EXTENSION, AND THE GENERATOR CROSS-ASSERTS THEM AGAINST THE MACROS THE REST OF THE PAPER USES, SO THE TWO COLUMNS ARE LIKE FOR LIKE. THE EXTENSION IS *not* POOLED INTO THE SAMPLE: IT WAS SEARCHED AFTER THE STATISTIC AND THE MASK WERE FIXED. THE LAST THREE LINES CARRY THE RESULT — THE ATTRIBUTED EVENTS ARE ALL THE β PICTORIS DISC, THE UNATTRIBUTED COUNT IS ZERO, AND THE ADD-ONE RANK IS DISPLACED FROM 0.5 BY ABOUT AS MUCH AS IN THE SAMPLE ITSELF, ON WINDOWS THE STATISTIC HAD NEVER SEEN.

	frozen sample	extension
execution blocks	404	115
windows	1651	463
Class A	402	67
stars	89	16
systems	82	16
threshold crossings	56	13
rank-first windows	14	4
stage-1 events	12	3
attributed to CO	10	3
unattributed	2	0
median add-one rank	0.485	0.409

count of unattributed events in this paper therefore remains 2 (Table 10).

An astrophysical positive control: β Pictoris CO

A search that finds nothing must show that it could have found something, and two demonstrations are needed. The injection campaign of Appendix G establishes that the pipeline recovers a drifting unresolved spectral carrier, which is the searched morphology. β Pictoris CO, resolved, several km s^{-1} wide and effectively undrifting, establishes the complementary property: a real, spatially localised, narrow spectral feature at a stellar position, recovered blind by the frozen pipeline, in two transitions, in 9 execution blocks, over 8.4 yr from 2013-10-06 to 2022-03-03, and at a drift consistent with none — the Band 3 crossings agree across blocks to 1.00 channel once each block’s own tuning is removed, which bounds any apparent drift far below the grid the window searches. It is therefore a positive control for localisation and recurrence, not for the carrier morphology. HD 48370 is the complementary case, an extended foreground line that fills the annulus and so is rejected by the same spatial statistic that promotes β Pictoris; CP–72 2713 and HD 23484 (Table 29) are the third regime, marginal single cells with no recurrence in the repeats available.

β Pictoris is also the site of a radio detection of a very different kind. Ortiz Ceballos et al. (2026) report recurring, highly circularly polarised bursts together with persistent emission from the planet β Pic b over 0.85–3.5 GHz with MeerKAT, which they attribute to the electron-cyclotron maser instability and use to infer a field of $\gtrsim 1.25$ kG at the planet. That emission is inaccessible to this survey through the physics of the mechanism rather than through our depth. Cyclotron maser emission cannot exceed the electron cyclotron frequency at its source, $2.8 \text{ MHz G}^{-1} \times B$; the quoted field reproduces their own 3.5 GHz upper edge, and reaching the lowest frequency searched here, 89.6 GHz, would require 32 kG, about 26 times the β Pic b field, with Band 7 requiring

99 times it. Their detection sits 1.4 decades below our coverage, so nothing in this paper bears on it and our null is not evidence against it. It does settle one argument of §4 in both directions. Strong circular polarisation is measurable from a planet-mass companion in one of the very systems searched here, so the polarisation discrimination we forgo is real rather than hypothetical; and it is not by itself a mark of artificiality, since a natural magnetosphere produces it. Polarisation would separate compact coherent emission from the thermal dust and line emission that fills this archive, not engineering from magnetospheres.

The working behind the attribution is as follows. The Band 6 window is a line tuning, 53.9 MHz wide at 15.26 kHz, so it holds 3533 channels like a wide window and searches 1944 trial drift rates against a survey median of 4, the largest trials load in the survey at $\eta_{\text{drift}} = 422$: this rank is harder to earn than a typical one, not easier. Carried through the frame chain of §F.1, the Band 3 crossing lies -0.39 and the Band 6 crossing -0.29 km s^{-1} from the stellar systemic velocity, in the two lowest CO rotational transitions of a star with mapped circumstellar CO (Matrà et al. 2017; Dent et al. 2014), and a third, unflagged window six years earlier shows the same component at -0.14 km s^{-1} ; all three lie within 0.39 km s^{-1} of systemic (Table 5). The full frame audit, the flux ratio and the excitation argument are in Appendix K, with the per-window numbers in Table 22.

Attributing the pair to CO does not diminish it. Against the survey background the Band 3 peak $T_{\star} = 30.76$ is matched or exceeded by 0 of the 1651 per-window control maxima, and the Band 6 peak $T_{\star} = 11.68$ by 14 of them. Those are ranks in a reference set this paper shows is not exchangeable with the stellar position, so they are quoted as ranks and not as p -values; the attribution rests on the velocity agreement and the recurrence, neither of which needs them, and the alternative reading needs a transmitter placed by coincidence at matched offsets from two CO transitions in one star.

A non-recurrence elsewhere means nothing unless the same machinery recovers a repeat when one exists, so the recurrence test was run on emission already believed in. Every β Pictoris CO feature this survey flagged is recovered in an independent execution block: the Band 3 unit is exhaustive at 4 of 4 blocks with the line present in every one, the Band 6 recurrence unit likewise, and the stage-1 Band 6 window recurs in its unit’s second block at $T_{\star} = 8.95$. Three things make this a recurrence and not a coincidence: the peaks agree to about one channel once each block’s own tuning is removed, the barycentric velocities agree to 0.7 km s^{-1} of systemic across the full baseline, and the lines are $10.8\text{--}15.2 \text{ km s}^{-1}$ wide, an order of magnitude broader than the single channel a carrier occupies. Appendix K gives the block-by-block numbers, including the warning that the spatial screen loses power when the CO belt fills the control annulus.

NOTATION

FREQUENCY CONVENTIONS, LINEWIDTH READING, AND SMEARING

Frequency and velocity reference frames

Every frequency axis here — searched spectra, tabulated ranges and crossing frequencies alike — is *topocen-*

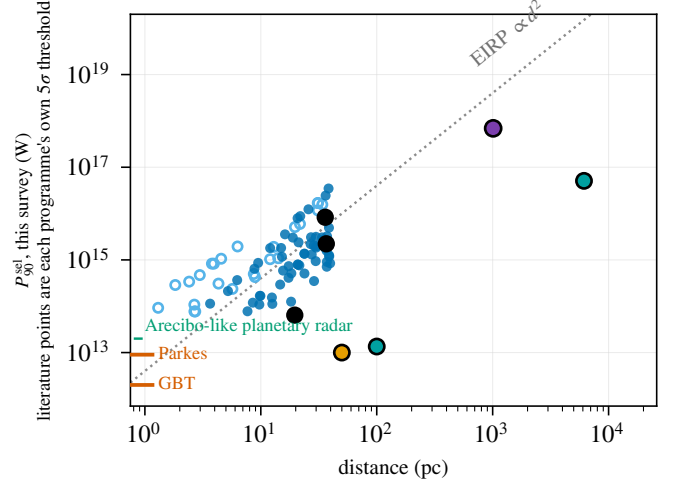


FIG. 8.— EIRP against distance for this survey and for published centimetre-wave searches. Blue: this survey, plotted as P_{90}^{sel} for the deepest window per star; orange: Enriquez et al. (2017); aqua: Margot et al. (2023); violet: Mason et al. (2025); left-edge ticks: Price et al. (2020). Grey dotted: $\text{EIRP} \propto d^2$ at constant received flux. Green dashed: the Arecibo 2.38-GHz benchmark, a power scale only. The vertical coordinates are commensurable as total powers for a sub-channel carrier; what differs between programmes is the detection threshold and the completeness level each adopts. This survey’s value is a 90 per cent recovery through the complete selection, a stricter convention than the nominal thresholds most of the comparison programmes quote.

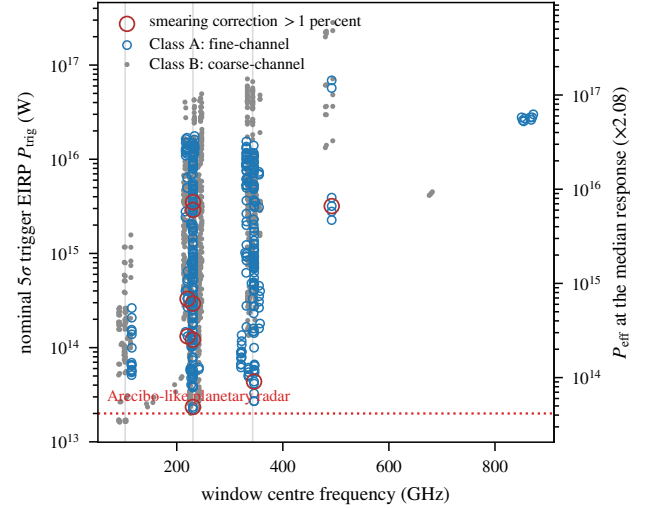


FIG. 9.— Two-dimensional sensitivity against window centre frequency for all 1651 searched windows, symbol by channelisation class, the axis along which completeness differs (Table 13). *Left axis*: the nominal trigger threshold P_{trig} , an internal detection statistic. *Right axis*: the effective threshold at the *median* response, $P_{\text{eff}} = 2.08 P_{\text{trig}}$, which is the power to compare with a transmitter (§4); the per-window ratio $P_{\text{eff},\text{total}}/P_{\text{trig}}$ runs 1.06–3.63, so a single window’s threshold should be read from the catalogue, not from this axis. Red rings mark the 9 windows whose smearing correction exceeds 1 per cent; dotted, the Arecibo 2.38-GHz benchmark. Note the two decades of scatter at fixed frequency: there is no single survey sensitivity.

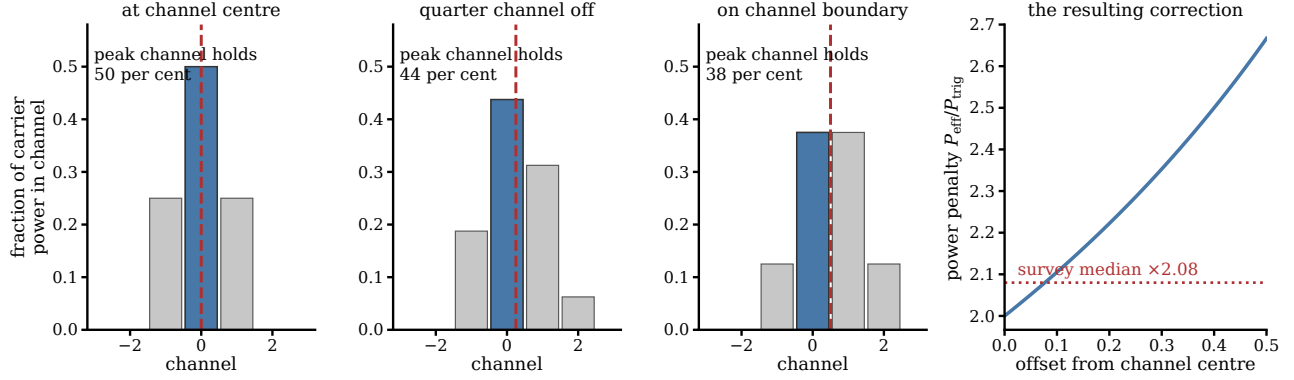


FIG. 10.— Why the nominal channel threshold is not the transmitter threshold. ALMA applies online Hanning smoothing by default, so an intrinsically unresolved carrier is spread over neighbouring channels and the peak channel holds only part of its power: half at a channel centre, 0.42 on a boundary (first three panels; blue is the peak channel, the dashed line the carrier). The search threshold is set in units of the channel noise, so the power a transmitter must radiate to reach it is larger than the nominal figure by the reciprocal of that fraction (right-hand panel). This is the whole of the difference between P_{trig} and P_{eff} in Table 11.

trig as delivered by the ALMA correlator, with no velocity-frame shift applied at any stage. Velocities enter only in the vetting stage’s line mask. The stacking is drift-aligned and inverse-variance weighted: coherent in frequency trajectory, not in electric-field phase.

The native channel width is also the linewidth convention for every EIRP threshold. Channel dilution spreads a sub-channel carrier’s total flux across the channel, so $\text{EIRP}_{\text{trig}}$ is the minimum *total* power of a channel-confined carrier, independent of the transmitter’s own width below the channel, and never a spectral power density. The Hz-resolution surveys’ limits are total powers in the same sense, each the best case of its own channelisation, which makes the axes of Fig. 1 commensurable. One number fixes the scale, illustratively and not achievably: the median Class A channel is 488 kHz, so rechannelising the same data to the 2.79 Hz of a dedicated narrowband search would, on $P_{\text{min}} \propto \sqrt{\Delta\nu}$, divide the threshold by 418 and put the median Class A selection power at $\sim 7.4 \times 10^{12}$ W. ALMA cannot do this: the correlator delivered the channelisation the archive holds, and no reprocessing recovers resolution that was never recorded.

Every spectral window of an observation enters the spectral-carrier search, capped at 8 per target to bound compute cost; the cap bound one target/band, 61 Vir Band 7, and three further targets are represented by their deepest single window and marked in the released catalogue. Each searched window yields its own $\text{EIRP}_{\text{trig}}$ row, and Figure 1 plots only the deepest per star.

Primary-beam response. Carrier-lane thresholds and continuum fluxes are corrected for ALMA’s primary-beam response at the star’s offset from the phase centre, negligibly for most targets (median factor 1.00002, 90th percentile 1.002), though 83 of the 1651 retained windows exceed 2 per cent and 34 exceed 10 per cent. A window is kept only where that response is at least 0.5, a correction of at most $\times 2$, since at low response a small error in the beam model becomes a large error in inferred flux. The offsets are bimodal and the floor sits in the gap: the last retained window has response 0.55 at $0.46 \theta_{\text{PB}}$ (LP 736-15 Band 7, $1.81\times$), the first window beyond it 0.35 at $0.62 \theta_{\text{PB}}$ ($2.88\times$), and nothing lies between, so the rule binds prospectively and excludes none of the 1651 released windows. Beyond it are the 12 ϵ Eri

Band 6 windows at $1.0\text{--}3.5\times$, also past the first sidelobe transition where the Gaussian form fails; the two criteria are independent and select the same windows. They are *withheld*, enter no number here and are absent from the released catalogue, although under the nominal rule they would have covered $3.7 \times 10^{13}\text{--}1.1 \times 10^{15}$ W at 3.22 pc, among the deepest thresholds in the survey but resting on a beam model that cannot be defended. The annulus geometry keeps the pipeline’s own $\theta_{\text{PB}} = 1.22 \lambda/D$, an Airy first-null coefficient; recomputing it on ALMA’s FWHM $1.13 \lambda/D$ at each window’s own dish diameter moves 75 of 1651 windows by more than 1 per cent, extremes -2.6 to $+10.3$ per cent, and no headline threshold moves.

Transmitter power benchmarks

The benchmark is frequency-matched and purely geometric. For a filled circular aperture of diameter D radiating P_{tx} at wavelength λ ,

$$G = \frac{4\pi A_{\text{eff}}}{\lambda^2}, \quad \text{EIRP} = P_{\text{tx}} G, \quad (\text{F1})$$

so a 12-m aperture radiating 1 MW at 230 GHz with a realistic aperture efficiency of 0.7 has $G = 5.9 \times 10^8$ and an EIRP of 5.9×10^{14} W; a perfect aperture, $\eta = 1$, would give 8.4×10^8 and 8.4×10^{14} W, and the realistic case is the one quoted. The median Class A effective threshold, 1.8×10^{15} W, misses such a transmitter by a factor 3.1.

The Arecibo-like planetary radar used for comparison (Drake 1974; Sheikh et al. 2025, $\text{EIRP } 2.0 \times 10^{13}$ W) is a 2.38-GHz *power* scale carrying no transmitter model. Gain scales as ν^2 , the same equation at fixed A_{eff} , so the same aperture and transmitter power at 230 GHz would radiate $2.0 \times 10^{13} \times (230/2.38)^2 \simeq 2 \times 10^{17}$ W. The comparison is between physical transmitter powers, so it uses P_{eff} and not the trigger: that difference carries the deepest window from below the numerical Arecibo figure to 3.4×10^{13} W above it (§6.1). **The frequency-scaled figure, $\sim 2 \times 10^{17}$ W at 230 GHz, is the physically appropriate comparison for this survey rather than the 2.38-GHz number**, because it is the same aperture and the same transmitter power radiating in our band; on that scale every window here is far below the benchmark. *And this paper’s own rule is that sensitivity means P_{90}^{sel}* , so a benchmark statement expressed in P_{eff} or P_{trig}

TABLE 11

NOTATION. EVERY POWER SCALE AND EVERY TERM THIS PAPER USES IN A PARTICULAR SENSE IS DEFINED HERE; LATER SECTIONS USE THEM WITHOUT RE-EXPLANATION. THE ABSTRACT AND CONCLUSIONS QUOTE P_{90}^{sel} ; THE RELEASED CATALOGUE CARRIES ALL OF THE POWER QUANTITIES PER WINDOW.

Quantity	Meaning and use
T_*	the search statistic at the stellar position (Eq. 2): a matched-filter amplitude divided by a noise estimate. Dimensionless, and <i>not</i> quoted in units of σ , its null distribution being neither assumed nor shown to be Gaussian
T_{trig}	the trigger, $T_{\text{trig}} = 5$: a fixed level on T_* , not a tail probability. Throughout, σ is reserved for genuine noise scales — injected amplitudes, image rms, fit uncertainties (§4.1)
<i>The power hierarchy, in order.</i> P_{trig}	<i>Only the last is a sensitivity.</i> the power at which the search statistic reaches T_{trig} in one native channel: an internal trigger, not a detection significance and not a completeness level (§4)
C_{resp}	spectral-response correction, $\times 2.08$ median, window-specific (Appendix F)
C_{smear}	intra-integration drift smearing, $1/\eta_{\text{smear}}$ (Appendix F)
P_{eff}	$P_{\text{trig}} C_{\text{resp}} C_{\text{smear}}$: the power a transmitter must radiate to reach the trigger (Fig. 10). Every EIRP quoted here is the <i>total</i> radiated power of an unresolved monochromatic carrier, not the power falling in the peak ALMA channel
P_{90}	the amplitude the <i>trigger</i> alone recovers nine-tenths of the time, measured end to end: $\times 3.0$ the nominal trigger for Class A and $\times 4.1$ for Class B. Completeness of one stage, not of the selection (§5.4, Appendix G)
P_{90}^{sel}	the amplitude the complete automated selection recovers nine-tenths of the time, trigger <i>and</i> the 512-control rank together, measured on the same injected tones: $\times 5.7$ the nominal trigger for Class A (§5.4). The headline sensitivity, in which every limit in the abstract, discussion and conclusions is expressed
Class A/B	fine-channel / coarse-channel windows; different experiments, never pooled in a completeness statement (§4, Table 13)
dwelt	recovery against the fraction of the track a signal is present (§G.1)
candidate	a stage-1 flagged window that also recurs or is independently confirmed; this survey reports none (§4.1)
pseudo-star	an inner-annulus control probe ranked against the others; it contains no star, so a first rank is a false alarm by construction (§C.1.2)
debit $m(0)$	the radial correction subtracted from the stellar statistic before ranking (§C.1.2)
stratum	a subset of windows sharing an instrumental property (band, array, channelisation) analysed separately (Appendix G)
ring max	the largest control statistic in a window’s annulus (§4.1)
R_σ	the star’s robust residual scale over the controls’ median; unity means no noise peculiar to the stellar position (§K.1.2)
hold-out	blocks withheld from the analysis by a rule fixed in advance, used only to check a threshold and never to set one (§C.1.1)
drift	the rate at which a carrier’s observed frequency changes, Hz s^{-1} (e.g. for line of sight

is an instrument-level quantity and is labelled as such: the corresponding completeness figure is a further $\times 5.7$ ($\times 3.7$ – $\times 5.9$) on the nominal Class A trigger (§5.4), and every comparison in this appendix should be read with that factor applied.

Reference frames. Two things must be kept apart. *The released line-offset columns, and the attribution the search applied, are topocentric:* all 1396 of them reproduce from the observed sky frequency against the frozen rest frequency, and none reproduces from a barycentric or stellar-frame evaluation (Appendix I). The *audit* chain of Table 22, applied by hand to the attributed windows, runs topocentric $\rightarrow$ barycentric (Earth ephemeris at the execution-block mid-time) $\rightarrow$ stellar rest frame (catalogued systemic radial velocity), with JPL/CDMS rest frequencies (Pickett et al. 1998; Müller et al. 2005); in the LSR frame it stops at the standard solar motion, not the systemic velocity (§5.2.6). The residual error is the catalogue radial-velocity uncertainty alone, a few km s^{-1} within 40 pc; second-order relativistic terms at the 50 km s^{-1} scale are $\lesssim 5 \text{ kHz}$ at 350 GHz, below the finest native channel searched (15.3 kHz). The β Pictoris attributions use this chain, so their offsets are comparable to the exclusion tolerance.

Channel widths, and the instrumental response. Channel widths come from the measurement sets’ own spectral window tables (CHAN_WIDTH). A missing “EFFECTIVE_BANDWIDTH” column is not evidence of an absence of online spectral averaging, MS v2 defining no such column, so the quoted channelisations are not claimed to be noise bandwidths.

ALMA applies online Hanning smoothing by default, so the spectral response is ~ 2 channels wide and neighbouring channels are correlated. For the 0.25/0.5/0.25 kernel the coefficients are $\rho_1 = 2/3 = 0.667$ and $\rho_2 = 0.167$, with $\sigma_{\text{smoothed}}/\sigma_{\text{raw}} = 0.612$: the correlation is not confined to adjacent channels, the lag-two term being a sixth. The peak-channel fractions 0.42–0.50 that follow, median 0.48, and the resulting $\times 2.00$ – $\times 2.36$ optimism of the nominal thresholds are given in §4; the released catalogue carries the nominal threshold and both corrected values per window, and the sensitivity summary quotes the worst case. The injections deposit an unsmoothed delta-in-channel tone, so the recovery curves of §G.1 and Appendix G carry the same bias and the end-to-end check (Eq. G1) cannot see it.

Two measurements fix the smoothing state. The retained per-integration residuals of the second-stage local null (§5) have a measured lag-one spectral autocorrelation of 0.65 against the Hanning prediction 0.667, so that correlation is instrumental and the residual spectra are smoothed and un-averaged. And the archive’s reported effective spectral resolution per window, against each measurement set’s own channel separation, is exactly 2.00 in 1505 of the 1651 windows, in both correlator modes and including 1217 of the 1249 coarse time-division-mode (TDM) ones; the other 146 report a smaller ratio, the signature of online channel averaging, where the peak-channel loss is smaller, and the 0 Band 9 and 10 windows, absent from the metadata snapshot, take ALMA’s default. The blanket factor is right for the great majority of both classes. Thresholds are not rescaled: each is nominal and up to $\times 2.36$ optimistic. The correlation model is truncated at lag

two, and the baseline step — a per-integration median in blocks of up to 65 channels — could imprint correlation on longer scales, which the released products retain no residual spectra to measure. Such a term would make the independent-trials count of Appendix J an over-estimate and the achievable per-window resolution coarser still.

Linewidth convention. The statistic is the single-channel flux density after drift correction, so the quoted $\text{EIRP}_{\text{trig}}$ applies to any transmitter narrower than the native channel and is flat in the assumed linewidth. Two sub-channel qualifications attach: flatness is exact only for a centred tone, a boundary-straddling one splitting its power between two channels at worst by a factor of two; and the channel-to-channel transition follows the correlator’s passband shape, which the frozen products do not retain. Thresholds therefore carry a sub-channel-position systematic of tens of per cent, and EIRP values are quoted to two significant figures.

Transmitters wider than one channel. The baseline filter (§4.1) is flat for features narrower than roughly half its effective width W and falls to zero above W , so the survey’s sensitivity is to channel-confined morphologies and W is the linewidth ceiling: 65–74 channels over the fine class and 30–59 over the coarse one, that is ≈ 32 MHz at 488 kHz and ≈ 0.9 GHz at 15.62 MHz. Within that ceiling, a top-hat transmitter spanning N_{ch} native channels has per-channel flux the total divided by N_{ch} at unchanged noise, so to first order

$$\text{EIRP}_{\text{trig}}(N_{\text{ch}}) \simeq N_{\text{ch}} \times \text{EIRP}_{\text{trig}}(1), \quad N_{\text{ch}} \lesssim W/2, \quad (\text{F2})$$

the rescaling the main text applies to signals broader than one channel.

Worked example: from flux to EIRP. Take UV Ceti (G 272-61B) Band 3, the deepest of its 4 windows in that band (execution block A002_X877e41_X14c1, lower edge 91.565 GHz). Its per-channel rms is 0.241 mJy, so $S_{\text{min}} = 5 \times 0.241 = 1.21$ mJy in one 15.625-MHz native channel, or $1.21 \times 10^{-29} \text{ W m}^{-2} \text{ Hz}^{-1}$ at 2.67 pc, and $\text{EIRP}_{\text{trig}} = 4\pi d^2 S_{\text{min}} \Delta\nu = 1.6 \times 10^{13} \text{ W}$.

Intra-integration smearing. The frequency excursion at the maximum searched drift rate across one integration, $\dot{\nu}_{\text{max}} t_{\text{int}}$, is at most 1.10 channels across the 1651 windows (median 0.001, 90th percentile 0.06). A linear drift smeared by Δ channels within an integration keeps a fraction $\eta_{\text{smear}} = \text{sinc}(\pi\Delta/2)$ of the peak-channel amplitude, so the effective threshold is $\text{EIRP}_{\text{trig}}/\eta_{\text{smear}}$; a pure occupancy argument would retain $\min(1, 1/\Delta)$ instead, giving 1.0 rather than 0.64 at $\Delta = 1$, so the sinc form is the conservative reading. The 9 windows with $\eta_{\text{smear}} < 0.99$, as star, band, (η_{smear} , threshold multiplier), are HIP 10679 Band 6 (0.574, $\times 1.74$), β Pic Band 6 (0.574, $\times 1.74$), HD 172555 Band 6 (0.574, $\times 1.74$), HIP 10679 Band 6 (0.606, $\times 1.65$), HD 172555 Band 6 (0.606, $\times 1.65$), η Crv Band 8 (0.865, $\times 1.16$), η Crv Band 7 (0.932, $\times 1.07$), HD 48370 Band 6 (0.979, $\times 1.02$) and CP-72 2713 Band 6 (0.979, $\times 1.02$); the remaining 442 agree with their nominal thresholds to better than 1 per cent. Compounding the worst smearing penalty with the worst Hanning factor gives $\times 4.6$ on the nominal threshold for those five windows, and no released column carries that product.

TABLE 12

SYSTEMATIC UNCERTAINTY BUDGET ON P_{90} . THE FIRST BLOCK ARE INDEPENDENT ERRORS ON THE QUOTED NUMBER AND COMBINE IN QUADRATURE; THE SECOND ARE CARRIED SEPARATELY: THE SUB-CHANNEL PHASE IS AN UNKNOWN ABOUT THE TRANSMITTER, THE DECORRELATION TERM IS A *one-sided bias* THAT MAKES P_{90} OPTIMISTIC AND CANNOT MAKE IT PESSIMISTIC, AND THE WINDOW-TO-WINDOW TERM IS A TRANSFER UNCERTAINTY FOR THE WINDOWS THE CAMPAIGN DID NOT INJECT INTO.

Term	Origin	ΔP_{90} (%)
Absolute flux scale	ALMA calibration, band-weighted	± 7
Distance	Gaia parallax, as d^2	± 0.2
Visibility calibration	block-to-block scale	± 4.5
<i>Combined</i>	<i>in quadrature</i>	± 9
Decorrelation	one-sided; injections suffer none	$+5/ +20$
Pointing	one-sided; worst case, $0.6''$ off axis	$+1.9$
Sub-channel phase	irreducible; median-corrected	$-12/ +17$
Calibration campaign	3530 tones, 18 configurations	$-36/ +4$

INJECTION-RECOVERY SENSITIVITY VALIDATION

Dwell-fraction completeness

How long a carrier occupies one native channel defines the searched class, so recovery is measured against it directly. A carrier was injected into the retained per-integration spectra of 12 real windows, six coarse and six fine, at amplitudes of 2–20 times the per-integration noise and dwell fractions $f_{\text{dwell}} = 0.10$ –1.00, six realisations each: 3888 trials, released with one row per injection (Table 13). The detection criterion, a logged per-integration S/N of five, re-derived from the released rows agrees with the logged flag 3888 times in 3888 trials, with 0 mismatches across a clean separating gap. Recovery rises monotonically with dwell: pooling amplitudes at or above 4σ , coarse windows recover 100 per cent of continuous carriers, 100 per cent at $f_{\text{dwell}} = 0.25$ and 83 per cent at 0.10, and fine windows 100, 91 and 71 per cent. Appendix G.3 gives the end-to-end check, the campaign’s scope and the representativeness of the injected windows.

The trigger acts on the stacked statistic and the de-drift step is an inverse-variance-weighted sum over integrations, so a carrier of per-integration amplitude A present for a fraction f_{dwell} of a track of N_{int} integrations accumulates to

$$a = A f_{\text{dwell}} \sqrt{N_{\text{int}}} \quad (\text{G1})$$

in units of the per-integration noise. Recovery depends only on a , so binning the released trials in a places all 12 injected windows on one axis; they span a factor of 63 in integration count over 11 targets, two resolution classes and Bands 3, 6 and 7. In the 6 windows whose amplitude grid straddles the trigger, the 50 per cent recovery point lies at $a = 5.3$ –8.3, the trigger itself, independent of channelisation, integration count and band. In the other 6 the weakest injection already stacks to $a = 3.4$, consistent but uninformative about the curve’s position. The pooled curves (Table 13) agree between classes within the sampling noise.

The drift ceiling carries one caveat. The searched value is 12.0 – $13.3 \text{ Hz s}^{-1} \text{ GHz}^{-1}$, line-of-sight $|a| = 3.60$ – 4.00 ms^{-2} , raised above that generic value only in the 56 windows where a known planet’s own bound requires it. It clears Earth-rotation and Earth-orbit drifts by two orders of magnitude but is marginal against a close-in planet on a TRAPPIST-1 b-like orbit, so the limits of §6

TABLE 13

COMPLETENESS MATRIX. (a) WHAT IS MEASURED FOR EACH SEARCHED MORPHOLOGY AND WHERE: ‘e2e’ IS INJECTION INTO VISIBILITIES THROUGH THE UNMODIFIED PIPELINE, ‘POST’ INJECTION INTO RETAINED PER-INTEGRATION SPECTRA AFTER THE BASELINE STEP, AND ‘CFG’ THE NUMBER OF WINDOW CONFIGURATIONS THE CELL SPANS. THE TWO CAMPAIGNS MEASURE DIFFERENT QUANTITIES: 1200 INJECTION TRIALS (APPENDIX G) AND 3 888 DWELL TRIALS OVER 12 REAL WINDOWS. THE 0 OF 500 COARSE-WINDOW ENTRY IS A PROPERTY OF THE RECOVERY CRITERION, WHICH REQUIRES THE RECOVERED DRIFT TO MATCH THE INJECTED DRIFT AND SO CANNOT BE SATISFIED ON A CHANNEL WIDER THAN THE WHOLE DRIFT TRACK (§G.1).

Morphology	e2e	cfg	thresh.	dwell	drift
Class A drifting	yes	1	yes ^a	no	yes
Class A near-static	post	6	no ^b	yes ^c	n/a ^c
Class B stationary	yes	6	yes ^d	yes ^e	none
Class B drifting	yes	6	yes ^d	yes ^e	none ^c
Intermittent, either class	post	12	no	yes ^e	by class

^a42 per cent at the trigger, 83 at twice it, one configuration only (AU Mic, Band 6, 488 kHz), transfer to the other 402 Class A windows measured at 0.64–1.13× on 28 configurations (Fig. 11; the breakdown by drift stratum is Table 27). ^bAfter the baseline step only. ^cSub-channel over the track by definition. ^dHalf recovery at stacked $a = 5.3$ –8.3 in the 6 windows straddling the trigger. ^e83–100 per cent (coarse), 71–100 per cent (fine), over $f_{\text{dwell}} = 0.10$ –1.00.

(b) completeness at the trigger, against stacked amplitude a (Eq. G1), pooled over the 12 injected windows; trial counts in parentheses.

stacked a	Class B (coarse)	Class A (fine)
< 3	0% (30)	0% (72)
3–4	0% (12)	7% (42)
4–5	3% (30)	12% (42)
5–6	46% (24)	33% (48)
6–8	58% (60)	70% (96)
8–12	100% (108)	99% (174)
> 12	100% (1680)	100% (1470)

are conditional on it (Figure 3).

Interference rejection is an operative gate. The highest frequencies allocated to any satellite constellation are the E-band feeder links of the second-generation systems, 71–76 and 81–86 GHz (SpaceX 2024), below the 89.6 GHz start of the searched union, and ITU Radio Regulations No. 5.340 prohibits all emissions in bands covering 13.3 GHz (11.2 per cent) of it. Allocation tables exclude neither harmonics nor intermodulation products nor near-field emitters, so no exclusion claim rests on one; the operative defences are the control ensemble, the visibility phase test and recurrence, measured in Appendix M.

What was searched, by class. Class A, the fine-channel drifting-carrier search: 26 frequency islands totalling 47.7 GHz between 114.0 and 873.1 GHz, channels 0.02–1.95 MHz, accelerations $|a| \leq 3.6$ –4.0 ms^{−2}. Class B, persistent channel-confined excess with no drift discrimination: 35 islands totalling $\Delta\nu_B = 90.5$ GHz between 89.6 and 682.8 GHz, overlapping Class A by 20.1 GHz and so adding $\Delta\nu_{B\text{-only}} = 70.4$ GHz, channels 7.81–31.25 MHz. Both cover only the archival epochs and only carriers one native channel or narrower, over 113.3 GHz of the 118.1 GHz union after molecular-line masking, at nominal trigger thresholds whose measured recovery is 42⁺¹⁶_{−14} per cent. Inside that box the non-detection is a constraint. Outside it, the exclusions are those of §5.4: the trigger is not a significance; the spatial screen is em-

pirically calibrated, and holds for isolated point sources rather than for beams filled by resolved circumstellar or foreground emission; the 1249 coarse windows carry no drift discrimination and no survey-threshold completeness curve; coverage is the 46 islands of Fig. 7 less the 4.1 per cent of the union removed by the ± 50 km s^{−1} line mask; drift above the ceiling and emission broader than one native channel lie outside the searched morphology; duty cycles below ~ 0.1 are essentially unconstrained; and the search is Stokes- I only, so a per-hand search of the 198 blocks that deliver both parallel hands would gain up to $\times 1.41$ on a polarised carrier.

Amplitude completeness from visibility-domain injections

Synthetic tones of known amplitude were injected into real visibilities, using the extraction code’s own mechanism as far upstream of the statistic as the retained products allow, and the unmodified pipeline’s return measured. Six configurations span three bands and both classes: three coarse Band 7 windows of AT Mic, one coarse window each in Bands 3 and 6 of AU Mic, and AU Mic’s 488-kHz flagband window with its 49-trial drift grid. Each receives a complex tone at the star position at 4–10× the statistic’s own noise, over drift fractions $f_{\text{drift}} = 0$ –1.0 of the ceiling, on- and half-grid rates and channel-centred and boundary placements: 200 trials per configuration, 1200 in all. The recovery criterion was frozen in advance: $T \geq 5$, peak channel within 3 of the injection, peak drift within one grid step.

Coarse windows recover 0 of 500 injections from $T_\star = 4$ to 10 under that criterion, which they cannot satisfy, every in-grid drift being sub-channel over the track so that the reported peak drift is arbitrary; the fine window fails it the same way. Scored on detection alone the same windows recover essentially everything (Appendix G.3). Class B limits therefore bound carriers of any dwell fraction, with the completeness of §G.1; what those windows lack is drift discrimination.

On the fine window, drifting carriers ($f_{\text{drift}} \geq 0.5$, pooled over grid phase and sub-channel placement) are recovered 17, 42, 58, 75 and 83 per cent at $T_\star = 4, 5, 6, 8$ and 10, saturating by twice threshold power (Fig. 11). With the tone injected after the baseline step the machinery alone is amplitude-faithful to within $-9/+14$ per cent for channel-centred tones. Boundary-straddling placement costs 0.6–0.8 in the statistic and recovers 0 per cent at the trigger against 83 per cent for centred tones, crossing 50 per cent near $1.4\times$ threshold amplitude. Survey completeness uses the placement-pooled curve, conservative for the boundary-straddling case: $\text{EIRP}_{50} = 1.10 \text{ EIRP}_{\text{trig}}$ and $\text{EIRP}_{90} > 2.0 \text{ EIRP}_{\text{trig}}$, a bound only.

That curve is measured on one window and is not a survey completeness function. Of the 402 Class A windows, 211 (52 per cent) differ from the calibration configuration on at most one axis, the mismatch dominated by band (199 windows) and drift-trial count (167) rather than channel width (30), integration time (87) or beam offset (13). Those are the axes the stratified campaign spans; other target environments and edge-phased drift grids are not spanned at all.

The stratified campaign. A drifting carrier was injected into the retained per-integration spectra of 28 fur-

ther windows, one per star, spanning Bands 3–10, 15.3–977 kHz channels, 45–485 integrations, 7–1186 drift trials and both the 12m array and the ACA, and pushed through the pipeline’s own baseline removal and de-drift stack on the search’s drift grid: nine amplitudes, 4032 trials. Recovery is a threshold crossing at the injected frequency and not the global maximum of the grid, since requiring an injection to beat a window’s brightest cell measures that window’s worst artefact. Pooled recovery is 34, 61, 89 and 99 per cent at $T_\star = 4, 5, 6$ and 8 , with $P_{50} = 4.6\sigma$ and $P_{90} = 6.1\sigma$ (Fig. 11). The spread across configurations sets the transfer bracket: P_{50} runs $0.64\text{--}1.13\times$ its median and P_{90} $0.48\text{--}1.35\times$ its own, so the assumed $0.5\text{--}2\times$ bracket is conservative. Band and array matter least, P_{50} varying only $4.4\text{--}4.8\sigma$ across bands (Table 27).

Drift rate, resolved. All 4032 trials carry a non-zero drift, drawn over $9\text{--}10344\text{ Hz s}^{-1}$, so the pooled curve is a drifting-carrier curve. A pooled curve could hide a deficit at the edge of the grid, where a carrier crosses the most channels and the de-drift stack has the least signal in any one of them, so Table 27 splits the campaign by drift rate as a fraction of each window’s own ceiling. P_{90} varies by 10 per cent across the three thirds with no trend, and the outer third is not the worst; at a fixed 6σ injection they recover 89, 84 and 93 per cent. The rate is recovered as well as the frequency: 92 per cent of detections above 6σ return the injected drift trial itself, median error 0.00 grid steps and 0.5 at the ninetieth percentile. Carrier sub-channel phase, stratified separately over 5 positions and 400 trials, is the one axis that does matter, and is why the threshold carries the $\times 2.69\text{--}\times 2.02$ response correction rather than a single factor.

How far the grid reaches. 28 configurations are a sparse grid for 402 windows. It spans Bands 3, 6, 7, 8 and 10, channel widths 15.3–977 kHz, 45–485 integrations and both arrays, and every Class A window lies in a band it samples. The configuration needed for the comparison — band, channel width and integration count together — is recorded for 131 of the 402; the peak-frequency freeze that carries the integration count does not cover the 271 blocks searched since, which are never assumed to be inside the grid. Of the 131 that can be judged, 112 lie inside the grid on all three axes at once, 85 per cent of them and 28 per cent of all 402. The remaining 19 are extrapolations, 18 in integration count alone and 1 in channel width. The transfer bracket is measured for the 112 windows and assumed for the rest. The five deepest Class A thresholds are all interpolated, so the headline sensitivity does not rest on an extrapolation.

The spectral response, computed from the correlator’s window and validated against the delivered data. The response correction is the largest single factor between the level at which the search fires and the power a transmitter must radiate, so it is computed from the instrument rather than assumed. ALMA applies its window in the lag domain, which is equivalent to the $0.25/0.5/0.25$ smoothing of the delivered spectrum that `hanningsmooth` performs, so the operation is not in doubt; what matters is the channel’s response to a monochromatic carrier. We build the lag function of a continuum source — a delta at zero lag, where the Hann taper is unity — and of an unresolved carrier at sub-channel position ϕ , transform both under one nor-

malisation, and take the ratio. The continuum reading fixes the flux scale, the same scale in which $S_{\min}$ is measured, and the carrier’s peak delivered channel is what the single-channel trigger sees. That channel retains a fraction 0.500 of the carrier at a channel centre, falling to 0.424 at a boundary, so $C_{\text{response}} = \times 2.00\text{--}\times 2.36$, median $\times 2.08$.

Six quantities, none of them used to construct the model, test it. The continuum flux scale is unchanged by the window. One channel’s frequency response has a full width at half maximum of 1.21 channel spacings unsmoothed and 2.00 with the default smoothing, against ALMA’s stated 1.2 and 2.0. The worst-case scalloping loss is 1.42 dB, the textbook figure for a Hann window. The effective noise bandwidth of a smoothed channel is 2.67 times its width, against the documented $8/3$. Strongest of the five, the archive reports an effective spectral resolution for every window in this survey, and all but one of those reports take two values: 2.000 for 1505 windows and 1.156 for the 145 delivered with online channel averaging by 2. The model reproduces both — the first from the taper alone, the second from the same taper followed by that averaging — so it is tested on the survey’s own data and not only against textbook values. The averaged windows carry a smaller correction, $\times 1.18\text{--}\times 1.96$, median $\times 1.33$, because their delivered channels are wider.

Sixth, and the only one of the six that can see the averaging factor: the delivered channel-to-channel noise covariance, measured on sky. The preceding checks cannot. The FWHM comparison is against archive metadata, and the scalloping figure is a ratio in which normalisation cancels, so neither can tell a window averaged by two from one averaged by four — both report an effective resolution smaller than the Hanning one. The noise covariance can: the Hann kernel summed over n raw channels and decimated by n predicts a lag-1 channel correlation of 0.667 for $n = 1$, 0.300 for $n = 2$ and 0.115 for $n = 4$, a factor 5.8 apart. Measured on control positions of 18 released windows, at a single drift row so that no maximisation enters, the three branches give 0.628–0.674 where the model says 0.667, 0.267–0.302 where it says 0.300, and 0.105 where it says 0.115. *Every window confirms its own key:* 18 agree and 0 disagree. Channel width alone would not have served — 244.1, 488.3 kHz each appear in both classes, because ALMA’s native channel count per spectral window depends on the correlator mode — which is why the keying is on the archive’s reported effective resolution and why the catalogue now publishes the averaging factor per window (`n_chan_avg`).

That measurement found one real defect in the 1651 released windows, which the catalogue now splits as $n = 1$: 1505, $n = 2$: 145, $n = 4$: 1. 51 Eri A002_Xb95160_X3e24, at 1953.1 kHz, reports an archive ratio matching the modelled $n = 4$ effective resolution 1.0143 rather than the $n = 2$ value 1.1560, and measures a lag-1 correlation of 0.105 against 0.115 predicted for $n = 4$ and 0.300 for $n = 2$; both routes say it is averaged by four, and earlier versions of this paper gave it the $n = 2$ response. Its C_{resp} is therefore 1.057 and not 1.333, so its published effective power was *overstated* by $\times 1.261$ (1.01 dB), in the conservative direction: $1.459 \times 10^{15}\text{ W}$ becomes $1.157 \times 10^{15}\text{ W}$, and

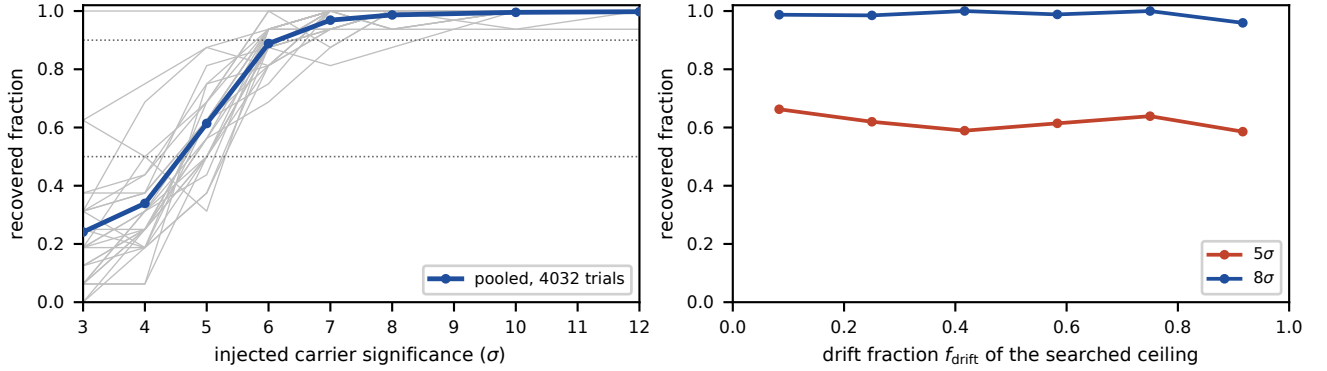


FIG. 11.— Stratified injection recovery. Grey, each of the 28 configurations separately; blue, the pooled curve over 4032 trials. Dotted lines mark the 50 and 90 per cent levels. The spread between grey curves is the completeness transfer, measured rather than assumed. Right, recovery against drift fraction of the searched ceiling at $T_\star = 5$ and 8.

because that is the deepest of 51 Eri’s 4 windows its system limit moves from 1.46×10^{15} to 1.16×10^{15} W. The window carries 0 threshold crossings, the count of systems reaching 10^{15} W is unchanged, and the per-window median effective power moves by 1.71 per cent, which is the median stepping to the adjacent window rather than any window’s power changing. The published C_{resp} values themselves do not move at all, because they come from the model and not from the window list.

Why the earlier checks could not catch it is the transferable part. The consistency check on the archive ratios asserted only that *two* values dominate, which allows one per cent of windows to be anything at all; and the flag selecting the response was a *boolean*, so it had no third state and could not fail for a window averaged by any factor other than the one it knew about. *A flag with too few states is a check that cannot fail.* The key is now the nearest modelled effective resolution among $n = 1, 2, 4$, and the build asserts both that every window’s archive ratio matches one of them and that the factor chosen is that nearest one — an assertion demonstrated failing on the superseded keying and passing on this one.

Two earlier estimates are superseded, in opposite directions. Modelling a channel as a top-hat, so that a carrier divides linearly between the two channels it straddles before the kernel mixes them, gives $\times 2.00$ – $\times 2.67$: a correlator channel is the transform of its lag window, not a two-point interpolation, and the linear split over-attenuates the channel edge by 13 per cent. Taking instead the peak *power fraction* of a Hann-windowed transform gives $\times 1.50$ – $\times 2.08$, which is 12 per cent too permissive at the edge, because dividing by the total power of the windowed series divides out the window’s coherent loss — the dominant part of the effect. The scalloping loss cannot distinguish the two. It is a ratio of the channel-centre to the channel-edge response, and an error in the absolute normalisation cancels in that ratio exactly, so a model can reproduce 1.42 dB and still be 12 per cent wrong in the number the paper needs. Adopting the validated value lowers the median correction by 9 per cent relative to the top-hat model, and every power quoted here follows it.

Dwell campaign: end-to-end check, scope and representativeness

A zero-drift, always-on tone injected into the calibrated visibilities of a coarse Band 7 window at 1, 3 and 10

times the per-integration rms returns $T_\star = 20.5, 65.3$ and 217.7 , each within one channel of the injection. The ideal stacking gain over its 639 integrations is $\sqrt{639} = 25$, and the measured 20.5 is that value less weighting and flagging losses. This is both the end-to-end check through the unmodified pipeline and what a coarse window returns when detection alone is scored: Class B detects excess power, and cannot tell whether the excess drifted.

Two distinctions apply. The dwell campaign injects into retained per-integration spectra, after the baseline step, while the end-to-end check injects into visibilities before it; and one cell, the through-pipeline recovery of near-static carriers on Class A windows, is measured only in part.

How representative the injected windows are of the survey is set out in §G.2: the end-to-end demonstration is made on one configuration, about half the Class A windows differ from it on at most one axis, and the stratified campaign measures the transfer to the rest and states which windows that bracket interpolates and which it extrapolates.

The injected carriers do not drift, so what is measured across 12 configurations is the persistent and partial-dwell non-drifting class. On Class B windows that is the searched class; on Class A windows the searched class is the drifting carrier, whose curve rests on the single calibration configuration of §G.2. The stacked amplitude a is a derived combination, justified by the form of the statistic and by the collapse of 12 configurations onto one curve, not measured independently at every (A, f_{dwell}) pair.

PER-TARGET RESULTS

The per-window results ship as the comma-separated catalogue of the Data Availability statement. Its $\text{EIRP}_{\text{trig}}$ column uses the definition of §4, on the same threshold and primary-beam correction as the continuum column, so a row’s two limits are comparable; its per-channel flux-density threshold uses the carrier lane’s per-channel noise rather than the tabulated rms_{mJy} , the two differing by $5 \times G(r)$. Bands 6 and 7 supply 91 per cent of the windows.

THE VELOCITY-SPACE LINE MASK

A one-channel line check is too tight for circumstellar gas: a native channel spans a fraction of a

TABLE 14

SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS: GAIA DR3 SOURCES WITH $\varpi > 0$, $\sigma_{\varpi}/\varpi \leq 0.1$ AND $1/\varpi \leq 40$ pc (17566 OBJECTS, OF WHICH 8594 CARRY A `TEFF_GSPPHOT` ESTIMATE, 49 PER CENT), AGAINST THE 89 STARS WITH AT LEAST ONE SEARCHED WINDOW. TEMPERATURES ARE AVAILABLE FOR 53 OF THE 89 (ALL 89 MATCHED TO THE 168-ENTRY ALMA-COVERED CENSUS, 56 BY EXACT NAME; 53 OF THOSE MATCHES CARRY A TEMPERATURE), SO THE TEMPERATURE PANEL IS NORMALISED OVER CLASSIFIED OBJECTS ONLY. “RATIO” IS THE SAMPLE COLUMN FRACTION OVER THE CENSUS COLUMN FRACTION; > 1 IS OVER-REPRESENTATION.

Property	Census	Searched	Ratio
Distance (all 89 searched stars; 17566 census objects)			
$d \leq 5$ pc	48 (0.3%)	12 (13.5%)	49.3
5–10 pc	240 (1.4%)	13 (14.6%)	10.7
10–20 pc	2069 (11.8%)	18 (20.2%)	1.7
20–30 pc	5391 (30.7%)	19 (21.3%)	0.7
30–40 pc	9818 (55.9%)	27 (30.3%)	0.5
T_{eff} class ^a (53 searched; 8594 census)			
A (≥ 7500 K)	24 (0.3%)	1 (1.9%)	6.8
F (6000–7500 K)	402 (4.7%)	9 (17.0%)	3.6
G (5300–6000 K)	860 (10.0%)	16 (30.2%)	3.0
K (3900–5300 K)	1400 (16.3%)	6 (11.3%)	0.7
M (< 3900 K)	5908 (68.7%)	21 (39.6%)	0.6
Other axes			
Known planet hosts ^b	20 (11.9%)	23 (25.8%)	2.2
Searched stars / systems	–	89 / 82	–

^aBins are temperature bins, not MK types: A ≥ 7500 , F 6000–7500, G 5300–6000, K 3900–5300, M < 3900 K. Gaia `teff_gspphot` is unavailable for about half the reference census, preferentially for the coolest and faintest objects, so the census M fraction is a lower limit and the earlier-type ratios are correspondingly upper limits. White dwarfs are not separable in this parameterisation and fall wherever their (unreliable) photometric temperature places them.

^bThe Gaia reference census carries no planet information; the census column here is instead the 168-entry ALMA-covered census, of which 20 entries are known planet hosts. The 89 searched stars include 23 hosts of 50 known planets.

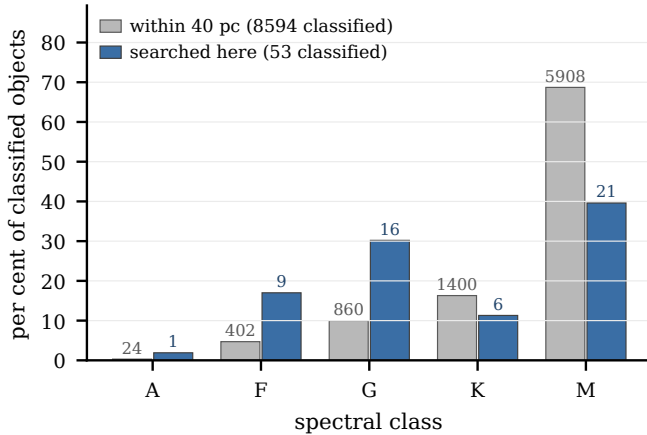


FIG. 12.— The archival selection, which no limit in this paper escapes. Grey: the spectral-class composition of the Gaia DR3 objects within 40 pc that carry a temperature. Blue: the same for the searched sample. Counts are printed above each bar. The sample is enriched in F and G stars by factors of four and three and depleted in M dwarfs by a factor two, so 21 of the 5908 classified M dwarfs within 40 pc carry a searched window. Because M dwarfs host most of the habitable-zone planets in the solar neighbourhood, this is the single most important limitation on what the null result means (§6.5).

TABLE 15

THE ARCHIVAL SELECTION BY SPECTRAL CLASS, INCLUDING THE COVERAGE THAT MATTERS. $N_{40\text{ pc}}$ COUNTS GAIA DR3 OBJECTS WITHIN 40 pc CARRYING A TEMPERATURE; N_{searched} THOSE WITH AT LEAST ONE SEARCHED WINDOW; AND THE LAST COLUMN THOSE WITH AT LEAST ONE *Class A* WINDOW, WHICH IS THE ONLY COVERAGE THAT CAN RESOLVE A DRIFTING CARRIER. 13 M DWARFS HAVE CLASS A COVERAGE, OF 5908 CLASSIFIED WITHIN 40 pc.

Class	$N_{40\text{ pc}}$	N_{searched}	f_{searched}	with Class A
A	24	1	0.0417	1
F	402	9	0.0224	4
G	860	16	0.0186	15
K	1400	6	0.0043	6
M	5908	21	0.0036	13

km s^{-1} , while real emission is shifted by many channels (§5.2). The mask tolerance is set by the kinematic budget of circumstellar gas: Keplerian motion where these molecules survive ($\lesssim 13 \text{ km s}^{-1}$ beyond 10 au) and linewidths $\lesssim 1 \text{ km s}^{-1}$; *The mask is evaluated topocentrically*, in the observed sky frame: all 1396 released `line_offset_MHz` values and all 1396 `line_offset_kms` values reproduce topocentrically against the frozen list and none reproduces from a barycentric or stellar-frame evaluation, and the stellar radial-velocity constant the code carries is read in two files and set in none. Earlier versions of this paper described the mask as evaluated in the stellar frame; that was a documentation error and not a computational one — no attribution changes between the frames at $\pm 50 \text{ km s}^{-1}$ (§F.1) — but it is corrected here rather than carried. The tolerance is wide enough to absorb the barycentric and systemic terms it does not apply. The result is empirically conservative: the largest attributed circumstellar component (β Pic, within 0.39 km s^{-1} of systemic) sits at 0.8 per cent of it. The wider tolerance buys robustness against radial-velocity catalogue error, and what narrowing it would cost is measured in §K.4.3 rather than argued here. The alternative selection quoted in §5.2.6 is every Splatologue transition over the 46 islands with $E_u \leq 150$ K and $\log A_{ij} \geq -5$: 5,114 transitions of 235 species; HCN, CS, SiO, CN and H_2CO are in the adopted list already. It masks 75.1 per cent of the union against the 4.1 per cent the adopted list masks, and changes no disposition, because no masked crossing exceeds its own control ensemble.

Rounding. The rounding defect is why the released CO(1 $\rightarrow$ 0) offset column, referred to the frozen mask’s rounded 115.271000 GHz entry, reads 0.202 MHz (0.53 km s^{-1}) smaller than the laboratory-referred value of Table 22. A cheaper rule exists: the archive records the rest frequency each execution block was tuned to, so masking each window’s own tuned line, inside an LSR-frame tube of a few $\times 10 \text{ km s}^{-1}$, catches the β Pictoris, HD 48370 and [C I] windows automatically.

The crossing ledger, once, and it closes: each of the 56 windows with an on-star crossing falls in exactly one of three bins. 12 also beat their control ring, of which 10 lie inside the $\pm 50 \text{ km s}^{-1}$ mask and 2 do not. 6 more lie inside the mask without beating their ring — 3 of them on HD 285968 alone, at -46.7 to -34.4 km s^{-1} from CO 2 $\rightarrow$ 1, each already rejected by its own ring, which reaches 7.22–8.12 against star peaks of 5.83–5.96. The remaining 38 are ordinary noise crossings: in each, the window’s own

TABLE 16

PER-BAND SURVEY CHARACTERISATION OF THE 1651 SEARCHED WINDOWS. “Sys.” COUNTS UNIQUE STELLAR SYSTEMS SEARCHED IN THE BAND, SO A SYSTEM OBSERVED IN TWO BANDS IS COUNTED IN BOTH ROWS (82 SYSTEMS, 89 STARS IN TOTAL); “EBs” THE CONTRIBUTING EXECUTION BLOCKS, ATTRIBUTED BY THE PROVENANCE OF THE DE-DUPLICATED WINDOWS; $\Delta\nu_{\text{ch}}$ THE NATIVE CHANNEL WIDTH IN MHz (MEDIAN, RANGE IN BRACKETS); “UNION” THE UNIQUE SKY FREQUENCY SEARCHED IN THE BAND AFTER MERGING OVERLAPPING WINDOWS; $\text{EIRP}_{5\sigma}$ THE MEDIAN NOMINAL THRESHOLD ACROSS THE BAND’S WINDOWS; $\dot{\nu}_{\text{ceil}}$ THE RANGE OF DRIFT-RATE CEILINGS IN Hz s^{-1} PER GHz OF CARRIER FREQUENCY. THE “ALL” ROW IS COMPUTED OVER THE WHOLE SAMPLE, NOT SUMMED DOWN THE COLUMNS: EBs AND SYSTEMS RECUR ACROSS BANDS AND THE FREQUENCY UNIONS ARE DISJOINT BY CONSTRUCTION.

Band	Sys.	EBs	Win.	$\Delta\nu_{\text{ch}}$ (MHz)	Union (GHz)	Med. $\text{EIRP}_{5\sigma}$ (W)	$\dot{\nu}_{\text{ceil}}$
3	8	27	112	15.63 (0.244–15.63)	20.6	1.3×10^{14}	12.0–13.3
4	1	1	4	15.63	6.9	2.6×10^{13}	12.0
5	1	1	4	15.63	6.9	4.4×10^{13}	12.0
6	55	265	1072	15.63 (0.015–31.25)	29.4	5.2×10^{14}	12.0–13.3
7	35	102	423	15.63 (0.061–15.63)	26.7	3.6×10^{15}	12.0–13.3
8	3	6	24	15.63 (0.061–15.63)	6.7	4.2×10^{16}	12.0
9	1	1	4	15.63	6.9	4.3×10^{15}	12.0
10	1	1	8	0.977	13.8	2.7×10^{16}	12.0
All	82	404	1651	15.63 (0.015–31.25)	118.1	7.1×10^{14}	12.0–13.3

TABLE 17

BAND-BY-BAND COST OF THE MOLECULAR-LINE EXCLUSION. UNION: UNIQUE SKY FREQUENCY SEARCHED IN THE BAND. MASKED: THE PART OF IT INSIDE A $\pm 50 \text{ km s}^{-1}$ STELLAR-FRAME TUBE AROUND A CATALOGUED TRANSITION OF ONE OF THE MASKED SPECIES. LEFT: WHAT A TECHNOSIGNATURE SEARCH STILL OCCUPY. THE MASK IS EXCLUDED SEARCH SPACE AND NOT A VETO (§5.2.6); THE LOSS IS CONCENTRATED IN BAND 6, WHERE $\text{CO } 2 \rightarrow 1$ SITS.

Band	Union (GHz)	Masked (GHz)	Left (GHz)	Lost (%)
3	20.64	0.272	20.37	1.3
4	6.94	0.049	6.89	0.7
5	6.94	0.065	6.87	0.9
6	29.45	1.374	28.07	4.7
7	26.74	1.589	25.15	5.9
8	6.68	0.164	6.52	2.5
9	6.94	0.454	6.48	6.5
10	13.80	0.844	12.96	6.1
All	118.13	4.811	113.31	4.1

control ring reaches or exceeds the stellar value (ring-maximum median 5.82 against a crossing median 5.18, interquartile 5.1–5.4), the expected Gaussian-tailed population of crossings with comparably strong peaks among 512 controls. None of them is a near miss on the mask: the closest lies 127 km s^{-1} from any catalogued transition of a masked species (LHS 1140, 2.5 times the adopted half-width), and 0 have no such transition in the window at all. But the transitions that make astrophysical false positives cheap are also natural reference frequencies for a deliberate communicator (Cocconi & Morrison 1959), so the released products keep a known-line-coincident list: every on-source threshold crossing inside the masked intervals, with its offset from each nearby transition.

FALSE-ALARM ACCOUNTING

This survey quotes *one* false-alarm expectation. It is built from the control statistics themselves, which are the only empirical null the data provide, and it imposes the condition a stage-1 event must actually satisfy: the position must reach the trigger *and* rank first. Drawing one control index per execution block — so that windows sharing weather, calibration and correlator setup are not treated as independent trials — and resampling blocks, the expectation over the 402 Class A windows is 0.89 unattributed stage-1 events against the 2 observed, and the probability of that count or more is $P(\geq 2) = 0.22$. Clustering changes the number little, because the *position* of a block’s control maximum is not shared across

its windows. Expectations that condition only on rank, without the trigger, predict a different quantity — the number of windows in which a no-signal position would rank first — and are pipeline diagnostics rather than a statement about candidates; they are not quoted as chance expectations for candidates.

The remainder of this appendix sets out what that expectation rests on: the conditioning, a second treatment of the same quantity that carries the measured non-exchangeability as a tail factor, the exchangeability of the ensemble itself, what its resolution is worth, and the trials budget.

What the expectation is conditioned on

First, which chain the expectation is conditioned on — and it is no longer the one this subsection was written for. A stage-1 event requires the stellar position both to reach the trigger and to outrank every control, and 0.89 is the expectation for exactly that. Under the chain the paper now states, however, the rank screen is a consistency check rather than a filter (§5.2.1), so the relevant expectation is conditioned on trigger, localisation and attribution instead. That is a different number and it is re-derived rather than relabelled, because the rank clause is a one-in-(512 + 1) filter and removing it cannot be a change of wording: 36.5 Class A windows are expected to trigger by chance (§5.2.2), the visibility fit localises 0.985 of events at $T_{\star} \approx 5$ — 0.977 through all three clauses, so the factor is the same either way — and the frozen mask occupies 4 per cent of the searched band, giving *34.4 expected unattributed Class A crossings against 35 observed*. That is a factor 39 larger than 0.89 and it accounts for the observed population outright. *0.89 remains correct for what it measures and is quoted below only for stage-1 events, which are defined by the rank screen.*

The stage-1 expectation must be conditioned on both trigger and rank: exactly one probe per window is the ensemble maximum, and the probability that the stellar position would have been that maximum, treated as exchangeable with the probes, is $1/(512 + 1)$ in windows whose control maximum reaches $T_{\text{trig}} = 5$ and zero in the rest. The condition is restrictive and informative: only 447 of the 1614 windows that retain a control vector — the subset of the 1651 searched for which the per-window ensemble was stored — have a control maximum reaching the trigger at all, and 390 of those are Class A. *This*

is why every stage-1 event in the survey is Class A, and the choice of reference class largely dissolves once the trigger condition is imposed. An expectation built on the rank alone predicts a different quantity, the number of windows in which the star would *rank first*, which is why rank-only figures are larger.

The same quantity admits a second treatment, which is quoted here for comparison and not as a second expectation. Resampling execution blocks with replacement, and additionally applying the measured non-exchangeability of the stellar position as a factor $\times 1.4$, gives 1.3 expected over the 1614 windows with stored ensembles, 95 per cent interval 0–4, against 0.89 without the tail factor. The difference is the tail factor and nothing else, and that factor is itself known only to 1.2–1.9; the probability of the observed count or more under it is 0.4 to one significant figure, and no inference is drawn from either. The count cannot be dismissed as an artefact of the screen’s position-dependent rank bias: that bias places the star *low* in its control distribution, and a bias in that direction suppresses star-first outcomes rather than manufacturing them. Neither number is evidence about any individual event.

An excess this size needs only a small unmodelled tail in the control ensemble, and §C.1.2 shows that one exists. The same noise model sets the recovery fractions of §G.1, however, so a tail large enough to remove the excess would make the quoted completeness optimistic by roughly the same factor; the two cannot be traded independently, and neither correction has been applied.

The control ensemble: its geometry, its calibration and its resolution

The screen’s geometry is set in §4.1. Its calibration, its resolution and what both cost are collected here.

Under that geometry — the repaired one, carried by 1053 of the 1054 ACA windows in the released catalogue — the uniform distribution exchangeability implies is not rejected in either stratum: $D = 0.025$ ($p = 0.85$) over 597 12 m windows and $D = 0.032$ ($p = 0.23$) over 1048 ACA windows, whose median rank is 0.478 against the expected 0.5. Both tests run on the released catalogue: 1645 of the 1651 windows, the remaining 6 having no stored control vector.

Two cautions keep that from being a proof of exchangeability. At these sample sizes the Kolmogorov–Smirnov test resolves only $D \gtrsim 0.042$, so a value below it is consistent with uniform rather than demonstrably uniform. And the diagnostic most directly sensitive to contamination of the inner controls — the excess of the innermost control bin over the outer annulus — is $+0.038$ with a bootstrap interval of -0.214 to $+0.282$, which spans zero and therefore neither confirms nor excludes a residual effect. The screen is used to decide which windows are examined and never as a significance.

Among the ACA windows the screen returns 3 stage-1 events. 2 are molecular — foreground carbon monoxide toward HD 48370 ($T_\star = 26.8$, -34 km s^{-1} from CO(2 $\rightarrow$ 1)) and the β Pictoris disc ($T_\star = 9.8$, -13 km s^{-1}), the survey’s positive control — and 1 is unattributed, 61 Vir ($T_\star = 6.0$, $+449 \text{ km s}^{-1}$ from the nearest catalogued transition). Every window retains the frequency and drift rate at which its statistic peaked, so each event can be re-tested at its own cell.

Neither form is a significance test. Ranking the star against 512 control positions drawn from the same field is a powerful way to decide which windows deserve examination, but $1/(512+1) = 1.95 \times 10^{-3}$ is the *resolution* of the ensemble, the smallest rank it can express, and not the probability that noise produced the window. Against a survey-wide Bonferroni scale

$$\alpha_{\text{survey}} = \frac{0.05}{1651} = 3.0 \times 10^{-5}, \quad \frac{1.95 \times 10^{-3}}{\alpha_{\text{survey}}} = 64, \quad (\text{J1})$$

that floor is 64 times too coarse (Appendix J), so a technosignature claim here requires independent recurrence or independent astrophysical confirmation and never rank alone.

Why no single window in this design can be significant

The rank floor of Eq. J1 is a structural limit on what any survey built this way can claim, and enlarging the control ensemble does not rescue it: the primary beam bounds how many independent control positions a field contains. That bound can be made explicit. The 512 controls are drawn uniformly in the annulus, so many of them land within one synthesised beam of a neighbour and measure the same noise. Requiring every control–control and control–star pair to be at least 2 synthesised beams apart — selecting greedily in the order the positions were drawn, which is fixed before any statistic is seen — leaves a median of 229 independent controls in the 12 m windows but only 13 in the ACA windows, whose beams are an order of magnitude wider (Table 18). The rank resolution these support is $1/(N_{\text{ctrl}}+1) = 0.004$ and 0.071 respectively, and the ACA figure is 37 times coarser than $1/(512+1)$ suggests.

Two things follow, and the second is the more useful. The reduced ensemble is exchangeable: the add-one rank over the 1644 windows it can be formed for has median 0.500 against the expected 0.5, with $D = 0.028$ ($p = 0.14$) once the rank is dithered within its own cell, which it must be — with a dozen controls the rank takes a dozen values, and comparing that to a continuous uniform manufactures $D = 0.052$ out of the quantisation alone. And the separated screen is *more* permissive, not less: 28 windows outrank all of their separated controls where 12 outranked all 512, 16 gained and 0 lost. Fewer controls resolve less, so a window clears them more easily: the ensemble that looks most significant is the one with the least power.

The survey’s own data show what a visibility-domain statistic buys instead. In the image plane the 12 stage-1 windows are indistinguishable: every one has zero of the 512 control positions at or above the star, so every one sits at rank 1 of $512+1$, equivalent to 2.9σ , with no dynamic range left to separate them. The visibility fit separates the same windows by 5.2σ , and its strongest reaches 9.5σ , a factor of 3.3 beyond anything the rank statistic can express whatever the data contain.

Hence the design adopted here, in which a technosignature claim rests on independent recurrence or independent astrophysical confirmation and the screen decides only which windows are examined. It also sets a target for successor surveys: raising the resolution floor requires either many more independent control positions per field than a single pointing affords, or a visibility-

TABLE 18

WHAT THE CONTROL ENSEMBLE IS WORTH ONCE ITS MEMBERS ARE REQUIRED TO BE INDEPENDENT.

CONTROLS ARE DRAWN UNIFORMLY IN THE ANNULUS, SO THE 512 POSITIONS OVERSAMPLE A FIELD THAT HOLDS FAR FEWER RESOLUTION ELEMENTS. SELECTION IS GREEDY IN DRAW ORDER, WHICH IS FIXED BEFORE ANY STATISTIC IS SEEN; GREEDY PACKING IS NOT MAXIMAL, SO EACH COUNT IS A LOWER BOUND. THE DENOMINATOR IS 1644 OF THE 1651 RELEASED WINDOWS — 593 12 M AND 1051 ACA AGAINST 597 AND 1054 IN THE SURVEY — THE REST NOT STORING THE SEED AND BOTH ANNULUS RADII NEEDED TO RECOVER THEIR CONTROL DRAW.

Array	Windows	Independent controls (median, of 512)	Rank resolution $1/(N_{\text{ctrl}} + 1)$
12 m	593	229	0.004
ACA 7 m	1051	13	0.071
All	1644	13	0.071

domain statistic, which does not inherit the floor at all.

Table 19 gives the trials budget in cells and decomposes the on-star excess. A window holds $177\text{--}6\,869\,740$ channel $\times$ drift cells (median 468), the survey 1.32×10^8 in total. Correlation cannot move the expected *number* above the trigger, which is Np whatever the correlation, only its clumping, so the excess over an iid-Gaussian budget decomposes into two multiplicative parts: cell multiplicity within a crossing window, a factor 1.4 which is what an over-count of 2.67 on the frequency axis predicts, one real threshold excursion being counted about three times; and the count of crossing windows, a factor 1.3 whose one-sided Poisson probability is 0.04. Once the 4 windows carried through the scramble validation below are set aside, the residual excess is consistent with chance at that level.

The control ensembles calibrate the budget without a Gaussian assumption. The same arithmetic predicts 476 of the 1651 windows carrying a control above T_{trig} and 455 do, agreeing to 4 per cent, with per-window control maxima tracking the iid-Gaussian prediction at a median ratio 0.995 (5th–95th percentile 0.91–1.11). Grid cells are not independent trials: Hanning smoothing correlates neighbouring channels at $\rho_1 = 2/3$ and $\rho_2 = 1/6$, and the drift grid is oversampled two to one, giving the combined over-count 5.3 of Table 19. Correcting for it leaves the observed control maxima +6 and +8 per cent above prediction in the fine and coarse classes independently. Non-Gaussianity of that size is why the operative statement is the empirical rank and not the Gaussian budget. The residual on-star excess is not zero, runs one way and carries the same sign at the controls, so it is reported as an open systematic: the crossing list is dominated by CO(2 $\rightarrow$ 1) at 230.5 GHz toward disc hosts, and a residual excess of chance crossings at the tens-of-per-cent level cannot be excluded.

All of this fixes what a single window can ever mean. The rank floor is $1/(N_{\text{ctrl}} + 1) = 1.95 \times 10^{-3}$ against a survey-wide Bonferroni scale of 3.0×10^{-5} , a factor 64, and correlation widens it: with $N_{\text{eff}} \simeq 30\text{--}43$ independent spatial trials the ensemble resolves only 0.023–0.032. Drawing more controls would not help, the primary beam bounding the annulus. This is the design’s central limitation, and is why authentication rests on recurrence and astrophysical attribution rather than on rank.

The *original* extraction placed the control annulus using a 12-m θ_{PB} for every window, so the 1054 ACA 7-m

TABLE 19

THE ON-STAR CELL EXCESS, DECOMPOSED. EVERY COUNT HERE IS TAKEN OVER THE 56 THRESHOLD CROSSINGS OF TABLE 1 WITH THE 12 STAGE-1 WINDOWS REMOVED, WHICH IS WHY THE CROSSING-WINDOW ROW READS 44 AND NOT 56. TOP: WHAT IS OBSERVED AGAINST AN IID-GAUSSIAN BUDGET. MIDDLE: THE TWO MULTIPLICATIVE PARTS OF THAT FACTOR. BOTTOM: THE SAME CORRELATION CORRECTION APPLIED TO THE CONTROL MAXIMA, WHICH ARE MEASURED IN THE SAME UNITS AND CALIBRATE THE BUDGET EMPIRICALLY.

Quantity	Value	Factor
On-star cells $\geq T_{\text{trig}}$	219	
in the stage-1 windows	206	
remaining, unattributed	13	
iid-Gaussian expectation (cells)	37.8	0.3
Cells per crossing window (obs./exp.)	1.6/1.1	1.4
predicted by Hanning correlation	2.67	
Crossing windows outside stage 1	44/33.2	1.3
(obs./exp.)		
one-sided Poisson p	0.04	
Frequency-axis over-count $1 + 2\rho_1 + 2\rho_2$	2.67	
Drift-grid oversampling	2	
Combined cell over-count		5.3
Control max., fine: pred. $\rightarrow$ corrected	5.74 $\rightarrow$ 5.45	
observed	5.77	+6 %
Control max., coarse: pred. $\rightarrow$ corrected	4.47 $\rightarrow$ 4.10	
observed	4.44	+8 %

TABLE 20

ARRAY-SPLIT ROBUSTNESS, AS A CHECK THAT NOTHING SURVEY-LEVEL RESTS ON THE STRATUM WHOSE CONTROL GEOMETRY THE ORIGINAL EXTRACTION PLACED WITH A 12-M θ_{PB} ; THAT GEOMETRY HAS SINCE BEEN REPAIRED FOR 1053 OF THE 1054 ACA WINDOWS IN THE RELEASE. EXPECTED RANK-FIRST COUNTS ARE $N_{\text{win}}/(512 + 1)$ UNDER EXCHANGEABILITY, AND ARE THEREFORE LOWER BOUNDS: THE MEASURED TAIL RATE IS A FACTOR 1.4 HIGHER (§K.1.3), WHICH BRINGS THE 12 M STRATUM’S 9 OBSERVED WITHIN ONE OF EXPECTATION. NO DISPOSITION AND NO HEADLINE THRESHOLD DEPENDS ON THE ACA WINDOWS.

	12 m	ACA 7 m
Windows	597	1054
Execution blocks	139	265
Stars; systems	69; 62	38; 38
Threshold crossings	21	35
Rank-first windows	9	5
expected	1.16	2.05
Stage-1 outliers	9	3
Stellar-rank $D(p)$	0.028 (0.72)	0.03 (0.186)

windows carried a geometric error. That error has since been removed at source rather than bounded: the affected blocks were re-fetched and re-searched on the geometry measured from each observation’s own antenna table and baselines (§K.4.2), and 1053 of the 1054 ACA windows in the released catalogue now carry the repaired control geometry, the remaining 1 (towards LHS 1140) falling outside the re-extracted set. The stratified figures below are quoted for completeness: on the true 7-m beam the original annuli spanned 0.08–0.46 of θ_{PB} at a median area-weighted gain of 0.75 against 0.47 for the 12-m windows, so the error ran in the conservative direction even before it was repaired. Table 20 gives the two strata side by side. The 12 m stratum holds 9 of the 12 stage-1 flagged windows, the one lost being HD 48370’s foreground CO, and the deepest trigger and the deepest P_{eff} are both 12 m windows, so no survey-level statement rests on the ACA windows either way.

Rank calibration and exchangeability

The drift grid and the thousands of channels apply identically at the star and at every control, so the per-

window trials factor is absorbed empirically: under exchangeability one designated position of 513 ranks first with probability exactly $1/513$ whatever the noise distribution, and a heavy tail cannot by itself produce an excess of rank-first windows. The survey expectation is 3.22 and 14 are observed; the execution-block clustered permutation returns $P(\geq 14) = 7.0 \times 10^{-6}$, against 7.6×10^{-6} for a plain Poisson of the same mean. *The two are computed independently, not copied from one another*: at this mean the clustered convolution and the Poisson tail agree to 8 per cent (ratio 0.92), which the regeneration script asserts, so block clustering does not change the answer.

The arithmetic that follows was wrong in earlier versions of this paper, and correcting it reverses what this paragraph concluded. Those versions stated that identified line emission accounted for 3 of the 14 rank-first windows and that the remaining 2 were “far too many for chance”, which neither subtracts correctly nor compares in the right direction. Recomputed from the released catalogue: 12 of the 14 rank-first windows also reach the trigger, and 10 of them carry an identified CO attribution at the stellar position (Table 5). Removing those leaves 4 against 3.22 expected, $P(\geq 4) = 0.40$ — *entirely consistent with chance*. **The rank-first excess is therefore explained by astrophysics, and is not by itself evidence that the rank is non-exchangeable.**

The claim that the rank is not exchangeable stands, but on evidence independent of the crossings and out of sample: on windows the frozen pipeline had never seen, the stellar add-one rank has median 0.405 where 0.5 is expected, and the displacement survives block clustering (Appendix C.1.1). The star-versus-control rank is an empirically useful candidate-prioritisation statistic, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection.

Ranking against 512 controls cannot resolve a probability below $1/513$, so a local scramble test carries 4 of the 12 stage-1 windows past that floor. It scrambles each integration independently in channel and re-runs the identical search with the same noise map, weights and drift grid, destroying anything coherent along a drift track while preserving each integration’s own spectral autocorrelation and trials factor. 3 of the 4 pass validation. The two β Pictoris windows behave as positive controls, no draw of 10^5 reaching $T_\star$ in either ($p < 1 \times 10^{-5}$ and $p < 5 \times 10^{-5}$); for CP–72 2713 the null resolves the value itself, 157 draws of 10^5 equalling or exceeding $T_\star$, with an extreme-value fit to that window’s own control maxima agreeing to a factor 1.3. HD 48370 Band 6 fails outright, 73 per cent of its real controls exceeding the maximum of 10^5 null draws, because the field carries extended emission and imaging residuals that are coherent across integrations and that the scramble destroys by design, so no scrambled-noise probability is quoted there. That failure follows from retaining full dynamic spectra for only 8 of the 512 control positions. No number quoted here rests on a parametric extrapolation of a null tail, and the diagnostic has not been re-run on the windows the completed sweep added (§L).

Exchangeability is the statistic’s central assumption.

For a debris-disc archive its dominant failure is resolved astrophysical emission filling the annulus, seen in HD 48370’s CO ring and in β Pictoris’ recurring Band 6 window, where the ring outshines the star by $1.5\times$. That suppresses a star-exceeds-ring outcome instead of manufacturing one, so it costs sensitivity and nothing else, and the repeat block of the *flagged* Band 6 window measures the cost (§D.2.2). The primary beam is settled by construction, the statistic being formed on *uncorrected* amplitudes whose thermal noise does not depend on position with $\sigma(t, \nu)$ one scale shared by the star and every control (§4.1). The 513 positions are therefore approximately exchangeable under a noise-only null, a statement that covers the thermal noise and the beam but not residual calibration or continuum systematics, which can occur preferentially where the pointed science target sits.

Four measurements bound what is left, and all four are null. Ranking each control against the other 512-1 by the rule used for the star gives 220 672 pseudo-star ranks, uniform to the resolution the ensemble can express ($p = 0.37$) in every band and at both channelisations. That geometry cannot by itself detect a centre-versus-annulus asymmetry, so the test that can is the two-sample comparison of the 1651 real stellar ranks against the pooled pseudo-ranks, which returns $p = 0.38$: on these data the stellar position behaves like one more position in its own annulus, though out of sample it does not (§C.1.1). The Spearman correlation of control statistic against reconstructed radius has median 0.002 over the 1187 crossing-free windows, bounding any systematic radial suppression of the controls below 10 per cent in the mean. And the star’s own residual variance is measured directly against the controls’ in §K.1.2. Per-control noise records and gain are absent from the frozen products, so those two stratifications are bounded only through these rank distributions.

One stratum departs. The fine-channel windows fail the $U(0, 1)$ rank test at $p = 0.016$ ($n = 402$), because 12 of them contain an on-star peak above their own control maximum against 0.78 expected, and those are precisely the flagged windows of Table 5. The coarse stratum, three times larger and predicting 2.43 such windows, contains 0: all 12 stage-1 flagged windows in the survey are drift-resolving, with $\eta_{\text{drift}} = 6.9\text{--}422$ (median 20), as are all 52 Class A threshold crossings. That is a measured statement about those two populations and not a property of the class: 13 of the 402 Class A windows have $\eta_{\text{drift}} < 1$ (§4). A 15.6–31.25 MHz channel dilutes a narrow circumstellar line, so the class that can resolve one is the class that finds it, and the same reasoning says a genuine unresolved spectral carrier would appear in Class A first. The departure is nonetheless genuine, and it survives removal of every fine window carrying a stellar crossing (Fig. 13), so fine-window rank probabilities are optimistic for the class as a whole and not only where a crossing sits. This is the quantitative reason the spatial comparison is used as a prioritisation and never as a test of the null (§4.1).

A fully symmetric region-max variant, which maximises identically at the star and at every control, can be run on only seven locally reprocessed windows, the frozen release not carrying the products it needs; in all seven the star is exchangeable with its ring (Appendix K).

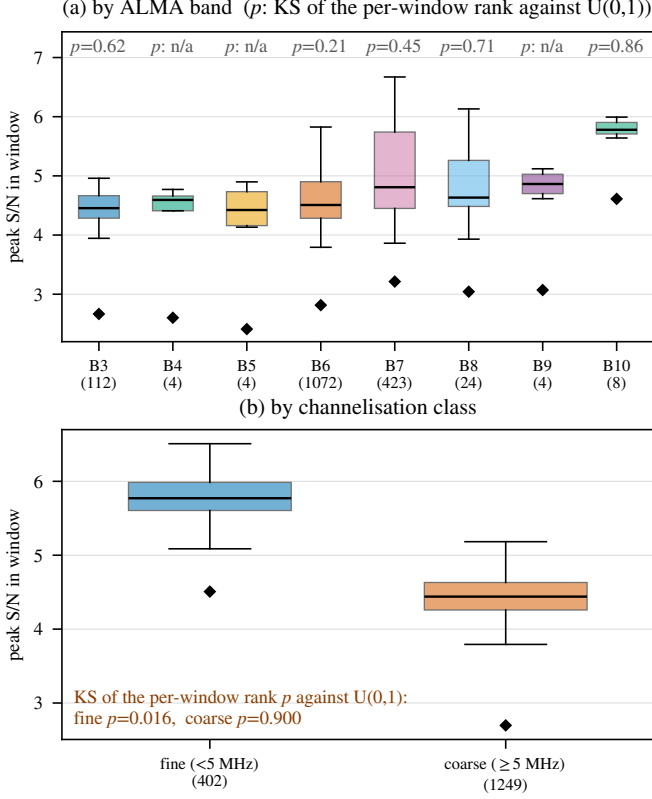


FIG. 13. — Exchangeability of the 512-position control ensemble by stratum, over 1651 windows. Boxes give the per-window control maximum, black diamonds the median on-star peak, and p the Kolmogorov–Smirnov test of the per-window rank against $U(0,1)$. (a) by ALMA band, with the window count under each label; (b) by channelisation class, fine (< 5 MHz) against coarse. The ensemble tracks the on-star statistic in every band, and the fine class sits higher than the coarse, 5.77 against 4.44, as its larger trial count predicts. The fine class departs from uniformity, $D = 0.077$ ($p = 0.016$); the coarse class does not, $D = 0.016$ ($p = 0.900$). The departure is not the flagged windows’ doing — removing all 52 fine crossings lowers the statistic only to 0.086 — and §C.1.2 identifies the radial noise gradient as the cause.

The single-position calibration above remains the release-wide criterion.

ANALYSIS PROVENANCE AND ROBUSTNESS CHECKS

The β Pictoris frame audit

Table 22 carries the frame chain for the line-attributed windows: frequencies match the catalogued rest values to the chain’s stated tolerance, velocities are concordant across epochs and transitions, the emission localises against the control ensemble, and the drift class is zero to within the grid. Two further properties fit CO and not a carrier. The stage-1 Band 6 window carries 114 threshold crossings where a channel-confined carrier admits one, a floor of 2.26 km s^{-1} on the feature’s extent; and the peak flux densities, 76 and 78 mJy per channel, are the right order for this belt’s spatially integrated CO, their ratio depressed below optically thin LTE by beam dilution on emission resolved in both beams and by the subthermal excitation of Matrà et al. (2017).

Stellar emission, the competing explanation towards an active young star, is suppressed twice: chromospheric and coronal gyrosynchrotron and free-free emission are broadband — the AU Mic flares of MacGregor et al.

(2020) are coherent across the full 8 GHz — whereas the block-median baseline of §4.1 removes everything broader than its 65-channel passband ($\sim 32 \text{ MHz}$ at the fine class), and the inverse-variance stack over all integrations dilutes time-confined flares. What survives is the multiplicative route of §4.1, a flare continuum times a spectral calibration residual, bounded there and measured for CP–72 2713 in §5.2.

The carbon tuning agrees by omission. All 24 Band 8 windows belong to 3 stars and are $[\text{C I}] \ ^3P_1\text{--}^3P_0$ (492.160651 GHz) tunings. In the corrected extraction no Band 8 window toward HD 48370 reaches the trigger, so the disc shows no line at the star in its carbon tuning, as Cataldi et al. (2023) report. The crossing peak in the flagged window is 0.82 Jy in one 122-kHz channel, with the ring at the same level.

The survey rescored under a repaired screen

Two corrections were built from the stored control vectors and rescored over the 443 windows of the 1651 searched that retain a control vector, and the 5 stage-1 windows inside them. *Radius matching* ranks the star against only the 54 probes with $u \leq 0.30$, at a cost in resolution; *detrending* keeps all 512 probes and subtracts the pooled radial profile $m(u)$ measured on the calibration windows, at unchanged resolution, its one free choice being the stellar debit $m(0)$ between 0.15 and 0.32σ . Table 25 therefore reports the debit each flagged window survives rather than adopting a value.

Neither correction is adopted, for two reasons. Neither promotes a window: under detrending the flagged set stands at 4 windows (Table 25), and the unattributed count is one for every repair and every debit between zero and 0.61σ . And neither makes the rank distribution uniform: detrending moves the calibration-sample median from 0.429 to 0.437 ($D = 0.091$ to $D = 0.081$) and the median over the windows it can be applied to from 0.485 to 0.499 ($p = 0.69$ to $p = 0.34$), while a leave-one-star-out fit over the 27 calibration stars returns median 0.437 with $D = 0.081$ against 0.437 and 0.081 in sample, so the profile is neither over-fitted nor sufficient to restore exchangeability. The survey’s false-alarm budget is therefore taken from the measured tail rate below.

Noise at the stellar position

The noise scale of Eq. 2 is built across the annulus, so it carries no variance peculiar to the stellar position, where residual continuum, deconvolution or self-calibration residuals and position-correlated calibration error would concentrate. That is the method’s largest untested assumption, and the retained products bound it. R_σ , the star’s robust residual scale over the 4 stage-1 windows that retain full control vectors, measured against the median of the 8 retained control positions, runs 0.995 to 1.001 : the star agrees with its ring to better than 0.7 per cent at 95 per cent confidence. That matters most for CP–72 2713, where an unmodelled broadband excess of 2.2 per cent at the star would push $T_\star$ below the ring maximum; the bound is a factor of three smaller, so that explanation is disfavoured without being excluded. On a seeded random sample of 25 *unflagged* windows the same test returns 0.997 ($0.968\text{--}1.004$), every one within 3 per cent of unity. It tests variances and not means, and is blind to the one- or two-channel artefact the search is

TABLE 21

WHEN EACH ANALYSIS DECISION WAS FIXED, AND WHAT HAD BEEN SEEN AT THE TIME. THE COLUMN THAT MATTERS IS THE LAST ONE. TWO DECISIONS WERE TAKEN WITH THE RELEVANT DATA ALREADY INSPECTED, AND ARE MARKED AS SUCH; THE RATES AND PROBABILITIES THIS PAPER QUOTES ARE THEREFORE *post hoc* THROUGHOUT, AND NO CLAIM OF PRE-REGISTRATION IS MADE FOR ANY OF THEM. THE HOLD-OUT IS THE ONE COMPONENT RESERVED BY A RULE WRITTEN AND COMMITTED BEFORE THE WINDOWS IT SELECTS WERE EXAMINED.

Decision	Fixed at	Data already seen when it was fixed
Detection statistic T_* : one position against 512 single-position controls	Before the survey search	None of this survey’s windows
Hold-out: 77 blocks reserved by a hash rule, committed c75069040eab on 2026 September 14	Before those blocks were extracted	The rule selects on block identifier alone and was committed before the selected blocks were searched
Trigger level $T_{\text{trig}} = 5$	Before the survey search	None of this survey’s windows
Molecular-line mask half-width $\pm 50 \text{ km s}^{-1}$	After inspection	The β Pictoris crossings had been examined (§5.2.6)
Control-annulus inner edge max($0.14 \theta_{\text{PB}}$, $2.0 \theta_{\text{syn}}$) and the criterion for accepting it	Written down before the affected blocks were re-extracted	The stellar-rank distribution of the <i>original</i> extraction
Radius-correction acceptance level $p \geq 0.05$	Before the out-of-sample test was run	The in-sample rank distribution
Injection ladder and the single recovery criterion of §5.4	Before the campaign was run	The windows’ noise, not their recovery

TABLE 22

LINE-ATTRIBUTION AUDIT THROUGH THE FRAME CHAIN OF §F.1: THE TWO LINE-ATTRIBUTED WINDOWS WHOSE FRAME CHAIN IS AUDITED HERE, AND A CLASS A β PICTORIS BAND 6 WINDOW THAT CARRIES NO STAGE-1 FLAG YET SHOWS THE SAME CO COMPONENT. AN AU MIC BAND 6 ROW APPEARED HERE IN EARLIER VERSIONS AT $+378.8 \text{ km s}^{-1}$ IN THE STELLAR FRAME; THAT IS FAR OUTSIDE THE $\pm 50 \text{ km s}^{-1}$ MASK, *no AU Mic attribution exists in the results*, AND THE ROW HAS BEEN REMOVED. THE 10 ATTRIBUTED STAGE-1 WINDOWS ARE LISTED IN TABLE 5; THIS TABLE AUDITS THE FRAME CHAIN ON TWO OF THEM AND IS NOT A SECOND LIST. TOPOCENTRIC OFFSETS ARE OBSERVED MINUS CATALOGUED LABORATORY REST FREQUENCY (PICKETT ET AL. 1998), SO A NEGATIVE OFFSET MEANS THE OBSERVED FREQUENCY LIES BELOW REST. THE BARYCENTRIC CORRECTION IS THE EARTH-EPHEMERIS TERM AT THE EXECUTION-BLOCK MID-TIME, SIGNED SO THAT THE STELLAR-FRAME OFFSET IS THE TOPOCENTRIC OFFSET *minus* THIS COLUMN *plus* v_{sys} . FOR β PICTORIS WE ADOPT $v_{\text{sys}} = +20.0 \pm 0.7 \text{ km s}^{-1}$ (MATRÀ ET AL. 2017) RATHER THAN THE *Gaia* DR3 VALUE OF $+16.84 \text{ km s}^{-1}$ (GAIA COLLABORATION ET AL. 2023), BECAUSE THE STAR IS AN A6V ROTATOR INSIDE THE HOT-STAR TEMPLATE REGIME IN WHICH DR3 RADIAL VELOCITIES CARRY A KNOWN SYSTEMATIC (BLOMME ET AL. 2023).

Flag	UTC start	f_{obs} (GHz)	Transition	Topo. offset (MHz; km s^{-1})	Bary. corr. (km s^{-1})	v_{sys} (km s^{-1})	Stellar-frame offset (km s^{-1})
β Pic B3	2022-03-03 21:49	115.260326	CO(1 $\rightarrow$ 0)	-10.88; -28.29	-7.90	+20.0	-0.39
β Pic B6	2019-03-12 00:14	230.516128	CO(2 $\rightarrow$ 1)	-21.87; -28.44	-8.15	+20.0	-0.29
β Pic B6 [†]	2013-10-06 06:23	230.528134	CO(2 $\rightarrow$ 1)	-9.62; -12.51	+7.63	+20.0	-0.14

[†]Class A Band 6 window of a different execution block, without a stage-1 flag (Appendix H), shown because it independently shows the same CO component.

TABLE 23

ACTIVITY AND MULTIPLICITY STATUS OF THE 2 HOSTS OF UNATTRIBUTED STAGE-1 EVENTS, FROM SIMBAD OBJECT-TYPE LISTS QUERIED 2026 SEPTEMBER 26; DISTANCES ARE INVERTED PARALLAXES. ACTIVITY INDICATORS ARE X-RAY, EMISSION-LINE, FLARE AND ROTATIONAL-VARIABILITY CLASSIFICATIONS; “NONE CATALOGUED” MEANS SIMBAD CARRIES NO SUCH FLAG, NOT THAT THE STAR IS KNOWN TO BE INACTIVE.

Host	Sp.	d (pc)	Activity	type	Mult.
61 Vir	G6.5V	8.5	rot. var.	PM*	none
CP-72 2713	K7Ve	36.7	em.-line, rot. var., X-ray	PM*	none

built to find, which the local null and recurrence address instead.

The measured false-alarm rate

Because the body of the rank distribution is displaced, $1/(512 + 1)$ is the rank *resolution* of the ensemble and not a per-window null probability, and the survey’s false-alarm budget is quoted from measurement. Two independent measurements agree: of the 599 calibration windows carrying no attributed line, 1 exceed all 512 of their controls, one in 599 or 0.9 times exchangeable, and the pseudo-stars, which cannot contain a signal at all, rank first in 0.28 per cent of out-of-sample trials against 0.19

TABLE 24

THE STAGE-1 STAR-BAND PAIRS, SCORED UNDER BOTH STATISTICS. S_{reg} : REGION MAXIMUM OF EQ. 2 OVER $n_{\text{src}} = 135$ POSITIONS; T_* : THE SYMMETRIC SINGLE-POSITION STATISTIC; ‘CTRL. MAX’: THE LARGEST CONTROL STATISTIC, WHICH IS THE SAME ENSEMBLE UNDER EITHER. ALL 5 ARE FLAGGED BY THE SYMMETRIC STATISTIC; THAT IS HOW THE ROWS WERE CHOSEN. ν IS EACH WINDOW’S *centre*; THE CROSSING FREQUENCY DIFFERS BY UP TO 0.85 GHz AND TABLE 5 CARRIES ν_{cross} . FOR CP-72 2713 THE REGION MAXIMUM ALREADY LAY ON THE STAR, SO THE TWO COINCIDE. THE REGION-MAXIMUM STATISTIC PROMOTES 34 FURTHER STAR-BAND PAIRS THAT THE SYMMETRIC STATISTIC DOES NOT, AND 0 IN THE OTHER DIRECTION.

Star	Band	ν (GHz)	S_{reg}	T_*	ctrl. max
β Pic	3	115.2604	36.46	30.76	7.34
HD 48370	6	230.7305	27.68	26.83	18.36
β Pic	6	230.5439	13.51	9.83	8.20
61 Vir	7	345.7926	6.20	5.96	5.65
CP-72 2713	7	345.1152	5.81	5.81	5.68

per cent exchangeable, a factor 1.4.

The uncertainty on that factor is not binomial, because the 16 inner probes of one window share its noise realisation and its execution block. Over all 1326 calibration windows the stored control vectors give 63 first ranks in 21 216 trials, a factor 1.5, and resampling whole windows, or whole execution blocks (327 of them), widens the 95 per cent interval barely at all, 1.2–1.9 and 1.2–

TABLE 25

THE SURVEY RESCORED UNDER BOTH REPAIRS. “SURVIVES” IS THE LARGEST STELLAR DEBIT $m(0)$, IN UNITS OF EACH WINDOW’S OWN CONTROL SCALE, AT WHICH THE STAR STILL EXCEEDS EVERY DETRENDED CONTROL; IT IS A PROPERTY OF THE WINDOW AND ASSUMES NEITHER DEBIT. FOR REFERENCE THE CONSERVATIVE DEBIT IS 0.15σ AND THE DEBIT THAT CENTRES THE CALIBRATION-SAMPLE RANK DISTRIBUTION IS 0.32σ , AND A WINDOW SURVIVES WHICHEVER DEBIT IS SMALLER THAN ITS ENTRY. NO WINDOW IS ADDED BY EITHER REPAIR.

Window	Survives (σ)	Rad.-m.	Detr.
β Pic B3	25.77	yes	yes
β Pic B6	10.80	yes	yes
HD 48370 B6	0.15	yes	yes
CP-72 2713 B7	0.61	yes	yes
Flagged, whole survey		6	4
Added		2	0
Expected by chance		8.1	0.9

1.9 against 1.2–1.9 binomial. The factor lies between 1.2 and 1.9, with the released-product value 1.4 inside that interval, and no disposition changes anywhere in it.

Propagated over the survey’s 1651 windows, the pseudo-star rate predicts 4.7 *rank-first* windows, $P(\geq 1) = 99$ per cent, against 3.22 under exchangeability. That is a rank-only rate: it does not impose the trigger, so it predicts rank-first windows and *not* candidates, and it is not this survey’s chance expectation for an unattributed stage-1 event, which is 0.89 (Appendix J). Every per-window probability in this paper should be read as multiplied by 1.4, and none was claimed significant after the survey-wide correction of Appendix J. The tail is mildly anti-conservative where the body is strongly displaced, and it is the tail the candidate screen uses. The displacement is not one configuration’s doing: 23 of 27 stars have a median rank below 0.5 (Fig. 14). The calibration sample covers 56 stars and windows within a block are not independent epochs, so the sizes above carry that caveat; at 578 null windows the test resolves a shift of this size ($D \geq 0.06$ against 0.08 observed).

Audit against a region-maximum statistic

The statistic of Eq. 2 evaluates star and controls identically. A region maximum over $n_{\text{src}} = 135$ positions at the star, ranked against single-position controls, is not exchangeable with them and ranks first with probability $n_{\text{src}}/(n_{\text{src}} + 512) = 20.9$ per cent rather than $1/(512 + 1)$, a property of the estimator class. All 1651 windows are scored under both forms and both columns are released per window. The symmetric criterion flags a strict *subset*, 12 windows against 68 in 40 star-band pairs, with no window flagged by the symmetric form alone (Table 24); those that drop out do so because the stellar position never exceeded its own control ensemble. AU Mic is the clearest case: the excess lies in the field around the star rather than at it.

50 stars hold further public blocks this survey did not take (§3), which §6 turns into an epoch-activity constraint.

Entries excluded from the released catalogue

Four classes of entry are excluded. (i) Four target/bands (Barnard’s Star [B7], Wolf 359 [B7], SCR J1845–6357, Wolf 358) yielded no carrier result, each keeping a valid continuum non-detection. (ii) Giclas 9–38A and B [B3], the pair’s only matched pointing lying ~ 52 arcmin from either star, $35\times$ its primary-beam

TABLE 26

THE CP-72 2713 REPEAT TEST, BOTH BLOCKS SIDE BY SIDE, SEPARATING “NOT DETECTED AGAIN” FROM “A PERSISTENT SIGNAL OF THE ORIGINAL STRENGTH WOULD HAVE BEEN RECOVERED”. THE REPEAT BLOCK IS THE DEEPER OF THE TWO, MATCHES THE FIRST IN TUNING, CHANNELISATION AND INTEGRATION, AND ITS FREQUENCY AXIS REGISTERS TO A FIFTH OF A CHANNEL. A SOURCE AT THE FIRST BLOCK’S FLUX WOULD HAVE RETURNED $T_* = 6.21$ THERE; THE MEASURED VALUE IS 4.70. INTERMITTENT EMISSION IS NOT EXCLUDED BY THIS TEST, AND NO STATEMENT HERE ASSUMES OTHERWISE.

	first block	repeat block
Execution block	Xff0235_X4a6d	Xff0235_X502d
Start-to-start separation (h)	2.05	
On-source time (s)	2994	2994
Channel width (kHz)	488.28	488.28
Channel rms (mJy)	2.097	1.964
Primary-beam FWHM (arcsec)	18.2	18.2
Frequency axes registered to	0.18 channel	
Test frequency (GHz)	344.2697	
Test drift (kHz s ⁻¹)	-2.06	
Drift track sweeps (channels)	15.8	
Flux at that cell (mJy)	11.3 ± 2.0	-1.0 ± 1.8
T_* in this window	5.81	4.70
Largest control statistic	5.68	5.93
T_* expected if persistent	–	6.21
Pair inconsistent at	4.6σ	
First flux taken as exact	6.8σ	

FWHM. (iii) 13 windows failing the noise test of §A.3, whose noise sits a factor 388–743 above that of the shallowest sibling of the same block and channel width. (iv) mislabelled entries such as “ γ Lupi”, which resolves to HD 139664 at 17.40 pc; these are relabelled and no measured value changes.

THE α CMA B LIMIT IS WITHDRAWN. Checking whether Sirius A falls inside the control annulus of the Sirius B extraction — it does, in 10 of the 15 windows, those in Bands 4 and 5, at 7.43–7.98 arcsec from the extracted position — exposed a larger error. The pipeline propagates Sirius B with the *system* proper motion only and does not integrate the 50.13-year visual orbit (semi-major axis 7.50 arcsec), so the extracted position lies 3.1–3.6 arcsec from where Sirius B actually was at the observation epochs — 3–5 synthesised beams. The search therefore ran on blank sky. **The α CMa B threshold quoted in earlier versions of this paper is withdrawn** and no limit is claimed for that star. 0 threshold crossings depend on it, so nothing else in the paper changes; the windows remain in the release, flagged, because they are valid measurements of the wrong position. The 12 Band 9/10 windows toward AU Mic and HD 61005 are included and none is a detection or spatial outlier ($T_* \leq 4.91$).

The archival sweep also searched 198 windows in 47 execution blocks toward 22 stars between 40.8 and 49.8 pc, queued by a beam-matching step that did not check distance. They lie outside the sample this paper reports and enter no number in it, but they went through the identical frozen pipeline and influenced no threshold, mask or statistic decision, which makes them an external null: their stellar add-one ranks have median 0.437 against the 0.5 expected under exchangeability ($D = 0.11$, $p = 0.02$), and they contain 0 stage-1 flagged windows against 0.4 expected.

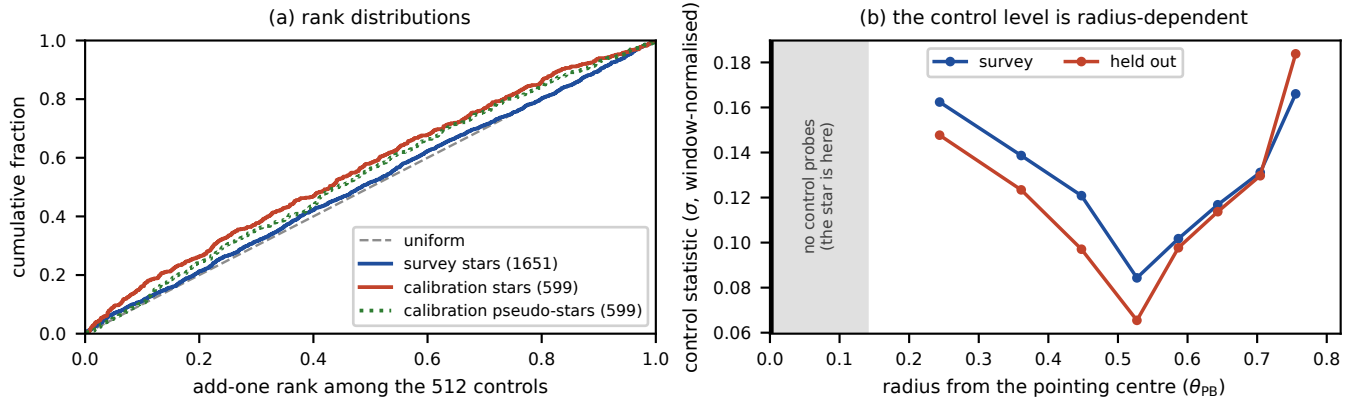


FIG. 14.— Why the spatial control ensemble is a prioritisation screen and not an exchangeable rank test. (a) add-one rank of the star among its 512 controls, for the survey and for the calibration windows carrying no attributed line: both lie above the uniform diagonal. The dotted curve is the *pseudo-stars*, control positions at the inner edge of the annulus ranked against the other controls, which contain no star and are displaced just as far (§C.1.2), so the displacement is instrumental. (b) the control statistic itself against radius from the pointing centre, in units of each window’s own noise scale. It is lowest at mid-annulus and rises towards both edges; the shaded strip inside $0.14\theta_{PB}$ holds no control probe at all, and that is where the star sits (median $0.0032\theta_{PB}$).

TABLE 27

THE INJECTION CAMPAIGN RESOLVED BY CONFIGURATION. P_{50} AND P_{90} ARE THE INJECTED AMPLITUDES, IN UNITS OF THE LOCAL NOISE, AT WHICH HALF AND NINE-TENTHS OF TRIALS ARE RECOVERED. NO AXIS SHOWS A DEFICIT LARGE ENOUGH TO MOVE THE QUOTED THRESHOLD OUTSIDE THE WINDOW-TO-WINDOW BRACKET OF $\times 0.48$ – $\times 1.35$; THE SUB-CHANNEL-PHASE ROWS ARE MEAN RESPONSE FACTORS, NOT RECOVERY AMPLITUDES, AND ARE THE REASON THE THRESHOLD CARRIES A RESPONSE CORRECTION.

Axis	Level	Trials	P_{50} (σ)	P_{90} (σ)
ALMA band	B3	432	4.8	6.7
	B6	1872	4.7	6.1
	B7	1152	4.4	6.0
	B8	432	4.4	6.0
	B10	144	4.5	5.9
Channel width (kHz)	15.3	144	4.2	5.8
	61.0	288	3.0	4.6
	122.1	432	4.2	5.8
	244.1	864	4.7	6.5
	488.3	1152	4.7	6.2
	976.6	1152	5.0	6.0
Array	12 m	3600	4.6	6.0
	ACA	432	4.8	6.7
Drift rate / ceiling	<1/3	1341	4.6	6.1
	>2/3	1410	4.5	5.9
	1/3–2/3	1281	4.6	6.5
Integrations	<150	720	4.6	6.4
	>350	1872	4.5	5.9
	150–350	1440	4.6	6.4
Sub-channel phase	0.000	80	2.0	–
	0.125	80	2.2	–
	0.250	80	2.3	–
	0.375	80	2.5	–
	0.500	80	2.7	–

Validation checks in detail

Each of the three checks summarised in §C.1.1, §C.1.2 and §5.2.6 is given here with the numbers behind it.

The hold-out reservation

The star-versus-control rank prioritises candidates and provides no detection p -value, because exact star/control exchangeability fails in the prospective hold-out sample. Every disposition in §5.2 therefore rests on astrophysical evidence or on recurrence, never on a rank alone.

The statistic and the mask were settled on the 1651 windows above and cannot also calibrate themselves. A hold-out was therefore reserved by a rule committed on 2026-09-14 (c75069040eab), before the archival search finished and before any reserved block was searched: a block is held out if and only if $\text{sha256}(\text{UID})_{[8]} \bmod 5 = 0$, subject to two guards that use identifiers alone — blocks in the previous release stay in the survey, and no system loses every block — so that the hold-out assignment can be recomputed from the UIDs alone. The three choices that can manufacture a candidate were fixed first: the detection statistic of Eq. 2 on 2026-09-09 (0c465f2661bd), the promote/retire criteria on 2026 September 11 (119a68a6ab50), and the hold-out rule 5 days after the statistic, with the first reserved block searched 3 days later. Because local commit timestamps can be rewritten, the dates are corroborated by GitHub server-side push times and by the Zenodo deposit’s minted date. The diagnostics built afterwards — the radius-corrected normalisation, the visibility-domain test, the drift-resolved injection analysis and the P_{90} budget — cannot promote a window.

Of 484 blocks processed, 77 are reserved (15.9 per cent), giving 315 windows toward 36 stars with no star

and no system lost from the headline sample. Four quantities are calibrated on those blocks and on nothing else: the rank displacement, the false-alarm tail factor, the radial behaviour of the control ensemble, and the claim that stage-1 status alone carries no candidate significance.

The ranks are displaced out of sample by very nearly the amount measured in sample: median add-one rank 0.405 against 0.5 (block-clustered 95 per cent 0.341–0.458), $D = 0.109$ at $p = 0.0011$ over 315 windows. The false-alarm tail factor, the measured pseudo-star first-rank rate divided by the exchangeable rate 1.95×10^{-3} , is 1.2 on 5040 trials, with a bootstrap clustered over the 77 reserved blocks giving 0.6–1.9, and the reserved blocks contain 0 stage-1 outliers against 0.61 expected. The standardised control level runs +0.12, +0.10, +0.12, +0.16 from the inner to the outer edge of the annulus, a mean per-window slope of $+0.067 \pm 0.033$: the displacement reproduces out of sample, but the steep inner-edge excess seen in sample does not, so the radial correction of §C.1.2 is one demonstrated mechanism for the displacement rather than the whole of it.

The older, processing-order calibration set is retained because it carries repeat observations the hold-out does not, and it must be read with one caveat stated plainly. **It was scored on 2026 September 17, by the pipeline as it then stood — that is, before the ACA control-annulus repair of §J.2.** Its 3 unattributed stage-1 outliers are therefore pre-repair outliers, and its measured tail rate, 3.7 unattributed events over 1322 windows, is a pre-repair tail rate. Because the repair *removes* outliers rather than creating them (below), that rate is an upper bound on the post-repair one, and it is used here only in that direction. The positive control recovers (β Pictoris CO at the stellar position in 4 of the 7 stage-1 windows), and Table 26 gives the CP–72 2713 repeat test in full.

WHICH EXTRACTION CHANGE REMOVED HD 14055 AND HD 23484. Earlier versions of this paper listed 3 unattributed outliers here beside 2 in the Results, and the two lists disagreed because they are different samples scored at different times, not because either count was mistyped. Table 29(b) resolves it by carrying both scorings of the same three windows. The repair re-extracted 1127 released windows in 278 blocks on the correct ACA primary beam, which moves each control outward to its intended radius; and because the noise scale σ is the median absolute deviation *across* the control positions, moving the controls moves the star’s own statistic as well as the ring’s. Over the 1127 windows scored both ways the repair is unbiased — median change -0.09 at the star and -0.00 at the ring — but it is not small: the typical move is 0.32 at the ring and 0.19 at the star, which is enough to change a rank decision in a window where the two were within a tenth of each other, as all three of these were. Under the repaired geometry HD 14055 and HD 23484 fall behind their own rings — 9 and 1 control positions respectively now exceed the star — and are no longer stage-1 events. 61 Vir survives, at $T_* = 5.96$ against a ring maximum of 5.65, which are the values Table 5 quotes. Recurrence rather than rank settles a marginal event in either scoring: none of the three recurs at its own tuning in a further block.

TABLE 28

DATA PARTITIONS USED IN THIS STUDY. “SEEN BEFORE THE STATISTIC FROZE” RECORDS WHETHER THE DATA WERE AVAILABLE TO THE CHOICES THAT DEFINE THE DETECTION STATISTIC AND THE MASKS; THE HOLD-OUT WAS RESERVED BY THE RULE OF §C.1.1, COMMITTED BEFORE ANY OF ITS BLOCKS WAS SEARCHED. ONLY THE SCIENCE SAMPLE CONTRIBUTES TO THE HEADLINE SEARCH RESULT.

Partition	Blocks	Windows	Stars	Purpose	Seen before the statistic froze?	In the headline result?
Science sample	404	1651	89	the search re-	yes	yes
Pre-specified hold-out	77	315	36	ported here calibrates rank, tail and radial behaviour	no	no
Out-of-sample, beyond 40 pc	47	198	22	external null	no	no
No surviving window	3	–	–	failed the quality cut	n/a	no

TABLE 29

THE EXTERNAL CALIBRATION SAMPLE AS SEARCHED AT 2026 SEPTEMBER 17 — *before* THE ACA CONTROL-ANNULUS REPAIR, WHICH IS WHY ITS COUNTS DIFFER FROM THE RESULTS. (a) WHAT THE FROZEN PIPELINE RETURNED ON 1322 WINDOWS IT HAD NEVER SEEN. (b) ITS 3 UNATTRIBUTED OUTLIERS, EACH SCORED TWICE ON THE SAME WINDOW: ONCE BY THE PRE-REPAIR EXTRACTION (THE FROZEN EXPORT) AND ONCE BY THE REPAIRED ONE (THE RELEASED CATALOGUE). “CONTROLS ABOVE” IS THE NUMBER OF THE 512 CONTROL POSITIONS EXCEEDING THE STAR AFTER THE REPAIR; A VALUE ABOVE ZERO MEANS THE WINDOW IS NO LONGER A STAGE-1 EVENT. GENERATED BY `COUNTS_V404.PY` FROM THE TWO EXPORTS, JOINED ON EXECUTION BLOCK AND WINDOW EDGES.

(a) Calibration sample, as searched

Execution blocks / stars	326 / 56
Spectral windows	1322
Median add-one rank of the star	0.419
against uniform	$D = 0.095$
Trigger crossings at the star	60
Stage-1 outliers	7
β Pictoris CO at the star	4
no astrophysical attribution	3
Expected unattributed at the measured tail rate	3.7

(b) Its unattributed outliers, before and after the ACA repair

Star	B	ν_{lo}	pre-repair			repaired	
			T_*	ring	T_*	ring	ring
61 Vir	7	344.872	6.16	5.88	5.96		5.65
HD 14055	7	330.387	6.03	5.96	5.57		5.99
HD 23484	6	229.610	5.30	5.25	5.52		5.55
Controls above the star after the repair, of 512:							
61 Vir							0
HD 14055							9
HD 23484							1

The rank distribution of the remaining 578 windows is adverse to the method. The star’s median add-one rank is 0.405 where 0.5 is expected, with 14 per cent of ranks below 0.1 and 6 per cent above 0.9 against 10 per cent each, so the distribution is displaced rather than growing a tail. The window-level figure ($D = 0.08$, $p < 0.01$) treats windows sharing a calibration as independent, but clustering removes neither the shift nor its significance: 99 of 149 blocks have a mean rank below 0.5 (sign test $p < 0.001$), and a star-clustered bootstrap over 578 windows places the median at 0.440 (95 per cent 0.42–0.46). The survey’s own 1651 windows lean the same way without resolving it (median 0.49, star-clustered 95 per cent 0.47–0.51) and the two distributions are indistinguishable ($D = 0.07$, $p = 0.03$), so the in-sample checks were underpowered rather than contradicted. The star-versus-control rank is an empirically useful candidate-prioritisation statistic, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection.

The radial gradient

The cause was the control geometry, not the primary beam. Earlier versions of this paper argued that primary-beam-corrected noise rising with radius left the probes at a different effective noise from the star. That explanation cannot be right: the statistic is formed on uncorrected amplitudes, whose thermal noise does not depend on position, and a primary-beam effect would act on both arrays, whereas the departure was confined to the ACA stratum. The original extraction code took the primary beam from a fixed 12 m dish for every window, so an ACA 7 m window had its control annulus placed at 58 per cent of the correct radius; at 230 GHz the inner edge landed at 3.8 arcsec instead of 6.6 arcsec, inside a typical ACA synthesised main lobe of 7.3 arcsec. Those controls carried the star’s own flux, stood high, and pushed the star low — the direction observed. The diagnosis below is of that original extraction; the repair, and what the released catalogue carries, are in §J.2.

Three tests establish it, all from the released products. The control probes are drawn once from a fixed seed, so fractional radii are identical in every window and each stored control statistic pairs with its radius. *First*, the excess is confined to the affected stratum and sits at the inner edge of the annulus: pooling by fractional radius, the innermost bin stands +0.11 above the outer half in the 529 ACA windows with stored control vectors, against +0.01 in the 297 12 m windows. Pooled over both arrays the profile runs +0.16 at $u = 0.24$, falls to +0.08 near mid-annulus and returns to +0.17 at the outer edge — the U shape that a monotonic radial noise gradient does not predict and that contamination of the innermost probes does. *Second*, matching on radius removes most of the departure: ranking the star against only the 54 probes with $u \leq 0.30$ moves the median over that restricted ensemble from 0.493 to 0.509 and $D = 0.022$ ($p = 0.54$) to 0.049 ($p = 0.005$). *Third*, a position with no star in it behaves the same way: an innermost probe treated as a *pseudo-star* and ranked against the other 512–1 probes is displaced as far as the real ones out of sample, median 0.443 against 0.429 ($p = 0.003$). The displacement is a property of *position in the field*, not of stars.

The repair belongs at the measurement stage: standardising each probe within its own radius bin, and the star on the innermost bin, removes the gradient by construction, with no free parameter. Run over all 1614 windows it changes the list — 2 windows lose the flag and 1 gains it, so the count falls from 12 to 11. We do not adopt the radius-corrected statistic as primary, and the reason is empirical: applied prospectively to 778 windows it played

no part in building, it leaves the stellar rank inconsistent with uniformity and moves the median further from the exchangeable value (§J.1). Since the correction removes the radial component by construction, what remains is larger than what it removed and is not radial. The frozen statistic therefore defines every count reported here; under the corrected one the survey would have 11 stage-1 windows, 10 attributed to CO and 1 unattributed — different windows, the same conclusion — and the released catalogue carries both.

The same mechanism touches completeness, which the injections alone would not reveal because they are placed at the stellar position only. Extrapolating the out-of-sample radial profile to the stellar radius gives an offset of $+0.08\sigma$, so an injected signal is divided by too much and the quoted recovery fractions are pessimistic by of order 2 per cent of the threshold: small, and in the safe direction, but not zero.

The general point is that spatial control positions prioritise candidates soundly in targeted interferometric data but cannot establish significance in them, because the field around a chosen target is not blank sky. The remedy is physical rather than statistical — a test for a point source at the stellar position, which the visibilities answer directly (§L).

The line mask

The searches as executed matched crossings to catalogued rest frequencies within one spectral channel at zero drift. The β Pictoris windows showed that inadequate: both fell outside that tolerance, by 10.6 and 21.9 MHz, while lying within 3.6 km s^{-1} of the systemic velocity of a resolved, mapped disc. Every crossing was therefore reclassified with a velocity-space mask, $\pm 50\text{ km s}^{-1}$ around each catalogued transition — a post-hoc classification criterion, never a detection threshold, whose half-width was adopted after the β Pictoris crossings had been seen. Reclassifying all 56 crossings at ± 20 to $\pm 100\text{ km s}^{-1}$ costs 2.3 to 8.7 per cent of gross bandwidth, and every crossing that narrowing the mask releases as unattributed is an identified carbon monoxide line of this survey’s own: the two rank-flagged unattributed crossings are the same pair at every half-width in that range. Widening it cannot suppress anything further, so the choice of ± 50 cannot have been tuned to remove a candidate.

The mask holds 17 transitions of 11 species at full JPL/CDMS precision (Pickett et al. 1998; Müller et al. 2005). *The search itself ran a shorter list*: 15 transitions of 9 species, rounded to 1 MHz, identical value for value to the veto list compiled into the search code; [C I](1–0) and H30 α were added to the paper’s mask at a later revision. Neither addition lies near a threshold crossing and no nearest-transition assignment changes between the two lists, so the difference is a documentation matter and not a correctness one; it is stated because one number was being used for two objects. The species criterion is the one objective rule statable in advance: the transitions ALMA’s band definitions are built around — the CO isotopologues, HCN, HCO $^+$, CS, silicon monoxide SiO, CN, H $_2$ CO and [C I] — which dominate the emission of disc-selected fields. The choice is consequential — the full Splatalogue selection of §I masks 75.1 per cent of the union against 4.1 per cent here — so “coincident with

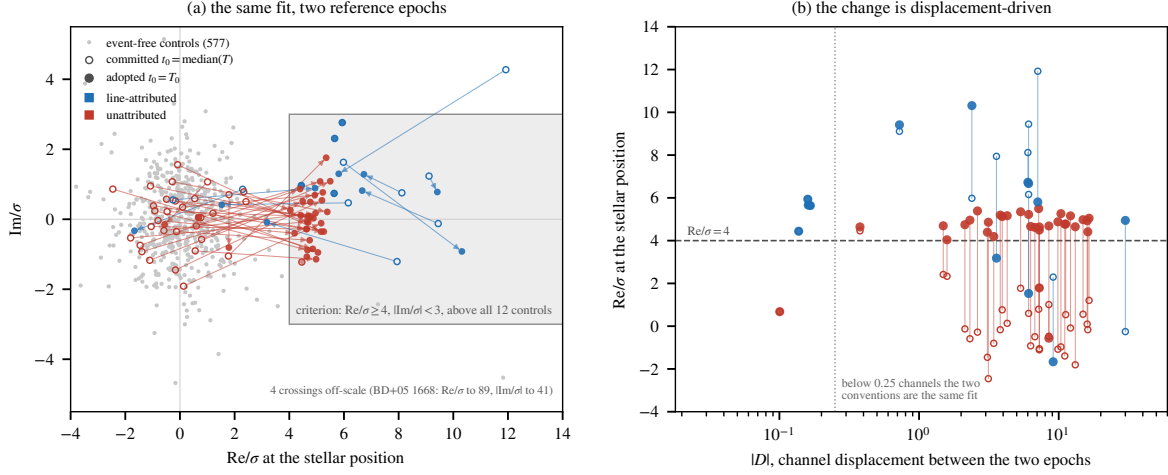
a known molecular line” is a statement about a catalogue rather than a binary property. Since no masked crossing exceeds its own control ensemble, no plausible widening or narrowing of the species list changes a disposition.

The cost is 4.1 per cent of the unique-frequency union and 4.7 per cent of window-summed gross bandwidth, CN and CO dominating. It is not uniform, so Table 17 gives it band by band: it reaches 6.5 per cent in Band 9 and stays below 1.5 per cent in Bands 3–5. Nor is it shared equally between the two experiments — 7.4 per cent of the Class A union against 3.1 per cent in Class B — so the fine-channel experiment pays 2.4 times the fractional price, its coverage being concentrated in the line-tuned windows a molecular mask is designed to remove. The two do not sum to the total because the classes overlap in frequency.

Masking is a design selection effect separating *discovery* from *classification*. A carrier at a line frequency is as detectable as anywhere else, but it cannot be classified, because the astrophysical prior at a catalogued transition is overwhelming. Removing those frequencies from the searched domain is the conservative choice, at the price that *we place no constraint at all on artificial emission lying inside the masked regions, at any signal strength*. A better design searches every frequency and attaches a *lower prior* to a line-coincident event instead, since these are exactly the frequencies a communicator wanting to be found near a natural reference might choose. The released channels support that second search, for which the β Pictoris crossings are the test case.

Additional robustness checks

The checks in Table 30 support the conclusions without changing them; the numbers behind each are in the released products, and the regeneration script reproduces them.



50 of 56 crossings fitted; 6 untested (fit outstanding, shown in the table, absent here). Localised: 11 committed, 40 adopted (38 with the epoch residual verified).

FIG. 15.— (a) The drift-following visibility fit for every crossing that could be fitted, in units of the measurement uncertainty: the real part of the visibility phase-rotated to the stellar position against its imaginary part. Open symbols are the committed reference epoch, filled the adopted one, joined for each crossing. Emission centred on the star makes the real part positive and significant and leaves the imaginary part near zero; emission displaced from the phase centre does not. Grey points are the control positions and control frequencies of the same fits, which carry no event by construction and show the null. The shaded region is the localisation criterion, fixed and committed at `edfcc0d3d7db` before any of these fits was run. (b) The same verdicts against $|D|$, the channel displacement between the two epochs: every crossing whose verdict moves does so above about one channel, and none moves below 0.25, which is where the injection campaign says the two conventions are the same experiment.

TABLE 30
FURTHER ROBUSTNESS CHECKS. EACH WAS RUN TO COMPLETION; NONE ALTERS A DISPOSITION.

Check	Outcome
R_σ residual scale at the unattributed positions	Consistent with the control median to better than 1 per cent; the stellar positions carry no noise peculiar to themselves
Fully symmetric region-maximum statistic	Flags a strict subset of the frozen screen's windows; no window is flagged by the symmetric form alone
Region-maximum audit, all windows	Both statistics released per window; the dependence of the candidate list on the choice is reproducible from the catalogue
Polarisation states	198 of 200 resolvable blocks deliver both parallel hands; a per-hand search remains available at no observing cost
CP-72 2713 diagnostics and second epoch	The event does not recur (§5.2); no diagnostic changes that
β Pictoris recurrence block by block	The disc CO recurs in every block whose tuning covers it, which is what makes it the survey's positive control

RECOMMENDATIONS FOR A NEXT-GENERATION
ALMA TECHNOSIGNATURE SEARCH

ALMA’s advantages here are its mm-wave sensitivity, an essentially unsearched tuning range, the spatial discrimination an interferometer supplies, and a public archive that can be reprocessed without new allocations. Six changes would enlarge what this experiment can say, given in priority order. The first two would change a conclusion; the rest widen a domain.

(1) *Use the visibilities.* A celestial point source at the stellar direction $\mathbf{x}_\star$ contributes

$$V_{ij}(\nu, t) = A(\nu, t) \exp(-2\pi i \mathbf{u}_{ij} \cdot \mathbf{x}_\star), \quad (\text{L1})$$

so rotating the phase centre to the star leaves a coherent signal entirely in the real part, with an imaginary component consistent with zero on every baseline. That asks whether the data contain a point source *where the star is*. The control ranking asks only whether one extraction is unusually large compared with its neighbours, which is a weaker question and one whose null depends on where in the field those neighbours sit. Applied to every window it would demote the control ensemble to a calibration layer and give each candidate a physically specific test rather than a rank. It requires retaining the visibilities, which this survey did not do.

(2) *Search the remaining public blocks these stars already hold.* §D.2.1 reports 115 of the 177 searched with the frozen pipeline, at no telescope cost. The gain is narrower than it sounds: every block is repeat coverage of a star and tuning already searched, so they add no star, no system and no frequency, and they extend only the time axis. They are also concentrated, attaching to 29 of the 89 stars. Of the 27 systems with a single searched epoch only 5 would gain a second block, and the extension moved 1 system out of the single-epoch class. A second block often supplies a baseline of hours rather than years, which one archive query settles before any download.

(3) *Retain more control spectra.* Full dynamic spectra survive for only 8 of the 512 control positions, which is what breaks the second-stage null in a structured field. Retaining all of them costs storage and nothing else.

(4) *Drift discrimination on coarse windows.* Every in-grid drift on a 15.625-MHz window is sub-channel over a track, so 1249 of 1651 windows measure amplitude but cannot distinguish a drifting carrier from a stationary one. Finer channelisation, or a drift model spanning several blocks, would recover that axis.

(5) *Extend the injection campaign and add polarisation.* The stratified campaign of Appendix G spans band, channel width, integration time and array; what remains unspanned is other target environments and edge-phased drift grids. A per-hand search buys partial discrimination against weakly polarised backgrounds at no sensitivity cost, but ALMA’s linear feeds mean it is blind to circular polarisation; a cross-hand extraction is what a circularly polarised carrier would require.

(6) *A boundary on drift linearity.* The search assumes linear drift over a track, and curvature matters once $\frac{1}{2}\ddot{\nu}T^2$ exceeds half a native channel, which only the shortest-period orbits reach: $P \lesssim 1.6$ d for a typical Class A window. Among the 50 known planets of the 23 hosts in the sample, 1 lies inside that boundary,

TRAPPIST-1 b at 1.51 d. The grid is built at constant acceleration, so the boundary is one number across the band, 3.60 ms^{-2} (Fig. 3); it clears the accelerations a transmitter on a rotating, orbiting planet cannot avoid by a wide margin, exceeding Earth’s equatorial rotational acceleration, 0.034 ms^{-2} , by a factor of 106 ($3.60/0.034$).

Finally, disc-CO false positives should be budgeted for, but the targets should not be pre-judged. 10 of the 12 stage-1 flagged windows are molecular emission toward disc hosts, arising in 2 of the 69 debris-disc systems searched: one flagged system in 34. Any future search on ALMA, NOEMA or the JVLA inherits that rate. The remedy is procedural — measure the continuum at each target *before* promoting a window, which would have caught HD 48370 and both β Pictoris windows at the first gate. We do *not* recommend lowering the threshold toward disc hosts: a detection statistic should not know whether its target is astrophysically interesting, or a real transmitter would be penalised for the company it keeps.

Sensitivity upgrades measured and declined

Three further improvements suggested by recent narrowband-search practice were implemented and measured against this survey’s own injection and completeness machinery. *All three are nulls, and are reported as nulls*; one real gain was measured and deliberately not applied.

Staged morphology gating in place of one amplitude threshold buys nothing, and the reason is measurable. The idea is to reject false alarms on their shape so the amplitude threshold can be lowered at fixed false-alarm rate. Our false-alarm rate has a measured logarithmic slope of -3.84 per σ at the trigger, so a veto must remove 62 per cent of false alarms to buy 5 per cent in flux, 85 per cent for 10 and 98 per cent for 20. Two discriminants we already own — feature width and time concentration, calibrated on the injection side alone as the loosest cuts retaining 95 per cent of injected carriers — remove 13.3 per cent, worth $5.00 \rightarrow 4.97$, a flux gain of $\times 0.994$, while costing 5–7 per cent of real carriers at every amplitude. Because P_{90}^{sel} is defined at 90 per cent recovery, a flat completeness cost is expensive there: τ_{90} moves $6.61 \rightarrow 7.67$ and the net factor is $\times 1.16$ ($1.07\text{--}1.21$), i.e. 16 per cent *worse*. The physical reason is that online Hanning smoothing correlates adjacent channels at $\rho = 0.667$, so a noise maximum is already two to three channels wide and looks like a carrier; the best of the four discriminants reaches only $\text{AUC} = 0.728$ (the shoulder ratio).

A cut that rejects nothing may simply be a broken cut, so the same cuts were run on signals they ought to reject. They reject a 20-channel resolved line 111/111 times and a 10-per-cent-duty-cycle transient 107/107, while retaining 96 per cent of continuous carriers. Both discriminants have full power against the morphology each was built for. The null is therefore a property of our false-alarm population — Gaussian-like stack maxima, plus the handful of broken windows of §5.2.2 — and not of a failed test. We keep both tests where they already are: width as the identification of resolved line emission, time concentration as a per-crossing check.

Recovering the drift-grid quantisation loss buys nothing either, because it is already booked. Interpolating around the grid maximum recovers $\times 1.0081$ on an in-

jected carrier, but the noise-side maximum inflates by $\times 1.011$ against the trigger and $\times 1.016$ against the rank screen, so the net is 0.997 ± 0.003 and 0.992 respectively — at or below unity. More importantly, *the completeness campaigns draw the injected drift rate and sub-channel offset continuously rather than on the grid*, so the average quantisation loss is already contained in the measured P_{90}^{sel} and *must not be corrected for twice*. Of the 6.7 per cent total alignment loss, 5.2 per cent is the integer-channel-shift ceiling of the architecture and only 1.49 per cent is grid and sub-channel misalignment.

The reported-yield artefact of top-hit deduplication is absent here, and of the opposite sign. Our search applies 0 deduplication windows, and over 12 real windows spanning a factor 8 in drift ceiling the number of channels above threshold is monotone non-decreasing in 10 of them: doubling the ceiling adds a median 8 per cent and quartering it removes 44 per cent, while the peak statistic moves by at most 3.4 per cent. Two documentation items follow and no published number changes: the released `hit_freqs_GHz` list is truncated at 50 entries in 3 windows, all of them resolved CO, and `drift_max_Hz_s` understates the largest drift actually searched by up to one grid step.

One gain is real, and is measured but not applied. The survey triggers on a single smoothed channel, into which the correlator delivers half of an unresolved carrier, and books the penalty in C_{resp} rather than filtering it away. A matched filter across the three-channel response recovers $\times 1.134$ at a channel centre, $\times 1.08$ averaged over sub-channel offset for one fixed filter, and $\times 1.17$ for a bank re-matched to the offset — which adds trials and must be paid for at the threshold in the manner of Eq. 4. It is *not* $\times 1.2247 = \sqrt{3/2}$, the figure a lag-1-only covariance gives: the Hann kernel’s own autocovariance has $\rho(1) = 0.667$ and $\rho(2) = 0.167$, verified on the data at 0.651 and 0.145 over 25 windows (18 within 0.08 of the lag-1 prediction). The gain comes entirely from whitening that correlation, which makes it sensitive to the response model and to residual spectral baseline structure, and it is $\times 0.972$ — a *loss* — for a carrier on a channel boundary. It also applies to the $n = 1$ (Hanning-only) population only, 91 per cent of the catalogue, since channel-averaged windows have nearly uncorrelated channels. We state it as the obvious next step and *not applied* here: it changes the search statistic, and with it the crossing list and the frozen catalogue, which is not a change to make inside a revision.

RADIO-FREQUENCY INTERFERENCE

Millimetre interferometry is an unusually poor environment for terrestrial interference, though not an empty one, and an earlier version of this statement overreached. The searched range is not above all allocated services: ITU Radio Regulations Article 5 carries active allocations throughout 90–275 GHz, and WRC-19 identified parts of 275–450 GHz for land-mobile and fixed use. The specific hazard in this band is the 94.0–94.1 GHz Earth exploration-satellite (active) allocation used by space-borne cloud radar, CloudSat and, since 2024, Earth-CARE, which is a documented contaminant of ALMA Band 3. 1 of the 1651 searched windows overlaps that band, and it carries 0 threshold crossings and 0 stage-

1 events, so no result here rests on it and no overpass timing needs to be checked. What does favour this band is that terrestrial transmitters are sparse compared with centimetre wavelengths, that the array sits at 5000 m on a site chosen partly for radio quietness, and, decisively, that an interferometer is not a total-power detector. A signal that is not in the far field at the phase centre does not fringe-track: it acquires a rapidly varying geometric phase across the array and is suppressed in the correlation rather than imaged.

What interference screening this survey actually has, stated with its coverage, because the coverage is uneven. Two screens are computed for every searched window from the released catalogue and are reproducible from it: the 94.0–94.1 GHz allocation test above, and the sky-frequency and intermediate-frequency clustering tests below. The 512-position rank screen asks whether an excess is at the star or everywhere in the field, which is an interference test as well as a prioritisation, and covers every window. Recurrence at the same tuning in an independent epoch covers the crossings for which a covering repeat block exists. The visibility fit of §5.2.1 asks whether an excess has the phase signature of a source at the phase centre, which a near-field emitter would not have, but it is a consistency check rather than a filter and it is not what disposes of anything here. Finally, the correlator flagging is the observatory’s own, replayed verbatim from each block’s delivered calibration script; it is not recorded per window in the release and cannot be reconstructed from it. *No per-channel interference flag of our own exists, and none can be recovered.*

Three measurements support that reasoning. First, interference is a property of the observatory and not of the star, so a terrestrial carrier should appear at the same *sky* frequency in every window that covers it. Of the 2 unattributed events, 2 have a released crossing frequency; 32 windows toward 8–24 other stars in 74 other blocks cover those frequencies, and 0 of them carries a crossing in the same channel. Second, the crossing population carries no residual clustering in sky frequency once the molecular attributions are accounted for. That statement replaces a weaker and wrongly-argued one, and the replacement is worth setting out, because the honest repair makes the evidence for interference *weaker* while the test that delivers it is much stronger.

Binned at 1 MHz, the largest multiplicity among the 1397 released per-window peak frequencies is 4 against a continuous-uniform null of 2.44 ($p = 0.0115$). Three things are wrong with that. Those 1397 values are not the survey’s 56 threshold crossings but every window’s own peak-SNR frequency, and 1067 of them are coarse continuum windows whose peak is a noise maximum. The null ignores the fact that a peak can land only on one of its window’s own ALMA channels, and that two windows sharing a standard tuning share an *identical* grid; drawn on the windows’ own grids the null mean is 5.05 and 4 is entirely typical ($p = 1.0000$). Every member of the crowded bins is a coarse 15.62 MHz continuum window sharing a tuning, and not one of them is a threshold crossing. And the statistic has no power against clustering wider than about one channel: against six, twelve and twenty-four events spread over 156 MHz it rejects 7.8 per cent of the time or less (Fig. 16a). Whatever its p -value, it was never evidence either way about the

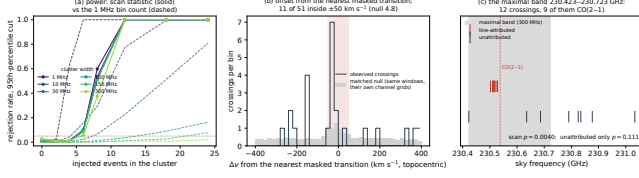


FIG. 16. — (a) Measured rejection rate of the 1 MHz multiplicity statistic (dashed) and of the scan statistic (solid) against the number of injected events, by cluster width, at a threshold fixed beforehand from the channel-grid null. The old statistic is blind to anything wider than a channel. (b) Each crossing's velocity offset from the nearest masked transition, with a null drawn from the same windows' own channel grids: 11 of 51 inside $\pm 50 \text{ km s}^{-1}$ against 4.7 expected. (c) The maximal band, 230.423–230.723 GHz, with its 12 members coloured by attribution and CO(2→1) marked.

structure at issue. It is not useless — against a cluster *exactly* one channel wide it beats the replacement outright, 100.0 per cent against 8.7 at six events — but it is a channel-coincidence detector and should be described as one.

A scale-aware scan statistic — the maximum count in any band of width 1 to 300 MHz, calibrated on the same channel-grid null — rejects 100.0 per cent at twelve injected events and is flat in cluster width. Applied to the 51 crossings with a released frequency it *does* find clustering, at $p = 0.0040$ (12 observed against 7.38 expected). The cluster is the survey's own molecular emission: its maximal band, 230.423–230.723 GHz, holds 12 crossings of which 9 lie within 50 km s^{-1} of CO(2→1), and removing the line-attributed crossings removes the clustering ($p = 0.111$ for the 40 unattributed). Equivalently, and with no choice of band or width at all, 11 of the 51 crossings lie inside the survey's own $\pm 50 \text{ km s}^{-1}$ tube against 4.7 expected under the matched grid null, $p = 0.0007$. *The crossing population is significantly concentrated on the masked transitions, and that concentration is the attribution rather than a residue left over after it.*

Third, an artefact of the signal chain would sit at a fixed intermediate frequency rather than a fixed sky frequency, and the crossings do not: their fractional positions within their own windows have median 0.50 and are uniform to $D = 0.022$ at $p = 0.49$ over 1397 crossings. CP–72 2713's crossing, the one with a released frequency, lies 35 channels from its nearest window edge.

Fourth, a cross-target occupancy screen over every searched window and not only the crossings. Interference is a property of the observatory, so a terrestrial carrier should recur at the same sky frequency toward *unrelated* stars. Over all 3027 searched windows toward 133 stars, with each window's peak randomised on its own channel grid, the number of cross-star coincidences within 5 MHz is 569 against 554 ± 13 expected ($z = 1.1$, $p = 0.15$); among the 109 windows whose peak exceeds 5σ it is 4 against 5.2 ± 1.9 , i.e. *below* the null ($z = -0.6$), and with the molecular-line mask applied 1 against 2.1 ± 1.3 ($z = -0.8$). There is no excess to explain. 35 of those

109 windows sit on a masked transition (CO(2–1) $\times 17$, CO(1–0) $\times 10$, CO(3–2) $\times 4$, $^{13}\text{CO}(2–1)$ $\times 3$, SiO(5–4) $\times 1$), which is the point: a CO line falls at almost the same *topocentric* frequency toward every nearby star, so cross-star frequency coincidence is exactly what real line emission produces, and the screen cannot distinguish it from interference unless the mask is applied first.

An earlier pass of this screen reported a population-level excess at $z = 3.5$, and that is withdrawn. Its key was the product directory name, which for 2529 of the 3027 records embeds the execution block, so a star's own repeat observations counted as different targets and every persistent feature — a real line most of all — entered as a cross-target coincidence. Keyed that way the statistic reproduces here at 680 against 625.0 ± 12.8 ($z = 4.3$), which is why its arithmetic looked sound; keyed on the star it is the null result above, and the direction of the error was always to inflate. An older stored report of the same module, dated 2026-09-01, records 0 coincidence clusters over 144 windows; that is a separate indexing defect which made the module incapable of reporting a coincidence at all, and it is withdrawn with the rest. *The statement this appendix makes is therefore now carried by a screen that could have contradicted it.*

None of this is a substitute for the ON/OFF cadence a dedicated search uses, and we do not present it as one. Accordingly we claim only that *no identified interference mechanism accounts for these events*; allocation tables cannot exclude harmonics, intermodulation products, unconventional emitters or near-field sources, and an archival dataset without a simultaneous off-source strategy cannot do better than that. What it establishes is that the crossing population as a whole carries none of the three signatures interference leaves — fixed sky frequency, fixed intermediate frequency, or presence in unrelated targets — which is a statement about the population and not a clearance of any individual crossing. One crossing is rejected as instrumental on these grounds: BD+05 1668, which appears in the same coarse channel toward unrelated stars and whose visibility fit places it far off the phase centre.

HD 48370, in one line. $T_{\star} = 26.83$ against a ring maximum of 18.36, a margin of 8.47; CO(2→1) at $v_{\text{bary}} = +41.2 \text{ km s}^{-1}$, matching the foreground cloud Cataldi et al. (2023) report; and the visibility fit puts the emission *off* the star, with an imaginary part 7.9σ from zero. *Disposition: foreground carbon monoxide, on three independent lines of evidence, none of them the rank.*

Three further checks are reported in Appendix D: a prospectively searched extension of 115 public blocks, which recovers the β Pictoris disc CO in three previously unsearched epochs and adds no unattributed event; the β Pictoris recovery itself, which is the survey's astrophysical positive control for both the spatial and the spectral machinery; and the absence of any shared data with the only prior ALMA-archival search, which leaves the null result resting on internal validation alone.

REFERENCES

- Astudillo-Defru N., et al., 2017, A&A, 602, A88
 Astropy Collab., 2022, ApJ, 935, 167
 Backus P. R., et al., 2002, BAAS, 34, 1173
 Blomme R., et al., 2023, A&A, 674, A7
 CASA Team, 2022, PASP, 134, 114501
 Burton K., 2024, PhD thesis, Univ. Colorado Boulder, ch. 4
 Cataldi G., et al., 2023, ApJ, 951, 111
 Choza C., et al., 2024, AJ, 168, 254
 Cocconi G., Morrison P., 1959, Nature, 184, 844
 Czech D. J., et al., 2021, PASP, 133, 064502
 Dent W. R. F., et al., 2014, Science, 343, 1490
 Drake F. D., 1961, Physics Today, 14(4), 40
 Drake F. D., 1974, in Sagan C., ed., CETI. MIT Press
 Enriquez J. E., et al., 2017, ApJ, 849, 104
 Gaia Collab., Vallenari A., et al., 2023, A&A, 674, A1

- Garrett M. A., et al., 2026, MNRAS, in press (arXiv:2605.10212)
- Gentile Fusillo N. P., et al., 2021, MNRAS, 508, 3877
- Ginsburg A., et al., 2019, AJ, 157, 98
- Harris C. R., et al., 2020, Nature, 585, 357
- Huang B.-L., et al., 2026, ApJ, 1006, 9
- Jacobson-Bell K., et al., 2025, AJ, 169, 233
- Johnson O. A., et al., 2023, AJ, 166, 193
- Li J.-K., et al., 2026, arXiv:2603.19023 (preprint)
- Ma P. X., et al., 2023, NatAs, 7, 492
- MacGregor M. A., et al., 2020, ApJ, 891, 80
- Manunza L., et al., 2025, AcAau, 233, 155
- Margot J.-L., et al., 2023, AJ, 166, 206
- Mason L. A., Garrett M. A., Wandia K., Siemion A. P. V., 2025, MNRAS, 536, 2127 (arXiv:2411.19827)
- Matrà L., et al., 2017, MNRAS, 464, 1415
- Morrison I. S., et al., 2023, PASA, 40, e038
- Müller H. S. P., et al., 2005, JMoSt, 742, 215
- Ortiz Ceballos K., et al., 2026, arXiv:2609.16720 (preprint)
- Pickett H. M., et al., 1998, JQSRT, 60, 883
- Poznanski D., et al., 2025, AJ, in press (arXiv:2505.03927)
- Price D. C., et al., 2020, AJ, 159, 86
- Sheikh S. Z., et al., 2025, AJ, 169, 118
- Sheikh S. Z., 2020, IJAsB, 19, 237
- Sheikh S. Z., et al., 2021, Nature Astronomy, 5, 1153
- SpaceX, 2024, FCC IBFS SAT-MOD-20230207-00021
- Tarter J., 2001, ARA&A, 39, 511
- Tremblay C. D., et al., 2024, PASA, 41, e004
- Vidal C., et al., 2026, arXiv:2605.21093 (preprint)
- Virtanen P., et al., 2020, NatMe, 17, 261
- White G. J., et al., 2026, MNRAS, submitted
- Wright J. T., et al., 2018, AJ, 156, 260
- Zhang Z.-S., et al., 2020, ApJ, 891, 174